

Trace-Tempered Choice

A Self-Contained Axiomatisation with Explicit Payoffs and Records

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Abstract

An action can matter twice: through the payoff it produces and through the visible trace it leaves. We separate these roles in the primitive domain. A finite channel maps actions into pairs consisting of a payoff-bearing outcome and an explicit record. Continuous weak order, coherent comparison environments, data processing on both sides of the channel, and a branchwise sure-thing principle characterize

$$\mathbb{E}[u(O)] + \lambda I(A; O, R), \quad \lambda > 0.$$

No primitive payoff–trace separability axiom is required. It is derived from the temporal asymmetry of the domain: records may arrive first, while payoff is terminal. The proof isolates that derivation in a calibrated branch-recursion lemma. A complete proof of the mutual-information characterization for the pure-trace restriction is included as an appendix, so the paper is logically independent of earlier versions of the traceability paper. Every primitive axiom is shown to be independent of the others.

1 Introduction

Shower upon him every earthly blessing . . . He would even risk his cakes and would deliberately desire the most fatal rubbish, the most uneconomical absurdity . . . simply in order to prove to himself—as though that were so necessary—that men still are men and not the keys of a piano. . . . If you say that all this, too, can be calculated and tabulated . . . then man would purposely go mad in order to be rid of reason and gain his point!

— Fyodor Dostoevsky, *Notes from the Underground*, Part I, Chapter VIII [22]

Dostoevsky’s provocation has the structure of a choice problem. The cakes represent material payoff, and the Underground Man is willing to risk them so that his conduct can attest to something: that it is his own rather than the mechanical implication of a table. Yet the proof he seeks is self-defeating. An act fixed in advance cannot provide it, and deliberate perversity offers no escape: rebellion, too, can be “calculated and tabulated.” His final resort is therefore paradoxical: he would “purposely go mad in order to be rid of reason”—not because madness is what he values, but because it is his last imagined way of preventing the table from closing before he acts. This paper takes that evidentiary idea literally, in a deliberately narrow statistical sense. In Dostoevsky’s example, conduct is observed directly: the act supplies its own evidence. We consider the more general case in which an act is observed only through a possibly noisy payoff-bearing outcome and an explicit visible record.

For visible consequences to carry information about an act, two conditions are needed. There must be uncertainty beforehand about which act will occur, and the eventual payoff–record pair must discriminate statistically among the possible acts. Either condition can fail independently.

A lottery over acts leaves no trace when every act generates the same distribution of visible consequences; conversely, even a consequence that perfectly reveals the act conveys no new information when the act was already known. We call the non-instrumental preference for this conjunction a preference for *traceable agency*. The paper axiomatizes it jointly with ordinary preferences over payoff risk.

The idea has a simple formal expression. Let A be a finite set of actions, O a finite set of payoff-bearing outcomes, R a finite set of explicit visible records, and $K : A \rightarrow \Delta(O \times R)$ the channel relating actions to visible consequences. An action lottery $q \in \Delta(A)$ gives the distribution of the realised action; conditional on that action, K gives the joint distribution of payoff and record. For each fixed channel K , the primitive is a weak order $\succeq_K$ over action lotteries. Thus $q \succeq_K p$ means that q is evaluated at least as highly as p in environment K . Record processing may rewrite what is visible while leaving payoff fixed, and action processing may coarsen the description of the realised action; neither operation can increase traceability. The remaining axioms impose order, continuity, and comparison coherence, together with a dominance condition for records that arrive in stages.

The main result is an if-and-only-if characterization. A family $\{\succeq_K\}_K$ satisfies the eight axioms if and only if there are a nonconstant payoff utility $u : O \rightarrow \mathbb{R}$ and $\lambda > 0$ such that, for every finite joint channel K and every $q, p \in \Delta(A)$,

$$q \succeq_K p \iff \mathbb{E}_{q,K}[u(O)] + \lambda I_{q,K}(A; O, R) \geq \mathbb{E}_{p,K}[u(O)] + \lambda I_{p,K}(A; O, R).$$

Reversing the three trace-facing clauses yields the trace-averse dual with $\lambda < 0$; replacing them by indifference yields expected utility, or $\lambda = 0$.

The information term has a simple interpretation. Observing (O, R) induces a posterior belief about the realised action. Its traceability is the expected reduction in uncertainty about A :

$$I_{q,K}(A; O, R) = H(q) - \mathbb{E}_{q,K}[H(q(\cdot \mid O, R))].$$

If visible consequences are independent of actions, the value is zero. If the action is known in advance because q is degenerate, the value is also zero. If each action in the support of q produces a distinct visible consequence, the value is $H(q)$ —Dostoevsky’s case, in which conduct is observed directly. Thus the criterion is not a taste for randomisation as such. Randomisation matters only because it creates uncertainty about agency for the visible consequence to resolve.

A more mundane example fixes ideas. A donor can route a gift to a named project or to a pooled fund; the material effect is payoff-bearing, while the public attribution is an explicit record. Holding the distribution of material effects fixed, if the two routes generate the same distribution over records, the record is independent of the act and its additional traceability is zero, however uncertain the giving. If the routes report differently, traceability is positive exactly when the realised route is genuinely uncertain beforehand—the donor wavers, or delegates the decision—and it reaches $H(q)$ when the visible consequence identifies the route. A donor certain to give by name leaves no trace in this sense: a perfectly attributed gift that was never in doubt yields a record that only confirms what was already known. If material consequences themselves differ across routes, they too form part of the trace. The criterion prices the news a visible consequence carries about conduct, not knowledge of it.

The example ranks lotteries within a single fixed environment; the representation also speaks across environments. The environment itself is never an object of choice: the theory begins neither with a preference over pairs (q, K) nor with a numerical index on channels. When lotteries carried by different channels must be compared, the channels are embedded as labelled blocks of a single larger environment, and the comparison is an ordinary one between two block-supported lotteries in a fixed problem. The theorem represents all such comparisons by the same u and λ on one scale: it does not merely say that each environment carries a ranking ordinally equivalent to the displayed criterion; it identifies common payoff and trace units across environments.

Traceable agency cannot be reduced to any preference over consequences alone, because here the action is part of the consequence: what is valued is the joint distribution of the action with the visible pair, not the distribution of the visible pair by itself. Consider a channel with three actions, a constant payoff, and two records. The first two actions produce different records with certainty, while the third produces either record with equal probability. A fair lottery over the first two actions and a certain choice of the third induce the same joint distribution of payoff and record. Yet in the first case the record identifies the realised action, whereas in the second it conveys no information about it. Every index of the induced consequence distribution—expected utility over payoffs, or over payoff–record pairs—is indifferent, while the traceability component assigns one bit to the first lottery and zero to the second.

This difference does not make traceable agency a substitute for expected utility. The point of the present theorem is to axiomatize their joint decomposition rather than assume it:

$$V(q, K) = \mathbb{E}_{q,K}[u(O)] + \lambda I_{q,K}(A; O, R).$$

The payoff–record distinction is essential to that identification. It permits records to be processed without permitting payoff to be garbled, and it gives branchwise continuation a temporal asymmetry: records may resolve first, whereas payoff is terminal. Those restrictions derive common payoff units, the additive decomposition, and matching neutrality. No primitive payoff–trace separability axiom is imposed.

A willingness to pay for decision rights and payoff autonomy is well documented. Bartling, Fehr, and Herz infer a positive intrinsic value of decision rights for 83 percent of participants assigned to the principal role; the average control premium is 16.7 percent and significantly positive in all ten parameterisations, and an independent replication closely reproduces their individual-level results [7, 13]. Owens, Grossman, and Fackler elicit beliefs separately: expected-money maximisation predicts betting on oneself in 56.4 percent of decisions, yet subjects do so in 64.9 percent, sacrificing 8–15 percent of expected earnings [41]. Neither design varies what consequences reveal about the act, so neither identifies the traceability channel.

Evidence closer to the action–consequence channel comes from two studies, the second incentivised, in which subjects prefer action menus with greater Jensen–Shannon divergence among consequence distributions even when current expected payoff and outcome entropy are fixed [39, 40]. Under their uniform prior over two actions, this index is exactly mutual information. Both designs, however, let token values change after the menu is chosen, so divergence also provides instrumental flexibility; a later model comparison finds that dynamic expected utility best explains the choices [37]. These studies therefore do not isolate a non-instrumental value of mutual information and, to our knowledge, no existing experiment does.

The model suggests the following direct test. Fix a channel with four actions, $a_1, \dots, a_4$. Every action yields the same monetary payoff together with a binary non-payoff record. Actions a_1 and a_2 each produce a different symbol; actions a_3 and a_4 each produce either symbol with equal probability. Participants choose between a fair coin flip over $\{a_1, a_2\}$ and one over $\{a_3, a_4\}$. The two flips induce the same joint distribution of payoff and record and involve the same degree of deliberate randomisation; only the first makes the record informative about the realised action. Traceable agency therefore predicts a strict preference for it. Because the record affects neither material payoffs nor any later decision, this comparison already excludes instrumental flexibility.¹

Let us now discuss the eight axioms. The structural axioms (A1)–(A3) provide the comparison framework. Each channel carries a continuous weak order over action lotteries. When lotteries associated with different channels must be compared, the channels are embedded as disjoint components of one larger environment; duplicating a component or adding components not involved in the comparison must not affect the ranking.

¹A garbling control completes the design. Repeat the choice in a second treatment with a channel that writes a fair record independently of the realised action: both flips then have zero traceability, and the preference should disappear. A preference surviving the garbling would indicate salience or framing effects rather than a value of trace.

Axioms (A4) and (A5) state the two sides of data processing. The agent never gains when her record is rewritten after the fact, holding payoffs fixed, nor when her action is reported noisily. Because action processing is stated on the joint law, reported actions reached with zero probability impose no constraint: their irrelevance is a consequence of the axiom rather than a convention appended to it.

Axiom (A6) is a sure-thing principle for records that arrive in stages. If one continuation is weakly preferred after every first-stage record that can occur, then the branch-by-branch replacement cannot make the compound environment worse; a strict improvement on a positive-probability branch makes it strictly better. This condition supplies the system’s cardinal discipline. It implies invariance to common public randomisation and calibrates continuation values across posterior action lotteries.

The two orientation axioms separate payoff from trace. Axiom (A7) requires some pair of payoff outcomes to be strictly ranked. Axiom (A8) requires full revelation of a genuinely uncertain action to be strictly preferred to an uninformative record when payoff is fixed. The first makes u nonconstant; the second selects $\lambda > 0$. Neither imposes a numerical scale.

A sketch of the proof follows the theorem in Section 4; complete arguments, including a self-contained proof of the pure-record characterisation on which the trace component rests, are in the Appendix.

2 Related Literature

The paper belongs to the decision-theoretic tradition that enriches the object of preference to capture motives that expected utility over final payoffs alone leaves out. Kreps [35] and Dekel, Lipman, and Rustichini [20] represent flexibility and subjective uncertainty about future tastes through preferences over menus, and the freedom-of-choice literature initiated by Pattanaik and Xu [42] ranks opportunity sets by the freedom they embody; here decision rights and feasible actions are held fixed, and what varies is ex-post authorship rather than ex-ante opportunity: whether the realised visible consequence statistically records the action. Gul and Pesendorfer [29] represent temptation and commitment; Kreps and Porteus [36] enrich the domain to temporal lotteries so that the timing of the resolution of uncertainty can itself carry utility; and Grant, Kajii, and Polak [28] characterise within that domain an intrinsic preference for information, valued independently of any planning use. The information valued here flows in the opposite direction—not what signals teach the agent about the world, but what visible consequences teach the world about the agent. Deliberate randomisation has also been axiomatised through a hedging motive [17]; unlike that motive, traceability depends on the observability of the action–consequence record. The nearest neighbours lie in the signaling tradition descending from Spence [48], in which actions are informative about hidden characteristics: social-image models explain attribution-seeking behaviour, such as named rather than anonymous giving, through observers’ beliefs entering utility [27, 10, 3]; self-image models direct the same inference inward, towards beliefs about one’s own latent traits [9]; and in Bernheim’s conformity model [11] agents *suppress* the legibility of their predispositions, an instrumental counterpart of the deniability orientation developed below. In all of these, what is inferred matters through its consequences—esteem, sanction, or self-regard; here no observer’s response enters utility, beliefs figure only as the statistical yardstick against which the trace is read, and the preference survives anonymity. Attribution, expressive value, and reputation are possible microfoundations for the behaviour the axioms describe, but they are not primitives of the theorem; what none of these literatures has made the object of preference is the trace itself.

The relation to axiomatic information theory is different in kind. Shannon [46] derived entropy from continuity, monotonicity on uniform distributions, and a grouping recursion; Khinchin [32] and Faddeev [24] progressively weakened these postulates, Faddeev reducing them to symmetry, continuity of the binary entropy, and the recursion that appears here as a derived

object. Rényi [44] showed that relaxing the grouping postulate to general quasi-arithmetic means, while retaining additivity over independent experiments, yields the one-parameter family bearing his name, in which Shannon entropy is the unique member still satisfying the recursion; Aczél and Daróczy [1] give a systematic treatment by functional equations, and Csiszár [19] surveys the field. Mutual information itself is typically characterised indirectly, as entropy reduction or as the relative entropy between the joint law and the product of its marginals: Hobson [31] characterises relative entropy by grouping and additivity conditions, and Baez and Fritz [6], extending the treatment of entropy as a measure of information loss in Baez, Fritz, and Leinster [8], derive it as the essentially unique functorial, convex-linear, lower semicontinuous invariant of Bayesian inference.

All of these characterisations take a numerical functional as primitive and impose structural equations on distributions, random variables, or information-loss maps. The present theorem starts instead from ordinal preference relations over lotteries in fixed joint environments and derives the functional: common cardinal structure, the expected-utility component, the information chain rule, and finally Faddeev’s recursion are consequences of behavioural axioms rather than postulates on a given function. The Shannon part of the exercise is thus the information-theoretic analogue of expected-utility theory, in which the numerical form of the criterion is the output of a representation theorem rather than its input; what is characterised is not an abstract measure of information but the preference of an agent for whom that measure is the value of traceable agency.

A smaller literature takes information structures themselves as the ranked domain. Gilboa and Lehrer [26] characterise which functions on partitions are values of information for some Bayesian decision problem, and Azrieli and Lehrer [5] extend the characterisation to stochastic structures; in both, value is instrumental by construction. Cabrales, Gossner, and Serrano [14] come closest to the present representation from the instrumental side: for ruin-averse investors, the ranking of information structures is represented by entropy reduction, so that mutual information emerges from a specific class of decision problems rather than from ordinal axioms. On the non-instrumental side, Caplin and Leahy [16] put beliefs into the utility function, and Ely, Frankel, and Kamenica [23] derive the entertainment value of suspense and surprise from preferences over the path of beliefs. The present paper combines the two strands: its domain consists of lotteries in fixed payoff–record channels, its trace valuation is non-instrumental, and, unlike either strand, the functional is not posited or induced by a decision problem but derived ordinally and pinned down jointly with material utility.

The closest formal literature axiomatises the cost of information. Pomatto, Strack, and Tamuz [43] take as primitive a cardinal cost functional on Blackwell experiments [12], defined without reference to a prior, and show that Blackwell monotonicity, continuity, additivity over independent experiments, and constant marginal cost characterise the log-likelihood ratio costs: weighted sums of Kullback–Leibler divergences between the state-conditional signal distributions. Caplin, Dean, and Leahy [15] and Denti [21] work from state-dependent stochastic choice data generated by rationally inattentive agents and characterise posterior-separable cost functions, with the Shannon cost singled out by invariance conditions; Frankel and Kamenica [25] characterise valid measures of information as expected reductions of measures of uncertainty, with entropy reduction selected by additivity requirements. Throughout this literature, information is instrumental: it concerns an exogenous state and is costly, or valuable, as an input into a downstream decision. This has two structural consequences. Blackwell monotonicity is a theorem or an unquestioned desideratum, since a decision-maker who can condition on a signal is weakly better off with a finer one; and the primitive object is cardinal, or is disciplined by an optimisation model that supplies cardinal structure.

The present paper shares the destination of that literature—coherence over mixing and sequential resolution forces the Shannon scale—but differs in primitive, in direction, and in what is derived rather than assumed. The primitive here is a family of ordinal weak orders over

lotteries on the agent's own actions; there is no downstream decision, and prior dependence is essential rather than excluded: the trace component is the whole map $(q, K) \mapsto I_{q,K}(A; O, R)$. Indeed, the two families are disjoint rather than nested: mutual information is prior-dependent and subadditive over independent experiments, so it violates the additivity that defines the log-likelihood ratio class, while fully revealing experiments, which have infinite log-likelihood ratio cost, receive the finite value $H(q)$ here. Because the primitive is ordinal and fibred by the prior, the cardinal structure that the cost literature postulates must here be constructed; branchwise continuation consistency performs that calibration jointly with the payoff scale. Finally, record and action data processing, Axioms (A4) and (A5), are primitive postulates of taste rather than theorems of instrumental rationality; as the sign trichotomy (Corollary 3) makes precise, their joint reversal is equally coherent and characterises a preference for deniability. Where the two traditions do meet—at the Shannon functional, reached from opposite ends of the channel—they axiomatise different objects: there, the price of learning about the world; here, a taste for being learned about, tempered by ordinary payoff utility.

3 The Model

Fix a finite set O of *payoff-bearing outcomes*, with $|O| \geq 2$. Let A be a nonempty finite set of *actions* and R a nonempty finite set of *explicit visible records*. A joint channel is a stochastic kernel

$$K : A \longrightarrow \Delta(O \times R).$$

For each finite joint channel K , the primitive behavioural datum is a binary relation

$$\succeq_K \quad \text{on} \quad \Delta(A).$$

The interpretation is that q is a lottery over actions used by the agent and $q \succeq_K p$ means that, in the fixed environment K , q is evaluated at least as highly as p . The lottery is the procedure being evaluated; the channel mapping actions into payoff–record pairs is not an object of choice.

Axioms are imposed on the family $\{\succeq_K\}_K$.

(A1) Weak order. For every finite joint channel K , $\succeq_K$ is complete and transitive on $\Delta(A)$.

(A2) Continuity. For each fixed triple of alphabets (A, O, R) , the set

$$\{(K, q, p) : q \succeq_K p\} \subseteq \Delta(O \times R)^A \times \Delta(A)^2$$

is closed.

Comparison environments. The remaining axioms compare a channel with a garbling of it and a compound experiment with its branches. Each relation $\succeq_K$ ranks only lotteries, so these comparisons need a domain in which two channels can appear in one preference statement. Labelled blocks provide it, much as objective lotteries enrich Savage's domain in Anscombe and Aumann [4], and without making the channel an object of choice.

For $K : A \rightarrow \Delta(O \times R)$ and $L : B \rightarrow \Delta(O \times S)$, let $K \sqcup L$ be the block-diagonal joint channel with action set $(A \times \{0\}) \cup (B \times \{1\})$, payoff set O , and record set $(R \times \{0\}) \cup (S \times \{1\})$, defined by

$$(K \sqcup L)(o, (r, 0) \mid (a, 0)) = K(o, r \mid a), \quad (K \sqcup L)(o, (s, 1) \mid (b, 1)) = L(o, s \mid b),$$

with all cross-block probabilities zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, let q^0 and p^1 denote their block-supported copies, defined on the tagged action set by

$$\begin{aligned} q^0(a, 0) &:= q(a), & q^0(b, 1) &:= 0, \\ p^1(a, 0) &:= 0, & p^1(b, 1) &:= p(b), \end{aligned} \quad (a \in A, b \in B).$$

When the blocks have the same action alphabet, q^0 and q^1 are the two tagged copies of the same lottery. For a finite family $\{K_i : A_i \rightarrow \Delta(O \times R_i)\}_{i \in I}$, write $\sqcup_{i \in I} K_i$ for the analogous block environment and q_i^i for the lottery that assigns $q_i(a)$ to (a, i) and zero to every other block. Write

$$(q, K) \succeq (p, L) \iff q^0 \succeq_{K \sqcup L} p^1.$$

Each instance of this shorthand is a single comparison inside the two-block environment $K \sqcup L$, which is itself built from the two pairs; $\succ$ and $\sim$ denote its asymmetric and symmetric parts. The pair (q, K) has no meaning in isolation.²

(A3) Block-comparison coherence. *Duplication:* for every $K : A \rightarrow \Delta(O \times R)$ and all $q, p \in \Delta(A)$,

$$q \succeq_K p \iff (q, K) \succeq (p, K).$$

Irrelevant blocks: for every finite block environment $\sqcup_{k \in I} K_k$, every ordered pair of distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\sqcup_{k \in I} K_k} q_j^j \iff (q_i, K_i) \succeq (q_j, K_j).$$

Block labels should not alter within-environment rankings, nor should unused blocks affect the comparison between active blocks.

Processing and compounding. The next three axioms use two ways of transforming an environment and one way of composing environments. A *record processor* is a stochastic kernel $T(r' \mid o, r)$ that may depend on the payoff but leaves it unchanged; it transforms $K : A \rightarrow \Delta(O \times R)$ into

$$(KT)(o, r' \mid a) := \sum_r K(o, r \mid a) T(r' \mid o, r). \quad (1)$$

An *action report* is a stochastic kernel $S : A \rightarrow \Delta(B)$; it transforms a lottery $q \in \Delta(A)$ into $(qS)(b) := \sum_a q(a) S(b \mid a)$ and K into any channel $\widehat{K} : B \rightarrow \Delta(O \times R)$ satisfying

$$(qS)(b) \widehat{K}(o, r \mid b) = \sum_a q(a) S(b \mid a) K(o, r \mid a) \quad \text{for every } (b, o, r). \quad (2)$$

If $(qS)(b) = 0$, equation (2) leaves the corresponding row of $\widehat{K}$ unrestricted, and every statement below applies to every consistent completion. Both transformations preserve the O -marginal. Finally, a first-stage record channel $P : A \rightarrow \Delta(Y)$ and a profile of continuation channels $K^y : A \rightarrow \Delta(O \times R_y)$ compose into the *compound environment*

$$(P \triangleright \{K^y\})(o, (y, r) \mid a) := P(y \mid a) K^y(o, r \mid a). \quad (3)$$

Given $q \in \Delta(A)$, a branch y is *reached* if $m(y) := \sum_a q(a) P(y \mid a) > 0$, and its branch posterior is $q_y(a) := q(a) P(y \mid a) / m(y)$.

²A complete numerical example. Let $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $O = \{o, o'\}$, and let the record alphabets be $R = \{r, r'\}$ and $S = \{s\}$. With rows indexed by actions and columns by payoff-record pairs,

$$K = \begin{array}{c|cccc} & (o, r) & (o', r) & (o, r') & (o', r') \\ \hline a_1 & 0.5 & 0 & 0 & 0.5 \\ a_2 & 0 & 0.7 & 0.3 & 0 \end{array} \quad L = \begin{array}{c|cc} & (o, s) & (o', s) \\ \hline b_1 & 0.3 & 0.7 \\ b_2 & 0.6 & 0.4 \end{array}$$

The block environment $K \sqcup L$ has tagged action set $\{(a_1, 0), (a_2, 0), (b_1, 1), (b_2, 1)\}$, payoff set O , and tagged record set $\{(r, 0), (r', 0), (s, 1)\}$; its kernel is block-diagonal, K on the first two rows, L on the last two, and every cross-block entry zero. For $q = (0.3, 0.7)$ and $p = (0.8, 0.2)$, the block-supported copies are $q^0 = (0.3, 0.7; 0, 0)$ and $p^1 = (0, 0; 0.8, 0.2)$, the semicolon separating the blocks. Then $(q, K) \succeq (p, L)$ abbreviates $q^0 \succeq_{K \sqcup L} p^1$: one comparison of two ordinary lotteries in this one environment. Both block-supported lotteries deliver o with probability 0.36, so under the representation of Theorem 1 the comparison turns on the trace terms alone—approximately 0.88 against 0.04 bits—and is strict.

(A4) Record data processing. For every $K : A \rightarrow \Delta(O \times R)$, every record processor T , and every $q \in \Delta(A)$,

$$(q, K) \succeq (q, KT).$$

The agent never gains when her record is garbled.

(A5) Action data processing. For every $K : A \rightarrow \Delta(O \times R)$, every action report $S : A \rightarrow \Delta(B)$, every $q \in \Delta(A)$, and every channel $\widehat{K}$ satisfying (2),

$$(q, K) \succeq (qS, \widehat{K}).$$

Nor does she gain when her action is reported noisily, holding the payoff lottery fixed.

(A6) Branchwise continuation consistency. For every first-stage record channel $P : A \rightarrow \Delta(Y)$, all continuation profiles $\{K^y\}$ and $\{L^y\}$, and every $q \in \Delta(A)$: if

$$(q_y, K^y) \succeq (q_y, L^y) \quad \text{for every reached } y,$$

then

$$(q, P \triangleright \{K^y\}) \succeq (q, P \triangleright \{L^y\}),$$

strictly if one reached branch comparison is strict.

In a compound environment the branching variable is a record, and payoffs arrive only in the continuations. The axiom is a sure-thing principle across the branches of an interim record: a profile weakly better on every reached branch is weakly better overall, with unreached branches unrestricted exactly as Savage's null events are.

(A7) Material relevance. For $o \in O$, let $K_o : \{*\} \rightarrow \Delta(O \times \{*\})$ deliver o with an uninformative record, $K_o(o, * \mid *) = 1$. For some $o^+, o^- \in O$,

$$(\delta_*, K_{o^+}) \succ (\delta_*, K_{o^-}).$$

The agent cares about material outcomes, and not only about their trace.

(A8) Positive trace orientation. For every finite A with $|A| \geq 2$, every full-support $q \in \Delta(A)$, and every $o \in O$,

$$(q, K_o^{\text{id}}) \succ (q, K_o^0),$$

where $K_o^{\text{id}}(o, a \mid a) = 1$, with record alphabet A , and $K_o^0(o, * \mid a) = 1$. At a constant payoff, a record that reveals a genuinely uncertain action is strictly better than one that reveals nothing: trace is sought, not shunned.

4 The Representation Theorem

For $q \in \Delta(A)$ and $K : A \rightarrow \Delta(O \times R)$, let

$$p_{qK}(o, r) := \sum_a q(a) K(o, r \mid a)$$

be the induced distribution of visible consequences, and define

$$I_{qK}(A; O, R) := \sum_{a, o, r} q(a) K(o, r \mid a) \log \frac{K(o, r \mid a)}{p_{qK}(o, r)}. \quad (4)$$

Summands with $q(a) K(o, r \mid a) = 0$ are set to zero, the logarithm has any fixed base, and expectations below are taken under the joint distribution $q(a) K(o, r \mid a)$.

Theorem 1 (Trace-tempered choice). *For a family $\{\succeq_K\}_K$ as described above, the following are equivalent.*

(i) *The family satisfies Axioms (A1)–(A8).*

(ii) *There are a nonconstant function $u : O \rightarrow \mathbb{R}$ and a number $\lambda > 0$ such that, for every nonempty finite A, R , every $K : A \rightarrow \Delta(O \times R)$, and every $q, p \in \Delta(A)$,*

$$q \succeq_K p \iff V(q, K) \geq V(p, K), \quad V(q, K) := \mathbb{E}_{q,K}[u(O)] + \lambda I_{qK}(A; O, R). \quad (5)$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff V(q_i, K_i) \geq V(q_j, K_j).$$

The proof of the equivalence of (i) and (ii) is in Appendix A; the moreover clause follows because a block-supported lottery has a constant block label.

Proof sketch. That the representation satisfies the axioms is a direct verification. The converse has four steps.

Step 1: reduction to terminal laws. Fix a full-support q . Two channels that induce the same joint law of realised payoff and posterior about the action are record processings of one another, so Axioms (A1), (A3), and (A4) rank them together (Lemma 19); Axiom (A5) deletes null actions, and applied to bijective reports in both directions it makes relabellings neutral. Each pair (q, K) is thereby identified with its *terminal law*, the joint distribution of realised payoff and posterior.

Step 2: cardinalisation. A public coin is a first stage that reveals nothing, so every branch posterior equals q . Axiom (A6) applied to such stages yields mixture independence over terminal laws (Lemma 21), and with continuity, Axiom (A2), the mixture-space theorem gives a continuous affine representation on each fixed-prior fibre. All cardinal content enters here: every other axiom is an ordinal, environment-by-environment condition.

Step 3: two scales. On singleton action sets the affine representation is expected utility, nonconstant by Axiom (A7); action reports transport the payoff scale across priors and alphabets, and block coherence splices the fibre representations into one common scale (Lemmas 22 and 23). On the constant-payoff restriction the axioms collapse to the pure-record conditions of Appendix C, whose characterisation identifies the residual order with mutual information; Axiom (A8) signs its weight (Lemma 24).

Step 4: assembly. Axiom (A6) enters once more: inserting continuations into a reached branch is an order isomorphism, and calibrating branches with terminal payoffs forces values to aggregate across branches at their reached probabilities (Lemma 25). Sequentialising an arbitrary channel—announce (o, r) as a first-stage record, deliver o terminally—is neutral by Axiom (A4), and the recursion returns $\mathbb{E}_{q,K}[u(O)] + \lambda I_{qK}(A; O, R)$.

Uniqueness To state uniqueness, fix one representation V supplied by Theorem 1. A real-valued index W on all finite lottery–channel pairs is a *common-scale representation* if, for every channel $K : A \rightarrow \Delta(O \times R)$ and every $q, p \in \Delta(A)$,

$$q \succeq_K p \iff W(q, K) \geq W(p, K),$$

and, for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff W(q_i, K_i) \geq W(q_j, K_j).$$

Call a common-scale representation W *branch-additive* if, for every first-stage record channel P , every $q \in \Delta(A)$, and every two continuation profiles $\{K^y\}$ and $\{L^y\}$ admitted in Axiom (A6),

$$W(q, P \triangleright \{K^y\}) - W(q, P \triangleright \{L^y\}) = \sum_{y:m(y)>0} m(y)[W(q_y, K^y) - W(q_y, L^y)]. \quad (6)$$

Thus continuation-value differences aggregate by their reached-branch probabilities; no arbitrary zero for material utility is required.

Proposition 2 (Uniqueness). *Suppose the equivalent conditions of Theorem 1 hold. For any real-valued index W on all finite lottery-channel pairs:*

- (i) *W is a common-scale representation if and only if $W = \varphi \circ V$ for a single strictly increasing $\varphi : [\underline{u}, \infty) \rightarrow \mathbb{R}$, where $\underline{u} := \min_{o \in O} u(o)$.*
- (ii) *A common-scale representation W is branch-additive if and only if $W = \alpha V + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.*

The primitive remains ordinal: part (i) permits any common increasing transformation. Branch additivity is an accounting requirement on an index, not another behavioural axiom; part (ii) selects the affine gauge. Consequently, if (u, λ) and $(\tilde{u}, \tilde{\lambda})$ are two trace-tempered representations, then

$$\tilde{u} = \alpha u + \beta, \quad \tilde{\lambda} = \alpha \lambda$$

for some $\alpha > 0$ and $\beta \in \mathbb{R}$. Once two payoff outcomes are assigned numerical utilities, λ is therefore uniquely measured in those payoff units per unit of information.

Sign Uniqueness concerns scale; a complementary observation concerns sign. Axioms (A4), (A5), and (A8) orient the trace component. Write $(A4^-)$, $(A5^-)$, and $(A8^-)$ for the clauses obtained by reversing their displayed comparisons, and write $(A4^0)$, $(A5^0)$, and $(A8^0)$ for the corresponding indifference clauses.

Corollary 3 (Sign trichotomy). *Suppose Axioms (A1)–(A3), (A6), and (A7) hold. Then:*

- (i) *under Axioms (A4), (A5), and (A8), the family has the representation in Theorem 1, with $\lambda > 0$;*
- (ii) *under $(A4^-)$, $(A5^-)$, and $(A8^-)$, it has the same representation with $\lambda < 0$;*
- (iii) *under $(A4^0)$, $(A5^0)$, and $(A8^0)$, it is represented by expected utility, equivalently by the same formula with $\lambda = 0$.*

Thus the trace component has exactly the positive, negative, and neutral orientations.

Proof. Part (i) is Theorem 1. For parts (ii) and (iii), the material calibration and branch-recursion arguments are unchanged. On the constant-payoff restriction, reversing both processing directions and full-revelation orientation gives $-I$, while replacing them by indifference gives zero, by Corollary 34. The converse follows directly from data processing and the two chain rules. $\square$

For $q \in \Delta(A)$ and $K : A \rightarrow \Delta(O \times R)$, write

$$\begin{aligned} \ell_{qK}(o) &:= \sum_r p_{qK}(o, r), \\ \pi_{or}^{qK}(a) &:= \frac{q(a)K(o, r | a)}{p_{qK}(o, r)} \quad \text{when } p_{qK}(o, r) > 0, \\ \mu_{qK} &:= \sum_{(o,r):p_{qK}(o,r)>0} p_{qK}(o, r) \delta_{\pi_{or}^{qK}}. \end{aligned}$$

Corollary 4 (Derived matching neutrality). *Suppose Axioms (A1)–(A8) hold. Fix A and $q \in \Delta(A)$. If channels K and L satisfy*

$$\sum_o \ell_{qK}(o)u(o) = \sum_o \ell_{qL}(o)u(o), \quad \mu_{qK} = \mu_{qL},$$

then $(q, K) \sim (q, L)$. Thus the matching of payoff labels to posterior atoms has no additional value.

Indeed, ℓ_{qK} is the payoff marginal, μ_{qK} is the law of reached posteriors, and

$$I_{qK}(A; O, R) = H(q) - \int H(\pi) d\mu_{qK}(\pi).$$

In words: the agent values what she obtains and what the visible consequence reveals about what she did, but not the coincidence of the two. Winning and being seen to act are priced separately; there is no premium for the good outcome arriving through the revealing record rather than alongside it. Any such premium would be a payoff–trace complementarity, and the corollary leaves no room for one. This is the behavioural face of the additive representation—the empirical content of the separability that the theorem derives rather than assumes. The separation is achieved in the domain and in Axiom (A6): compound environments may branch on interim records, never on payoffs.

Proposition 5 (Independence of the axioms). *For each $i \in \{1, \dots, 8\}$ there is a family satisfying the other seven axioms and violating Axiom (Ai).*

The eight families are constructed in Appendix A. Three mark the substantive frontier of the system. A conditional-entropy penalty satisfies every axiom except record data processing and values the removal of extraneous record noise. A Gini posterior-reduction criterion satisfies every axiom except action data processing. The satiation criterion $\mathbb{E}[u(O)] + \min\{I(A; O, R), c\}$, for $c > 0$, satisfies every axiom except branchwise continuation consistency and ceases to value trace once c units of information are secured.

Taken together, the results separate orientation, scale, and behavioural content. The sign trichotomy determines whether trace is sought, avoided, or ignored; branch additivity selects the affine gauge; and no displayed axiom is redundant. Unlike the pure-record model, the magnitude of λ is not merely a choice of information unit once payoff utility has been calibrated: it is the identified payoff–trace trade-off.

5 Choice implications

This section works out what an agent with the representation of Theorem 1 does: which finite designs falsify the model directly (§5.1), when she deliberately randomises (§5.2), what she will pay for changes to the record technology (§5.3), and what her behaviour can never disclose about the environment (§5.4). Throughout, λ is measured in payoff units per unit of information in the fixed base. Proofs for this section, together with full statements of the supporting results that the text presents in summary form, are collected in Appendix B; only one-line verifications are kept here.

5.1 The trace test

The functional V reads the joint distribution of the action and its visible consequence. Existing theories of choice read less: the distribution of consequences, the lottery, or both. The following experiment separates them.

In the first treatment the agent chooses between two fair coins over actions. One flips between two actions that leave distinct marks; the other flips between two actions that produce noise.

Both coins yield the same payoff for certain and the same fifty–fifty distribution over records; only the first coin’s record identifies the realised action. In the second treatment the record technology is disabled: the same choice is offered in a channel where every action produces noise.

Proposition 6 (The trace test). *Four actions pay the same outcome o_0 . Two of them leave distinct marks: a_1 always produces record r_1 , and a_2 always produces the distinct record r_2 . The other two are indistinguishable: a_3 and a_4 each produce r_1 or r_2 with equal probability. Call this channel K , and let q' be the fair coin over $\{a_1, a_2\}$ and q'' the fair coin over $\{a_3, a_4\}$.*

(i) *The two coins induce the same joint distribution of visible consequences, $p_{q'K} = p_{q''K}$; yet*

$$V(q', K) - V(q'', K) = \lambda \log 2 > 0, \quad \text{so} \quad q' \succ_K q''.$$

(ii) *Let K° be the channel in which every action pays o_0 and produces r_1 or r_2 with equal probability. Then $q' \sim_{K^\circ} q''$.*

The prediction is strict preference for the revealing coin in the first treatment and exact indifference in the second. Part (i) alone rules out every consequentialist theory. The two coins induce the same distribution over payoff–record pairs, so any criterion that evaluates only that distribution is indifferent between them: expected utility over payoffs, expected utility that values records directly as consequences through some $v(o, r)$, and every non-linear functional of $p_{q'K}$. What separates the coins is the joint distribution of the action with its visible consequence, and that is precisely what consequentialist theories do not read.

Part (i) restricts only how a theory reads consequences. A theory that also reads the lottery can match part (i). Existing theories of deliberate randomisation are of this kind: in the hedging model of Cerreia-Vioglio, Dillenberger, Ortaleva, and Riella [17], and in the preference for randomisation documented by Agranov and Ortaleva [2], the value of mixing depends only on the induced distribution and the lottery. Call an index W *channel-blind* if it reads only these two objects: there are functions F and $h : \Delta(A) \rightarrow \mathbb{R}$, fixed across channels, such that

$$W(q, K) = F(p_{qK}, h(q)).$$

The second treatment rules these out. Nothing a channel-blind index reads changes between the treatments: in all four lottery–channel combinations the distribution of visible consequences is uniform on $\{(o_0, r_1), (o_0, r_2)\}$, and the lotteries are the same two lotteries. A channel-blind index therefore separates the coins in both treatments or in neither, so none reproduces the pattern of Proposition 6: strict preference under K and indifference under K° .³

The test also rejects. The first treatment separates the three cases of Corollary 3: the revealing coin if $\lambda > 0$, the other coin if $\lambda < 0$, indifference if $\lambda = 0$. In the second treatment every representation assigns both coins the same value, whatever u and λ , so a strict preference there rejects the model.

5.2 Deliberate randomisation

A degenerate lottery leaves no trace, so an agent with $\lambda > 0$ may deliberately randomise: she gives up expected payoff to make her act genuinely uncertain, because only then can its consequence reveal it. In a payoff-based theory a sufficiently large payoff advantage makes a single action optimal. Here it need not: an action that leaves a mark no other action can leave is never wholly abandoned, however poorly it pays, because the first grain of mass on it is infinitely informative per unit, and no finite payoff difference outweighs that. The next proposition states

³If h does not depend on how the actions are labelled—an entropy bonus, say—the first treatment is enough: the permutation that swaps a_1 with a_3 and a_2 with a_4 turns q' into q'' and leaves p_{qK} unchanged, so $W(q', K) = W(q'', K)$. The second treatment is needed for arbitrary h .

this, and gives the exact terms on which a pure action survives: its payoff advantage must cover, for every rival, the price of the rival's distinctiveness from it.

Some notation. Let $Z := O \times R$ and write $K_Z(z | a) := K(o, r | a)$ for $z = (o, r)$; let

$$\bar{u}(a) := \sum_{o,r} K(o, r | a) u(o), \quad m_{q,K}(z) := \sum_a q(a) K_Z(z | a)$$

be the expected payoff of an action and the distribution of visible consequences induced by a lottery. The Kullback–Leibler divergence $D_{\text{KL}}(P \| Q) := \sum_z P(z) \log(P(z)/Q(z))$ between distributions on Z measures how sharply consequences drawn from P stand out against a background of Q ; it equals $+\infty$ when P puts mass where Q does not.

Proposition 7 (Exercised distinctiveness). *Let $Z_K := \{z \in Z : K_Z(z | a) > 0 \text{ for some } a \in A\}$.*

- (i) *Every optimal lottery q^* satisfies $m_{q^*,K}(z) > 0$ for every $z \in Z_K$. Consequently, if every action a has a private consequence—some z_a with $K_Z(z_a | a) > 0$ and $K_Z(z_a | b) = 0$ for all $b \neq a$ —then every optimal lottery has full support.*
- (ii) *The pure lottery δ_a is optimal if and only if*

$$\bar{u}(b) + \lambda D_{\text{KL}}(K_Z(\cdot | b) \| K_Z(\cdot | a)) \leq \bar{u}(a) \quad \text{for every } b \in A,$$

and uniquely optimal if these inequalities are strict for $b \neq a$. In particular, if some rival reaches a consequence outside the support of $K_Z(\cdot | a)$, then δ_a is not optimal, however large its payoff advantage.

With two actions, the threshold below which the agent mixes is a case of Proposition 7(ii). Let $A = \{a, b\}$, write $K_a := K_Z(\cdot | a)$ and $K_b := K_Z(\cdot | b)$, and let $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$ be the payoff advantage of a . Then δ_a is optimal if and only if $\Delta \geq \lambda D_{\text{KL}}(K_b \| K_a)$. The threshold is directional: what protects the pure action is the detectability of the rival against the background of its own signature. It is infinite when K_b puts mass where K_a does not—the two-action face of Proposition 7(i).

Below the threshold the agent mixes (Proposition 26, Appendix B). The optimal mix is unique, and it puts more mass on the inferior action the larger is λ and less the larger is Δ . Mixing therefore stops exactly at $\Delta = \lambda D_{\text{KL}}(K_b \| K_a)$, and since the divergence is known once the channel is designed, a single switching experiment identifies λ . At equal payoffs the mix is the capacity-achieving input of the channel, whatever λ : an agent who values trace at all, however little, randomises at zero stakes so as to maximise the information the visible consequence carries about her act. The comparative statics restate the wedge of the trace test (§5.1): randomisation here is instrumental to trace, so it strengthens with the informativeness of the channel and vanishes on uninformative ones, whereas a taste for mixing as such is channel-blind.

With more than two actions, the same logic is captured by a first-order condition. The objective $q \mapsto V(q, K)$ is concave in q , so an optimum exists and Kuhn–Tucker conditions characterise it: at an optimum, every action in use offers the same total of expected payoff and priced distinctiveness.

Proposition 8 (Distinctiveness condition). *Let $\lambda > 0$. A lottery q^* is optimal if and only if there is a constant c such that*

$$\bar{u}(a) + \lambda D_{\text{KL}}(K_Z(\cdot | a) \| m_{q^*,K}) = c \quad \text{for } a \in \text{supp}(q^*), \quad \leq c \quad \text{for } a \notin \text{supp}(q^*).$$

The proposition gives a *distinctiveness premium*. Each action is credited, on top of its expected payoff, with λ times how far its visible consequences depart from the average visible behaviour that the lottery itself generates; at an optimum this total is equalised across the actions in use. An action can therefore receive more probability because it pays more or because

its consequences are more distinctive. The condition mirrors rational-inattention first-order conditions with the channel direction and sign reversed [47, 38].

Which lotteries are optimal depends on the coupling of actions to consequences, not on consequences alone; and optimal lotteries need not be unique. Yet every optimal lottery induces the same distribution of visible consequences. The model therefore predicts the record an observer sees, not the mix that generated it.

Proposition 9 (Optimal marginal and multiplicity). *Write $P_a := K_Z(\cdot | a)$ and $C^*(K) := \max_{q \in \Delta(A)} V(q, K)$. There is a unique $m^* \in \Delta(Z)$ such that $m_{q^*, K} = m^*$ for every optimal lottery q^* . With*

$$\mathcal{A}^* := \{a \in A : \bar{u}(a) + \lambda D_{\text{KL}}(P_a \| m^*) = C^*(K)\},$$

the set of optimal lotteries is exactly

$$\{q \in \Delta(A) : m_{q, K} = m^*, \text{supp}(q) \subseteq \mathcal{A}^*\},$$

and the optimum is unique whenever the rows $\{P_a : a \in \mathcal{A}^\}$ are affinely independent. If actions are partitioned into classes with identical rows, $V(q, K)$ depends on q within each class only through the class's total mass: every redistribution of an optimal lottery within such a class is optimal, and the optimal lotteries are exactly the lifts of the optimal lotteries of the channel that collapses the classes.*

The proposition's final claim—that moving mass between actions with identical rows changes nothing—is a test. Split an action into two copies with the same row. The record cannot tell the copies apart, so the value, the induced marginal m^* , and the pair's total probability do not depend on how the mass is divided between them. Any model that gives each action its own weight, as multinomial logit does, predicts instead that duplication raises the pair's combined share. Trace-tempered choice predicts exact invariance.

Remark 10 (Logit boundary). Suppose every action leaves its own mark: the supports of the rows are pairwise disjoint, so the visible consequence identifies the action and $I_{qK}(A; O, R) = H(q)$. The unique optimum is then multinomial logit,

$$q^*(a) = \frac{\exp(\bar{u}(a)/\lambda)}{\sum_{a'} \exp(\bar{u}(a')/\lambda)},$$

with $1/\lambda$ as the payoff sensitivity (information in natural logarithms; a change of base rescales λ). If instead no action leaves any mark—identical rows—every lottery is optimal. Between the two extremes shares solve the equalisation condition of Proposition 8, in which distinctiveness is measured against the average behaviour the optimum itself generates; there is no closed form. Multinomial logit is thus the fully revealing special case of trace-tempered choice, not its general prediction. In rational inattention the same formula arises for the opposite reason, as the optimal tilting of a chosen channel at a fixed prior over states [38]; here the channel is fixed and the source is chosen.

5.3 The price of being on the record

Because Theorem 1 represents comparisons across environments by one V , the model prices not only lotteries but changes to the record technology itself: what is it worth to the agent that her coin is observed, that her record is coarsened, that a further signal about her is disclosed? Each answer below is an exact amount of information multiplied by λ , and each corresponds to a block comparison in the primitive domain. The first result concerns the mixing device.

Corollary 11 (The recorded coin). *Let $K, L : A \rightarrow \Delta(O \times R)$ and $t \in (0, 1)$. Let $tK \oplus (1 - t)L$ be the public mixture that draws an action-independent coin Y with $\Pr(Y = 0) = t$, applies K if $Y = 0$ and L otherwise, and includes the coin in the record; let $M_t := tK + (1 - t)L$ be the rowwise mixture, which hides it. Then, for every $q \in \Delta(A)$,*

- (i) $V(q, tK \oplus (1-t)L) = tV(q, K) + (1-t)V(q, L)$;
- (ii) $V(q, tK \oplus (1-t)L) - V(q, M_t) = \lambda I(A; Y \mid O, R) \geq 0$, with equality if and only if A and Y are conditionally independent given (O, R) .

Proof. Part (i) is the chain rule with an action-independent first stage: $I(A; Y) = 0$ and each reached branch reproduces the joint law of the corresponding constituent. Part (ii) follows from Proposition 12 below, since deleting the coin coordinate is a deterministic payoff-preserving record processor carrying the public mixture to M_t . $\square$

Part (i) is the value-level face of derived public-coin independence (Lemma 21): the agent has no intrinsic taste for or against randomising over environments, so a publicly randomised environment is never ranked strictly above or below both of its constituents. Part (ii) says what she does care about: whether the coin is on the record. Hiding the randomisation device costs $\lambda I(A; Y \mid O, R)$, which is strictly positive whenever knowing the operating regime would sharpen inference about the action given the visible consequences—even though the coin is independent of the action *ex ante*. A trace-tempered agent weakly prefers her mixing device publicised; theories that attach value to randomisation as such are silent on this visibility margin.

Behind part (ii) is the exact price of any record garbling. Axiom (A4) asserts only a weak preference against garbling; the representation upgrades the axiom to an exact price and characterises when the price is zero.

Proposition 12 (Price of a record garbling). *Let $K : A \rightarrow \Delta(O \times R)$, let $T(r' \mid o, r)$ be a record processor as in (1), and write $L := KT$. Fix $q \in \Delta(A)$ and let (A, O, R, R') carry the joint law $q(a)K(o, r \mid a)T(r' \mid o, r)$. Then*

$$V(q, K) - V(q, L) = \lambda I(A; R \mid O, R') \geq 0,$$

and the following are equivalent: (a) $V(q, L) = V(q, K)$; (b) A and (O, R) are conditionally independent given (O, R') ; (c) $\mu_{qL} = \mu_{qK}$.

Rewriting the record costs λ times the information about the action that the original record retains beyond the rewritten one, and a garbling is exactly free when it pools only visible consequences that already induced the same posterior about the action: an agent with a trace motive does not pay for gratuitous detail, only for detail that discriminates among her own actions. Symmetrically, appending to the record any further signal S drawn given (a, o, r) raises $V(q, \cdot)$ by exactly $\lambda I(A; S \mid O, R) \geq 0$: more evidence about oneself never hurts, a conclusion the deniability dual of Corollary 3 reverses sign for sign. Since the visible pair always includes the payoff coordinate, the total value of trace in any environment is bounded by $\lambda \min\{H(q), \log(|O| |R|)\}$.

These prices invite measurement: how much payoff would the agent give up for a more revealing record? The difficulty is that the domain has no money—the finite outcome set contains no outcome worth $u(o) + \tau$ —so a compensating offset must be manufactured from the two outcomes that Axiom (A7) strictly ranks, using comparisons that stay inside the primitive domain. Let o^+ and o^- attain the maximum and minimum of u , write $\Delta u := u(o^+) - u(o^-) > 0$, and for $\beta \in [0, 1]$ let $w(\beta) := \beta u(o^+) + (1 - \beta)u(o^-)$.

Definition 13 (Sweetened dilution). For $K : A \rightarrow \Delta(O \times R)$, $\varepsilon \in [0, 1]$, and $\beta \in [0, 1]$, let $P_\varepsilon : A \rightarrow \Delta(\{0, 1\})$ be the public coin $P_\varepsilon(1 \mid a) = \varepsilon$, and let $B_\beta : A \rightarrow \Delta(O \times \{*\})$ have the constant rows $B_\beta(o^+, * \mid a) = \beta$ and $B_\beta(o^-, * \mid a) = 1 - \beta$. The (ε, β) -sweetened dilution of K is the compound channel

$$D_{\varepsilon, \beta} K := P_\varepsilon \triangleright \{K, B_\beta\},$$

as in (3), listing the branch-0 and branch-1 continuations in order.

Proposition 14 (Willingness to pay for trace). *Fix $q \in \Delta(A)$, $p \in \Delta(B)$, and channels $K : A \rightarrow \Delta(O \times R)$ and $L : B \rightarrow \Delta(O \times S)$.*

(i) *For every $\varepsilon \in [0, 1)$ and $\beta \in [0, 1]$,*

$$V(q, D_{\varepsilon, \beta} K) = (1 - \varepsilon) V(q, K) + \varepsilon w(\beta).$$

(ii) *If $\varepsilon \in (0, 1)$ and $\beta, \beta' \in [0, 1]$ satisfy $(q, D_{\varepsilon, \beta} K) \sim (p, D_{\varepsilon, \beta'} L)$, then*

$$\frac{\varepsilon}{1 - \varepsilon} [w(\beta) - w(\beta')] = V(p, L) - V(q, K).$$

When the material parts agree, $\mathbb{E}_{q, K}[u(O)] = \mathbb{E}_{p, L}[u(O)]$, the elicited offset equals $\lambda[I_{pL}(B; O, S) - I_{qK}(A; O, R)]$: information is priced linearly at rate λ .

(iii) *Such a pair (β, β') exists if and only if $|V(p, L) - V(q, K)| \leq \frac{\varepsilon}{1 - \varepsilon} \Delta u$; a boundary response therefore brackets the value difference instead of measuring it.*

The public coin is incentive-neutral: it rescales both blocks' values identically, so the elicited odds gap measures only the value difference, whatever the dilution. The price it recovers is, moreover, unique and portable.

Proposition 15 (Single information price). *Write $E_{qK} := \mathbb{E}_{q, K}[u(O)]$ and $\phi(q, K) := (I_{qK}(A; O, R), E_{qK})$, and let $\underline{u} := \min_o u(o)$ and $\bar{u} := \max_o u(o)$.*

(i) *The image of ϕ over all finite pairs (q, K) is exactly $[0, \infty) \times [\underline{u}, \bar{u}]$, and the indifference classes of the block-comparison relation are, in these coordinates, the nonempty intersections of the lines $E = c - \lambda I$ with that strip.*

(ii) *If $(q, K) \sim (p, L)$ and $I_{qK} \neq I_{pL}$, then*

$$\frac{E_{pL} - E_{qK}}{I_{qK} - I_{pL}} = \lambda,$$

and λ is the unique number with this property: for every $\lambda' \neq \lambda$ there are finite pairs $(q, K) \sim (p, L)$ with $E_{qK} + \lambda' I_{qK} \neq E_{pL} + \lambda' I_{pL}$.

At any fixed calibration of payoff units, the cross-environment indifference map in the information–payoff plane consists of parallel straight lines of slope $-\lambda$: bits carry one price, the same at every prior, every information level, and in every finite environment, as Proposition 2 transports it. This is the sharpest testable fingerprint of branchwise continuation consistency. The satiation criterion of Proposition 5, which fails exactly Axiom (A6), produces kinked indifference curves whose measured slope collapses to zero beyond the cap; constancy of the elicited slope across designs with different priors or alphabets is therefore a direct over-identification test. Appendix B records a complete two-treatment protocol implementing this elicitation (Remark 29), with the information gap fixed by design and the garbling control built in.

Choices, and not only indifferences, bound the same price. Two behaviours ordered by λ are ordered in what they choose: a higher weight buys more information and less payoff, and the rate of exchange between the two brackets the weight itself.

Remark 16 (Trace demand). *Fix K , write $e(q) := \mathbb{E}_{q, K}[u(O)]$ and $i(q) := I_{qK}(A; O, R)$, and let q_j maximise $e + \lambda_j i$ for $0 < \lambda_1 < \lambda_2$. Comparing each optimum at the other weight gives $i_1 \leq i_2$, $e_1 \geq e_2$, and, whenever behaviour responds,*

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2 :$$

the payoff surrendered per additional unit of information brackets the trace weight, whatever selection among optima the data reflect. The indirect value is convex in λ , with the chosen information as its derivative wherever it is differentiable, and its extreme regimes are expected utility ($\lambda \downarrow 0$, with information breaking material ties) and maximal distinctiveness ($\lambda \uparrow \infty$, with payoff breaking information ties). Exact statements are Proposition 27 and Corollary 28 in Appendix B.

5.4 What preferences cannot reveal

Suppose an agent ranks environment K above environment L at *every* lottery. Has she thereby revealed that L is a coarsening of K —that some garbling maps the one record technology into the other? For a Blackwell decision-maker the answer would be yes: dominance in every decision problem is equivalent to garbling [12]. Here the answer is no, and the gap can be stated exactly.

Definition 17. Let $K : A \rightarrow \Delta(O \times R)$ and $L : A \rightarrow \Delta(O \times R')$ have identical payoff kernels, $\sum_r K(o, r \mid a) = \sum_{r'} L(o, r' \mid a)$ for every (a, o) . Write $K \succeq_g L$ if $L = KT$ for some record processor T as in (1), and $K \succeq_c L$ if $I_{qK}(A; O, R) \geq I_{qL}(A; O, R')$ for every $q \in \Delta(A)$; the latter is the *more capable* order on the visible-consequence channels [34].

Proposition 18 (Uniform dominance is more-capable, not garbling). *Let K and L have identical payoff kernels.*

- (i) $K \succeq_g L \Rightarrow K \succeq_c L$, and $K \succeq_c L$ holds if and only if $V(q, K) \geq V(q, L)$ for every $q \in \Delta(A)$.
- (ii) *The converse of the first implication fails: there are channels with identical payoff kernels such that $V(q, K) \geq V(q, L)$ for every q and every representation (u, λ) , yet $L \neq KT$ for every record processor T .*

The reason Blackwell’s equivalence fails here is that his test family is rich—all state-dependent utilities—while the axioms pin the attitude to records down to the single functional $\mathbb{E}[u] + \lambda I$, so environments are assessed only through mutual information at priors. The revealed informativeness order is then exactly the information theorists’ more-capable order, strictly coarser than garbling. And the counterexample is utility-robust—the payoff terms cancel identically—so observing that an agent always ranks one environment above another licenses no inference that the second is a censored version of the first.

6 Interpretation and scope

Evidence, not authorship. The name promises agency; the functional rewards uncertainty. An agent certain of her act earns zero trace value, however perfectly the record attributes it, while one who hands the decision to a fair coin earns the most the channel allows. We concede what this invites: the theory prices evidence—the news a visible consequence carries about conduct—and news requires antecedent doubt, which is Dostoevsky’s own logic. The zero at certainty is a discriminating prediction rather than an embarrassment. Image models value the level of an observer’s regard, so anticipated attribution keeps its worth for an agent certain of her act; here trace dies with the doubt. Two readings of q are consistent with this: a mixing device the agent operates, which makes the randomisation results of Section 5 literal, or an observer’s prior or population frequency against which a realised act is evaluated. Under the second reading q is exogenous, and the theorem is silent on how the observer forms it.

Trace or control. Mutual information is symmetric, and the domain inherits the symmetry. A taste for visible consequences that reveal the act is behaviourally indistinguishable from a taste for acts that statistically shape visible consequences: the theorem pins down the value of the coupling between action and consequence, and trace and control are two readings of that one object. No data in a channel-indexed domain can separate them, much as Savage’s data cannot separate belief from state-dependent taste. Separation would require interventional primitives—an audience that can be removed while the channel is left intact—and these lie outside the theory. Every choice implication in Section 5 therefore tests both readings at once.

Empowerment. At constant material utility, maximising V yields the Shannon capacity of the visible-consequence channel: the empowerment index of artificial-intelligence and complex-systems research [33, 45]. The connection is stronger than a shared optimum. The theorem gives a preference foundation for the entire map $(q, K) \mapsto I_{q,K}$, so it covers constrained behaviour and choices tempered by material payoffs, and it identifies λ , the payoff price of a unit of empowerment. The connection is also narrow in an instructive way: the criterion depends only on actions assigned positive probability, so it captures control that is exercised, not options held in reserve—a genuine option value would require preferences over menus, outside this domain. Each side gains something: agency research acquires a behavioural test for its objective, and decision theory acquires an objective already in use in another field.

A belief-based utility, located. At a fixed prior the trace term is a belief-based utility: the expected reduction in the entropy of the posterior about the act, a posterior-separable felicity in the manner of [16]—though the belief concerns the agent’s own conduct, not an external state. What the axioms add is the location of the functional form in behaviour. Entropy is nowhere assumed; branchwise continuation consistency carries the whole cardinal content, and it is a substantive claim, not a convenience. Its fingerprint is that bits carry one price: cross-environment indifference curves in the information–payoff plane are parallel straight lines (Proposition 15). An agent whose taste for trace satiates violates exactly this axiom, produces kinked curves, and satisfies every other. The theorem’s scope is agents for whom that constancy of price holds.

One-shot, with no audience in the model. Beliefs enter only as the yardstick for how much the realised consequence narrows uncertainty about the action. No observer responds: the representation predicts neither praise nor sanction nor any return to reputation, and the staged records of Axiom (A6) introduce sequencing of evidence, not repeated interaction. This is the right first step, not an oversight: it isolates the intrinsic value of trace, so that a strategic or dynamic extension can add an audience’s response without confusing it with the motive identified here.

The split is scaffolding. The payoff–record split is scaffolding, not substance. Identification needs two independently variable directions and a temporal order: records may be garbled while payoff stays intact, and terminal payoff anchors a common scale. But the object valued is the undivided pair; the theorem separates O from R only to recombine them in $I(A; O, R)$. The Underground Man, on this reading, was not asking to be rid of reason: he was asking for a lottery genuinely open beforehand and a consequence able to tell his acts apart, and the cakes he stood ready to risk are what λ measures.

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Proofs of the main results

A Proof of the representation theorem

The sufficiency proof is modular. The first lemma removes inessential record labels. The second records the precise consequence of Axiom (A6) that was an axiom in many earlier formulations. The next lemmas calibrate material payoffs, import the pure-trace engine, and align continuation scales. Sequentialising an arbitrary joint channel then yields the representation in one line.

Proof notation. For a prior q and joint channel K , define the *terminal law*

$$\eta_{qK} := \sum_{(o,r): p_{qK}(o,r) > 0} p_{qK}(o,r) \delta_{(o, \pi_{or}^{qK})}. \quad (7)$$

It records the payoff and the posterior about the realised action after everything visible has been observed; its posterior marginal is μ_{qK} , as defined immediately after the main theorem. Bayes plausibility gives

$$\int_{O \times \Delta(A)} \pi d\eta_{qK}(o, \pi) = q.$$

Let $\mathcal{E}_q$ be the set of finite-support laws η on $O \times \Delta(A)$ satisfying this barycentre condition. Every $\eta = \sum_i m_i \delta_{(o_i, \pi_i)} \in \mathcal{E}_q$ is implementable: take records i and, for $q(a) > 0$, set

$$K_\eta(o, i \mid a) := \mathbf{1}_{\{o=o_i\}} \frac{m_i \pi_i(a)}{q(a)}. \quad (8)$$

Bayes plausibility makes the rows sum to one; null rows may be completed arbitrarily. Thus terminal laws range over the entire finite Bayes-plausible domain.

For channels K, L on the same action set and $t \in [0, 1]$, let $tK \oplus (1-t)L$ denote the channel that draws a public coin independent of the action, applies the selected channel, and includes the coin in the record. Then

$$\eta_{q, tK \oplus (1-t)L} = t\eta_{qK} + (1-t)\eta_{qL}. \quad (9)$$

A.1 Terminal-law sufficiency and a derived public coin

Lemma 19 (Terminal-law sufficiency). *Assume Axioms (A1), (A3), (A4). Fix a full-support prior q on A . If $\eta_{qK} = \eta_{qL}$, then $(q, K) \sim (q, L)$. Boundary priors satisfy the same conclusion after deleting null actions under Axiom (A5).*

Proof. For every reached (o, r) , replace r by the posterior π_{or}^{qK} . This is a deterministic record processor allowed to depend on o . The resulting canonical channel is

$$K^\eta(o, \pi \mid a) = \frac{\eta_{qK}(o, \pi) \pi(a)}{q(a)}.$$

Conversely, conditional on (o, π) , the canonical record can draw an original record r with probability $p_{qK}(o, r) / \eta_{qK}(o, \pi)$ among those records inducing π . This draw is independent of a : Bayes' formula gives $q(a)K(o, r \mid a) = p_{qK}(o, r)\pi(a)$. Hence K and K^η are record processings of one another. Axiom (A4) ranks each weakly above the other; weak order and block coherence give indifference. The same canonical channel is obtained from L when the terminal laws agree. For a boundary q , Axiom (A5) projects to $\text{supp } q$ and embeds back, making null rows irrelevant. $\square$

Corollary 20 (Derived pure-action consequentialism). *Let $q = \delta_a$ and let $\ell_a(o) := \sum_r K(o, r \mid a)$. Then (δ_a, K) is indifferent to the one-action, uninformative-record channel that pays according to ℓ_a . Thus no separate pure-action axiom is needed.*

Proof. Axiom (A5) first restricts the problem to the singleton support $\{a\}$ and embeds it back. On a singleton, Axiom (A4) may delete the record conditional on o ; in the reverse direction it redraws r from the conditional distribution induced by $K(\cdot | a)$. The two record processings and the two action maps give indifference by weak order and block coherence. $\square$

Lemma 21 (Derived public-coin independence). *Assume Axioms (A1), (A3), (A4), (A6). For a full-support q , channels K, L, M on A , and $t \in (0, 1)$,*

$$(q, K) \succeq (q, L) \iff (q, tK \oplus (1-t)M) \succeq (q, tL \oplus (1-t)M).$$

Proof. Use as first stage an action-independent binary record with probabilities $(t, 1-t)$. Both branch posteriors equal q . Put K versus L on the first branch and the common continuation M on the second. Axiom (A6) gives the forward implication. The compounds have the terminal laws in (9), so Lemma 19 identifies them with the stated public mixtures.

For the converse, if $K \not\succeq_q L$, completeness gives $L \succ_q K$. Apply the strict clause of Axiom (A6) with the continuations exchanged. It makes the reverse public mixture strictly preferred, which contradicts the displayed weak comparison. $\square$

This is the old public-coin axiom derived from the continuation axiom. It is not a ninth primitive.

A.2 Material and product calibration

Lemma 22 (Common material scale and dummy neutrality). *Assume Axioms (A1)–(A7). There is a nonconstant vNM index $u : O \rightarrow \mathbb{R}$ such that every channel whose payoff lottery ℓ is independent of the action and whose record is uninformative is ranked by $\sum_o \ell(o)u(o)$, independently of its prior and action alphabet.*

Moreover, independent dummy actions are neutral. If $p \in \Delta(B)$ and $K^\uparrow(o, r | a, b) := K(o, r | a)$, then

$$(q \otimes p, K^\uparrow) \sim (q, K). \quad (10)$$

Proof. At a singleton action, Lemma 21 gives mixture independence over payoff lotteries. The public coin is visible, but Lemma 19 flattens it to the ordinary mixed payoff lottery because every terminal posterior is the singleton prior. Axiom (A2) supplies the Archimedean condition, so the finite mixture theorem gives expected utility. Axiom (A7) makes the index nonconstant.

For arbitrary priors q and p , let an action report ignore its input and draw the reported action from p . Applied to an action-independent payoff channel, (2) produces the same payoff lottery at prior p . Axiom (A5) gives one weak comparison; repeating the construction from p to q gives the reverse. Thus the payoff-lottery order is independent of the prior. Block coherence and two-way relabelling transport it across finite action alphabets. Inserting two payoff lotteries into the same reached branch and using the weak and strict parts of Axiom (A6) shows that the insertion is an order isomorphism. The two distinct payoff anchors supplied by nonconstant u will later fix its cardinal multiplier.

For dummy neutrality, project (a, b) to a . Equation (2) and Axiom (A5) give $(q \otimes p, K^\uparrow) \succeq (q, K)$. In the other direction, report (a, b) from a by independently drawing $b \sim p$; the Bayesian pushforward is $K^\uparrow$, so Axiom (A5) gives the reverse comparison. Weak order and block coherence yield (10). $\square$

Lemma 23 (Affine terminal representation and block bridge). *For every full-support prior q , preferences on $\mathcal{E}_q$ have a continuous affine representative W_q . These representatives may be normalized so that*

$$W_q\left(\sum_o \ell(o)\delta_{(o,q)}\right) = \sum_o \ell(o)u(o). \quad (11)$$

With this normalization, dummy lifting preserves numerical value and the family $\{W_q\}$ represents every block-supported comparison, including comparisons whose priors or action alphabets differ.

Proof. Terminal-law sufficiency turns the order at q into an order on the mixture set $\mathcal{E}_q$. Public-coin independence is mixture independence there. Along any mixture segment the union of the two finite supports is fixed, so Axiom (A2) gives mixture continuity. The mixture-space theorem therefore supplies a continuous affine representative directly on all of $\mathcal{E}_q$. Normalize it by the two distinct payoff values in Lemma 22, which gives (11).

Dummy lifting is an affine ordinal isomorphism between a q -face and its copy in the $q \otimes p$ -fibre. Affine uniqueness initially permits a transformation $x \mapsto cx + d$. The two payoff anchors have the same numerical values on both faces by (11); hence $c = 1$ and $d = 0$.

Finally, lift an alternative at $q \in \Delta(A)$ by an independent $p \in \Delta(B)$ and lift an alternative at p by an independent q . Both lifted alternatives now have prior $q \otimes p$ on $A \times B$. Dummy neutrality, block coherence, and transitivity in a common multiple-block environment allow each original alternative to be replaced by its lift. Their block comparison is therefore represented by the single affine index $W_{q \otimes p}$, whose two restrictions equal W_q and W_p . Boundary priors follow by support restriction and a nondegenerate dummy lift. $\square$

A.3 The pure-trace engine

Lemma 24 (Pure-record reduction and affine scale). *Fix $o_0 \in O$ and restrict attention to channels*

$$K_P(o, r \mid a) := \mathbf{1}_{\{o=o_0\}} P(r \mid a).$$

Under Axioms (A1)–(A6), (A8), every within-channel and block-supported comparison on this restriction is ordered by $I_{qP}(A; R)$. After the common material scale and dummy normalisation of Lemmas 22–23, the affine value of this restriction is

$$W(q, K_P) = u(o_0) + \lambda I_{qP}(A; R) \tag{12}$$

for one $\lambda > 0$, common to all priors and finite action sets.

Proof. On a constant-payoff restriction, Axioms (A1)–(A6), (A8) induce Conditions (P1)–(P5) of the pure-record model developed in Appendix C: weak order and positive orientation, continuity, block coherence, record and action data processing, and branchwise continuation. The self-contained pure-record characterization, Proposition 32, therefore gives the ordinal conclusion $I(A; R)$.

By Lemma 23, $W - u(o_0)$ is a common-scale affine representation of the ordinal I order on the constant-payoff domain, so it is $\varphi(I)$ for a strictly increasing φ with $\varphi(0) = 0$. Publicly mixing two experiments at the same prior makes both their posterior laws and their I values mix linearly; affinity gives

$$\varphi(tx + (1 - t)y) = t\varphi(x) + (1 - t)\varphi(y).$$

Erasure channels at the uniform prior on n actions attain the whole interval $[0, \log n]$. These intervals are nested and unbounded as n grows, so continuity gives $\varphi(x) = \lambda x$ on $\mathbb{R}_+$. Axiom (A8) gives $\lambda > 0$, proving (12). $\square$

A.4 Calibrated branch recursion

Lemma 25 (Calibrated branch recursion). *The common-scale representative W of Lemma 23 satisfies*

$$W(q, P \triangleright \{K^y\}) = \lambda I_q(A; Y) + \sum_{y: m(y) > 0} m(y) W(q_y, K^y). \tag{13}$$

For a continuation that pays o and reveals nothing further, $W = u(o)$.

Proof. Hold all continuations except one reached branch y fixed. Axiom (A6) and its strict converse say that branch insertion is an order isomorphism. Between affine faces, the change in compound value must therefore equal a positive multiplier c_y times the change in continuation value.

This multiplier is independent of the fixed continuations in the other branches. Indeed, if their terminal sublaw is η_0 and the reached branch has mass $m = m(y)$, affinity gives

$$W_q(\eta_0 + m\eta_K) - W_q(\eta_0 + m\eta_L) = m L_q(\eta_K - \eta_L),$$

where L_q is the linear part of W_q ; the background η_0 cancels.

By the background-independence just shown, reset every other continuation to the deterministic payoff o_0 fixed in Lemma 24 and an uninformative record. Now choose in branch y two continuations with that same payoff and different, nontrivial record experiments. Both the local and aggregate comparisons lie in the pure-record restriction. Equation (12) and the mutual-information chain rule give

$$\Delta W_{\text{compound}} = \lambda m(y) \Delta I_{\text{branch}} = m(y) \Delta W_{\text{branch}}.$$

Thus $c_y = m(y)$. The multiplier belongs to the insertion map, so it applies to every pair of continuations, including payoff changes and mixed payoff–record changes. If q_y is degenerate and has no nontrivial trace direction, append an independent nondegenerate dummy action, use the preceding calibration there, and project it away using (10); the conclusion is unchanged.

Changing branches one at a time now shows that the two sides of (13) differ by a term depending only on the fixed first stage. Evaluate that term with every continuation paying o_0 and revealing nothing. The compound then lies in the pure-record restriction and has value $u(o_0) + \lambda I_q(A; Y)$, while the weighted continuation values sum to $u(o_0)$. The missing term is therefore exactly $\lambda I_q(A; Y)$. $\square$

A.5 Completion of the representation

Proof of Theorem 1. Assume Axioms (A1)–(A8) and fix an arbitrary joint channel K . Write $Z := O \times R$ and sequentialise K as follows. The first-stage record channel is

$$P_K(z \mid a) := K(o, r \mid a), \quad z = (o, r).$$

After the record $z = (o, r)$, the continuation pays o for sure and reveals nothing further. This does not mean that payoff is paid early: the first record announces the payoff that the terminal continuation subsequently delivers.

The compound channel is indifferent to K . Starting from K , a record processor can copy the already observed pair (o, r) into the compound record; starting from the compound, another processor deletes the copy. Both processors leave the payoff coordinate fixed, so Axiom (A4) applies in both directions. The calibrated branch recursion now gives

$$W(q, K) = \lambda I_q(A; Z) + \sum_z p_{qK}(z) u(o_z) = \lambda I_{qK}(A; O, R) + \mathbb{E}_{q, K}[u(O)].$$

This is (5), including block comparisons by Lemma 23. Axiom (A7) makes u nonconstant and Axiom (A8) makes $\lambda > 0$.

Conversely, let preferences be represented by (5). Weak order and continuity are immediate. Block tags are constant under a block-supported lottery, so they change neither expected utility nor mutual information, proving block coherence. Record processing preserves the payoff marginal and weakly reduces mutual information by data processing. Equation (2) also preserves the payoff marginal, and the Markov chain $(O, R) - A - B$ gives $I(A; O, R) \geq I(B; O, R)$. Thus Axioms (A4) and (A5) hold.

For a compound channel, the two chain rules are

$$\begin{aligned}\mathbb{E}[u(O)] &= \sum_y m(y) \mathbb{E}_{q_y, K^y}[u(O)], \\ I(A; Y, O, R) &= I(A; Y) + \sum_y m(y) I_{q_y, K^y}(A; O, R).\end{aligned}$$

The first-stage term $\lambda I(A; Y)$ is common to the two continuation profiles. Their aggregate value difference is therefore the positive-probability weighted sum of their branch value differences. This proves both clauses of Axiom (A6). Nonconstant u gives Axiom (A7), and $\lambda > 0$ gives Axiom (A8). $\square$

Proof of Proposition 2. Choose $o_* \in O$ with $u(o_*) = \underline{u}$. First note that the range of V is $[\underline{u}, \infty)$. The lower bound is immediate. For $q \in \Delta(A)$ and $t \in [0, 1]$, let a constant- o_* record channel reveal the realised action with probability t and otherwise report $*$:

$$E_t(o_*, a' \mid a) = t \mathbf{1}_{a'=a}, \quad E_t(o_*, * \mid a) = 1 - t.$$

Then $V(q, E_t) = \underline{u} + \lambda t H(q)$. At the uniform lottery on n actions these values fill $[\underline{u}, \underline{u} + \lambda \log n]$; the intervals are unbounded.

For part (i), let W be a common-scale representation and define $\varphi(v) := W(q, K)$ for any pair with $V(q, K) = v$. This is well defined: equality of the two V -values gives indifference in their two-block environment by Theorem 1, and the common-scale property gives equality of the two W -values. A strict inequality between V -values similarly gives a strict inequality between W -values. Thus φ is strictly increasing and $W = \varphi \circ V$. The converse is immediate from Theorem 1.

For part (ii), the expected-utility and mutual-information chain rules give, for any two continuation profiles,

$$V(q, P \triangleright \{K^y\}) - V(q, P \triangleright \{L^y\}) = \sum_{y: m(y) > 0} m(y) [V(q_y, K^y) - V(q_y, L^y)].$$

Hence every $\alpha V + \beta$, with $\alpha > 0$, is common-scale and branch-additive.

Conversely, let the branch-additive common-scale representation be $W = \varphi \circ V$ and define

$$\psi(x) := \varphi(\underline{u} + x) - \varphi(\underline{u}), \quad x \geq 0.$$

Fix $n \geq 2$ and let q be uniform on n actions. For $x \in [0, \lambda \log n]$, choose a constant- o_* erasure channel whose value is $\underline{u} + x$. In the first continuation profile, use this channel after the first branch of a public coin with probabilities θ and $1 - \theta$, and use the uninformative constant- o_* channel after the second. Let the comparison profile use the uninformative channel after both branches. The two compound values are $\underline{u} + \theta x$ and $\underline{u}$, so (6) gives

$$\psi(\theta x) = \theta \psi(x) \quad (0 \leq \theta \leq 1).$$

Taking $x = \lambda \log n$ shows that ψ is linear on $[0, \lambda \log n]$. These intervals are nested and unbounded, so $\psi(x) = \alpha x$ on $[0, \infty)$ for one $\alpha > 0$. Therefore $\varphi(v) = \alpha v + \beta$ on the range of V .

Finally, any second trace-tempered index $\tilde{V}$ is itself branch-additive. Applying part (ii) and then restricting to deterministic payoff outcomes and to constant-payoff record channels gives $\tilde{u} = \alpha u + \beta$ and $\tilde{\lambda} = \alpha \lambda$. $\square$

Proof of Corollary 4. The result follows from Theorem 1 and $I_{qK}(A; O, R) = H(q) - \int H(\pi) d\mu_{qK}(\pi)$. $\square$

A.6 Independence of the axioms

Proof of Proposition 5. Fix a nonconstant u and write $U(q, K) := \mathbb{E}_{q,K}[u(O)]$, $Z := (O, R)$, and $I := I(A; Z)$. All scores below are extended to block and compound channels by the same formula.

Failure of Axiom (A1). Use the Pareto relation on the two coordinates (U, I) :

$$(q, K) \succeq (p, L) \iff U(q, K) \geq U(p, L) \text{ and } I(q, K) \geq I(p, L).$$

It is transitive but incomplete whenever one alternative has more payoff and the other more information. Its graph is closed. Both data-processing axioms are coordinatewise monotonic, and both chain rules make branchwise weak and strict improvements aggregate. The two relevance axioms and block coherence hold.

Failure of Axiom (A2). Order alternatives lexicographically by (U, I) , with payoff first. This is a weak order and satisfies all structural, processing, continuation, and relevance requirements. It is not continuous even within one fixed channel. Let a three-action channel reveal the action fully and let only action 2 pay the high outcome. Put

$$p = (1/4, 1/2, 1/4), \quad q = (0.49, 0.50, 0.01), \quad q_n = (0.49 - \varepsilon_n, 0.50 + \varepsilon_n, 0.01),$$

where $\varepsilon_n \downarrow 0$. The lotteries p and q have the same expected utility but $H(q) < H(p)$, so $p \succ_K q$. Every q_n has strictly higher expected utility than p , so $q_n \succ_K p$. Thus the graph in Axiom (A2) is not closed.

Failure of Axiom (A3). Treat block labels as syntactic tags, with the outermost tag taking precedence. Put $w(a^0) = 1$, $w(a^i) = 0$ for $i \neq 0$, and $w(a) = 0$ on untagged actions. For $\varepsilon > 0$, use

$$V_\varepsilon(q, K) := U(q, K) + I(q, K) + \varepsilon \sum_a q(a)w(a).$$

Every record- or action-processing comparison places the original alternative in block 0, so the data-processing inequality receives the additional ε . If Δ_y is the base-score difference in continuation branch y , its block comparison is weak exactly when $\Delta_y + \varepsilon \geq 0$, while the aggregate block difference is

$$\sum_y m(y)\Delta_y + \varepsilon = \sum_y m(y)(\Delta_y + \varepsilon).$$

Thus Axiom (A6), including strictness, holds; the remaining axioms are immediate. Duplication coherence fails. Take constant payoff, K revealing a binary action, a degenerate q , and a nearby nondegenerate p with $0 < H(p) < \varepsilon$. Then $p \succ_K q$, but $q^0 \succ_{K \sqcup K} p^1$.

Failure of Axiom (A4). For $\varepsilon > 0$, use

$$V(q, K) := U(q, K) + I(A; Z) - \varepsilon H(Z | A). \tag{14}$$

This score is continuous and branch-additive, since both mutual information and conditional entropy obey the chain rule. Under action processing $A \rightarrow B$, the score difference between the original and processed alternatives is

$$[I(A; Z) - I(B; Z)] + \varepsilon [H(Z | B) - H(Z | A)] = (1 + \varepsilon)I(A; Z | B) \geq 0,$$

so (14) satisfies Axiom (A5). It also satisfies both relevance axioms. Record data processing fails: with constant payoff and a fair record independent of A , the original value is $-\varepsilon$; deleting that noise gives value zero.

Failure of Axiom (A5). Let $G(q) := 1 - \sum_a q(a)^2$ be Gini entropy and define its posterior reduction

$$J_G(q, K) := G(q) - \int G(\pi) d\mu_{qK}(\pi), \quad V := U + J_G.$$

Concavity of G gives record data processing, and the potential-difference form gives the exact branch chain rule. Full revelation is strictly valuable. Action processing can nevertheless increase J_G . Take $q = (1/8, 1/8, 3/4)$, a binary record with

$$\Pr(0 \mid a_1) = \Pr(0 \mid a_2) = 0, \quad \Pr(0 \mid a_3) = 1/4,$$

and merge a_1, a_2 into one reported action. Direct calculation gives $J_G = 9/416$ before the merge and $J_G = 3/104$ afterwards. Hence Axiom (A5) fails.

Failure of Axiom (A6). Use $V := U + \min\{I, 1\}$, with information measured in bits. It is continuous, respects both data-processing axioms, and has both strict orientations. Let q be uniform on four actions. A first-stage record reveals which of two pairs contains the action. In every branch, revealing the action within the pair strictly improves the continuation from zero to one bit. The two compounds have respectively one and two bits before capping and are therefore indifferent. The strict clause of Axiom (A6) fails.

Failure of Axiom (A7). Use $V := I$. All other axioms hold, but all payoff lotteries are indifferent.

Failure of Axiom (A8). Use $V := U$. All other axioms hold, but revealing the action at a fixed payoff is valueless. $\square$

B Proofs of the choice implications

Results stated in the text only in summary form (Propositions 26 and 27, Corollary 28, and Remark 29) are restated here in full. Proofs appear in an order that front-loads shared machinery.

Proof of Proposition 6. (i) Under either lottery, each visible pair (o_0, r_j) , $j = 1, 2$, has probability $\frac{1}{2}$: under q' because each a_j produces (o_0, r_j) for sure, under q'' because each of a_3, a_4 produces either pair with equal probability. Hence $p_{q'K} = p_{q''K}$. On $\text{supp}(q')$ the rows are degenerate and distinct, so the visible pair reveals the action and $I_{q'K} = H(q') = \log 2$; on $\text{supp}(q'')$ the rows are identical, so the visible pair is independent of the action and $I_{q''K} = 0$. The payoff terms agree because the payoff marginals do, giving the displayed gap. (ii) All rows of K° are equal, so $I_{q,K^\circ} = 0$ for every q ; since the payoff is constantly o_0 , the representation gives indifference. $\square$

Proof of Proposition 8. Expected utility is linear in q , while mutual information is concave in the input distribution of a fixed channel [18]. Continuity on the compact simplex gives existence. Using

$$I_{q,K}(A; Z) = \sum_a q(a) D_{\text{KL}}(K_Z(\cdot \mid a) \parallel m_{q,K}),$$

its directional derivative in the mass on action a equals the displayed divergence up to a term common to every action. The result is the Kuhn–Tucker condition for a concave programme. At constant u it reduces to the classical capacity condition. $\square$

Proof of Proposition 9. The entropy identity gives

$$V(q, K) = \lambda H(m_{q,K}) + \sum_a q(a) [\bar{u}(a) - \lambda H(P_a)].$$

If q_0 and q_1 are optimal with $m_{q_0,K} \neq m_{q_1,K}$, strict concavity of Shannon entropy and linearity of $q \mapsto m_{q,K}$ and of the remaining sum give $V(tq_0 + (1-t)q_1, K) > C^*(K)$ for $t \in (0, 1)$, a contradiction. Hence every optimum induces the same m^* . By Proposition 8, an optimum q^* satisfies the displayed equalities at some constant c ; averaging them under q^* gives $c = V(q^*, K) = C^*(K)$, so $\text{supp}(q^*) \subseteq \mathcal{A}^*$. Conversely, if $m_{q,K} = m^*$ and $\text{supp}(q) \subseteq \mathcal{A}^*$, then

$$V(q, K) = \sum_a q(a) [\bar{u}(a) + \lambda D_{\text{KL}}(P_a \parallel m^*)] = C^*(K).$$

Affine independence makes the representation $m^* = \sum_a q(a)P_a$ with $\text{supp}(q) \subseteq \mathcal{A}^*$ unique. Identical rows share $\bar{u}(a)$ and $H(P_a)$, so both $m_{q,K}$ and the displayed expression for V are unchanged by redistribution within a class; collapsing the classes and lifting aggregate masses proves the last claim. $\square$

Proof of Proposition 7. Throughout, m^* denotes the common optimal marginal of Proposition 9.

(i) Suppose $m^*(z) = 0$ for some $z \in Z_K$ and choose a with $K_Z(z | a) > 0$. Then $D_{\text{KL}}(K_Z(\cdot | a) \| m^*) = +\infty$, which is incompatible with the finite on-support equality if some optimum uses a and with the off-support inequality of Proposition 8 if none does. Under the private-consequence condition, an optimum with $q^*(a) = 0$ would force $m^*(z_a) = 0$.

(ii) For $q = \delta_a$ the induced marginal is $K_Z(\cdot | a)$ and the on-support constant is $\bar{u}(a)$; the displayed system is the off-support condition of Proposition 8, necessary and sufficient by concavity. Under strict inequalities, $\mathcal{A}^* = \{a\}$ in Proposition 9, so every optimum is δ_a . $\square$

Proposition 26 (Optimal mixing). *Let $A = \{a, b\}$ with $K_a := K_Z(\cdot | a) \neq K_b := K_Z(\cdot | b)$ and $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$. For $x \in [0, 1]$ let q_x put mass x on b , so that $m_{q_x, K} = (1-x)K_a + xK_b =: m_x$, and for $x \in (0, 1)$ define*

$$g(x) := D_{\text{KL}}(K_b \| m_x) - D_{\text{KL}}(K_a \| m_x).$$

(i) *g is continuous and strictly decreasing, with $g(0^+) = D_{\text{KL}}(K_b \| K_a)$ and $g(1^-) = -D_{\text{KL}}(K_a \| K_b)$ as extended-real limits. If $\Delta < \lambda D_{\text{KL}}(K_b \| K_a)$, the optimum is unique and interior, $q^* = q_{x^*}$ with x^* the unique solution of $\Delta = \lambda g(x)$.*

(ii) *The optimal mass $x^*(\Delta, \lambda)$ on the inferior action is nonincreasing in Δ and nondecreasing in λ , strictly so on the mixing region; at $\Delta = 0$ it is the capacity-achieving input of the two-row channel, independently of λ .*

Proof. Write $f(x) := V(q_x, K) = \bar{u}(a) - x\Delta + \lambda I(x)$ with $I(x) = H(m_x) - (1-x)H(K_a) - xH(K_b)$. Since $K_a \neq K_b$, $x \mapsto m_x$ is affine and injective, so strict concavity of H [18] makes f strictly concave on $[0, 1]$, and a unique maximiser exists. On $(0, 1)$ every $z \in \text{supp } K_a \cup \text{supp } K_b$ has $m_x(z) > 0$ and, using $\sum_z (K_b(z) - K_a(z)) = 0$,

$$I'(x) = H(K_a) - H(K_b) - \sum_z (K_b(z) - K_a(z)) \log m_x(z) = g(x),$$

so $f'(x) = -\Delta + \lambda g(x)$, with g continuous and strictly decreasing as the derivative of a strictly concave function. As $x \downarrow 0$, $m_x(z) \rightarrow K_a(z)$; if $\text{supp } K_b \subseteq \text{supp } K_a$ every summand converges and $g(0^+) = D_{\text{KL}}(K_b \| K_a)$, while if some $\bar{z}$ has $K_b(\bar{z}) > 0 = K_a(\bar{z})$ the summand $-K_b(\bar{z}) \log(xK_b(\bar{z}))$ diverges to $+\infty$, matching the extended-sense divergence. Exchanging the roles of a and b gives the limit at 1. By concavity and the mean value theorem, δ_a is optimal if and only if $\lim_{x \downarrow 0} f'(x) \leq 0$, i.e. $\Delta \geq \lambda D_{\text{KL}}(K_b \| K_a)$; this is the threshold stated in the text, and Proposition 7(ii) with two actions. If $\Delta < \lambda D_{\text{KL}}(K_b \| K_a)$, the maximiser is not $x = 0$ by the threshold and not $x = 1$ because $f'(1^-) = -\Delta - \lambda D_{\text{KL}}(K_a \| K_b) < 0$; it is therefore the unique root of $\Delta = \lambda g(x)$, which exists because g decreases strictly between its limits. This proves (i). The comparative statics of (ii) follow by inverting the strictly decreasing g at Δ/λ : raising Δ raises the target, raising λ lowers it; at $\Delta = 0$ the condition $g(x) = 0$ is the capacity condition of Proposition 8 at constant payoff. $\square$

Proposition 27 (Trace demand). *Fix K and write $e(q) := \mathbb{E}_{q,K}[u(O)]$, $i(q) := I_{qK}(A; O, R)$,*

$$V^*(\lambda) := \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}, \quad Q(\lambda) := \arg \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}.$$

(i) V^* is convex on $(0, \infty)$, i is affine on the convex compact set $Q(\lambda)$, and

$$(V^*)'_-(\lambda) = \min_{q \in Q(\lambda)} i(q), \quad (V^*)'_+(\lambda) = \max_{q \in Q(\lambda)} i(q);$$

at every point of differentiability, $(V^*)'(\lambda) = i(q)$ for every $q \in Q(\lambda)$.

(ii) Let $0 < \lambda_1 < \lambda_2$ and $q_j \in Q(\lambda_j)$, writing $e_j := e(q_j)$, $i_j := i(q_j)$. Then $i_1 \leq i_2$, $e_1 \geq e_2$, and

$$\lambda_1(i_2 - i_1) \leq e_1 - e_2 \leq \lambda_2(i_2 - i_1).$$

Hence either $(e_1, i_1) = (e_2, i_2)$, or $i_1 < i_2$ and $e_1 > e_2$ and

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2.$$

Proof of Proposition 27. (i) V^* is a pointwise maximum of functions affine in λ , hence convex and, since $0 \leq i \leq \log |A|$, finite. On the convex compact set $Q(\lambda)$ the objective is constant and e is linear, so $i = (V^* - e)/\lambda$ is affine there. For $q \in Q(\lambda)$ and $\mu > 0$, $V^*(\mu) \geq V^*(\lambda) + (\mu - \lambda)i(q)$, so each such $i(q)$ is a subgradient. For the right derivative, choose $q_h \in Q(\lambda + h)$; then

$$\max_{q \in Q(\lambda)} i(q) \leq \frac{V^*(\lambda + h) - V^*(\lambda)}{h} \leq i(q_h),$$

the first inequality by the subgradient bound, the second by optimality of $Q(\lambda)$ at λ . By compactness and continuity, accumulation points of $(q_h)_{h \downarrow 0}$ lie in $Q(\lambda)$, so the right-hand side accumulates in $i(Q(\lambda))$ and the right derivative equals the maximum. The left derivative is symmetric, and a convex function on an interval is differentiable off a countable set.

(ii) Optimality at the two weights gives $e_1 + \lambda_1 i_1 \geq e_2 + \lambda_1 i_2$ and $e_2 + \lambda_2 i_2 \geq e_1 + \lambda_2 i_1$. Adding yields $(\lambda_2 - \lambda_1)(i_2 - i_1) \geq 0$, hence $i_2 \geq i_1$; the two inequalities rearrange to the displayed bracket, which gives $e_1 \geq e_2$ and, when the pairs differ, the final bound. $\square$

Corollary 28 (Extreme trace weights). *Let $e^0 := \max_q e(q)$, $i^0 := \max\{i(q) : e(q) = e^0\}$, $C := \max_q i(q)$, and $e^C := \max\{e(q) : i(q) = C\}$. If $\lambda_n \downarrow 0$ and $q_n \in Q(\lambda_n)$, every accumulation point of (q_n) maximises i among the maximisers of e ; if $\lambda_n \uparrow \infty$, every accumulation point maximises e among the maximisers of i . Moreover*

$$V^*(\lambda) = e^0 + \lambda i^0 + o(\lambda) \quad \text{as } \lambda \downarrow 0, \quad V^*(\lambda) = \lambda C + e^C + o(1) \quad \text{as } \lambda \uparrow \infty.$$

Proof of Corollary 28. For $\lambda_n \downarrow 0$, comparing q_n with an e -maximiser attaining i^0 gives

$$0 \leq e^0 - e(q_n) \leq \lambda_n(i(q_n) - i^0).$$

Boundedness of i forces $e(q_n) \rightarrow e^0$, and the same inequality gives $i(q_n) \geq i^0$; continuity delivers the accumulation claim. Dividing by λ_n ,

$$\frac{V^*(\lambda_n) - e^0}{\lambda_n} = i(q_n) - \frac{e^0 - e(q_n)}{\lambda_n},$$

where the last fraction lies in $[0, i(q_n) - i^0]$ and vanishes, proving the first expansion. For $\lambda_n \uparrow \infty$, comparing q_n with an i -maximiser attaining e^C gives $0 \leq \lambda_n(C - i(q_n)) \leq e(q_n) - e^C$, and the symmetric argument completes the proof. $\square$

Proof of Proposition 12. The processor leaves the payoff coordinate unchanged, so the expected-utility terms coincide and the difference is $\lambda[I_{qK}(A; O, R) - I_{qL}(A; O, R')]$. Write $Z := (O, R)$ and $Z' := (O, R')$. By construction $A \rightarrow Z \rightarrow Z'$ is a Markov chain, so expanding $I(A; Z, Z')$ in both orders gives

$$I(A; Z) - I(A; Z') = I(A; Z \mid Z') = I(A; R \mid O, R'),$$

the last equality because O is part of Z' . Conditional mutual information is nonnegative and vanishes exactly under the conditional independence in (b) [18], proving the identity and (a) $\Leftrightarrow$ (b). For (b) $\Rightarrow$ (c): the Markov property gives $\Pr(A = \cdot \mid Z, Z') = \pi_Z^{qK}$ almost surely, while (b) gives $\Pr(A = \cdot \mid Z, Z') = \pi_{Z'}^{qL}$; the two random posteriors coincide almost surely, so their laws are equal. For (c) $\Rightarrow$ (a): by the identity displayed after Corollary 4, the information term depends on (q, K) only through μ_{qK} , and the payoff terms already agree. $\square$

Proof of Proposition 18. (i) If $L = KT$, Proposition 12 gives $I_{qK} \geq I_{qL}$ at every q . Because the payoff kernels coincide, $V(q, K) - V(q, L) = \lambda[I_{qK} - I_{qL}]$ with $\lambda > 0$, so uniform value dominance and $K \succeq_c L$ are the same statement.

(ii) Let $A = \{0, 1\}$, fix $o^* \in O$, and let both channels pay o^* surely. Let K carry the binary erasure record with erasure probability $\varepsilon = \frac{1}{2}$ and L the binary symmetric record with crossover $p = \frac{1}{5}$:

$$K(o^*, a \mid a) = 1 - \varepsilon, \quad K(o^*, e \mid a) = \varepsilon, \quad L(o^*, a \mid a) = 1 - p, \quad L(o^*, 1 - a \mid a) = p.$$

The payoff kernels agree row by row, so it suffices to compare informations, and the comparison is invariant to the base; compute in bits with h the binary entropy. For $q = (1 - t, t)$, grouping records gives

$$I_{qK} = (1 - \varepsilon)h(t), \quad I_{qL} = h(p * t) - h(p), \quad p * t := p(1 - t) + (1 - p)t.$$

By Mrs Gerber's Lemma [49], $v \mapsto h(p * h^{-1}(v))$ is convex on $[0, 1]$, with h^{-1} valued in $[0, \frac{1}{2}]$; since $h(p * t)$ is invariant under $t \mapsto 1 - t$, take $t \leq \frac{1}{2}$ and $v = h(t)$. The chord between $v = 0$ and $v = 1$ gives

$$h(p * t) \leq (1 - v)h(p) + v, \quad \text{hence} \quad I_{qL} \leq (1 - h(p))h(t) \leq (1 - \varepsilon)h(t) = I_{qK},$$

the last step because $\varepsilon = \frac{1}{2} \leq h(\frac{1}{5})$. Thus $K \succeq_c L$. Now suppose $L = KT$ for a processor T , and write $t_r := T(0 \mid o^*, r)$ for $r \in \{0, 1, e\}$. Matching the probability of the processed record 0 in the two rows,

$$(1 - \varepsilon)t_0 + \varepsilon t_e = 1 - p, \quad (1 - \varepsilon)t_1 + \varepsilon t_e = p,$$

and subtracting gives $t_0 - t_1 = (1 - 2p)/(1 - \varepsilon) = \frac{6}{5} > 1$, impossible. Compositions of record processors are record processors, so no chain of garblings produces L from K either. $\square$

Proof of Proposition 14. (i) The material part of the compound is $(1 - \varepsilon)\mathbb{E}_{q,K}[u(O)] + \varepsilon w(\beta)$. For the trace part, the coin is action-independent, so both branch posteriors equal q and the chain rule for the compound gives an information term of $(1 - \varepsilon)I_{qK}(A; O, R)$, the sweetener branch contributing zero. (ii) By the moreover clause of Theorem 1, the indifference holds if and only if $(1 - \varepsilon)V(q, K) + \varepsilon w(\beta) = (1 - \varepsilon)V(p, L) + \varepsilon w(\beta')$; rearrange, and substitute the representation when the material parts cancel. (iii) $w(\beta) - w(\beta')$ is continuous on $[0, 1]^2$ with range $[-\Delta u, \Delta u]$; apply the intermediate value theorem to the equation in (ii). $\square$

Proof of Proposition 15. (i) Inclusion is immediate. Conversely, fix (t, w) in the strip, choose n with $\log n \geq t$, and on $A = \{1, \dots, n\}$ use continuity of entropy along a segment from a point mass to the uniform lottery to find q_t with $H(q_t) = t$. Let $K(o^+, a \mid a) = \beta$ and $K(o^-, a \mid a) = 1 - \beta$ with β chosen so that $w(\beta) = w$. The payoff lottery is action-independent with mean w , and the record reveals the action, so $\phi(q_t, K) = (t, w)$. The description of the indifference classes then reads the representation through ϕ , using the moreover clause of Theorem 1. (ii) The identity rearranges $E_{qK} + \lambda I_{qK} = E_{pL} + \lambda I_{pL}$. For uniqueness, let $\lambda' \neq \lambda$, put $t := \frac{1}{2} \min\{1, (\bar{u} - \underline{u})/\lambda\} > 0$, and use (i) to realise pairs with ϕ -coordinates $(t, \bar{u} - \lambda t)$ and $(0, \bar{u})$; they are indifferent, while $E + \lambda' I$ differs across them by $(\lambda - \lambda')t \neq 0$. $\square$

Remark 29 (Eliciting λ). Normalise $u(o^+) = 1$ and $u(o^-) = 0$. All comparisons below are block comparisons, hence within the primitive domain, and no information quantity is estimated: the information gap is an accounting identity of the design. *Step 1.* For each $o \in O$, elicit β_o with $(\delta_*, K_o) \sim (\delta_*, B_{\beta_o})$, where K_o is the degenerate one-action channel of Axiom (A7); one-action channels carry zero information, so $u(o) = \beta_o$. *Step 2.* Fix $o \in O$, $n \geq 2$, the uniform q on $\{1, \dots, n\}$, a dilution $\varepsilon \in (0, 1)$, and the constant-payoff channels K_o^{id} and K_o^0 of Axiom (A8). Elicit β with

$$(q, D_{\varepsilon, \beta} K_o^0) \sim (q, D_{\varepsilon, 0} K_o^{\text{id}}); \quad \text{then} \quad \lambda = \frac{\varepsilon}{1 - \varepsilon} \cdot \frac{\beta}{\log n},$$

by Proposition 14(i) applied to $V(q, K_o^{\text{id}}) = u(o) + \lambda \log n$ and $V(q, K_o^0) = u(o)$. Such a β exists if and only if $\lambda \log n \leq \varepsilon/(1 - \varepsilon)$; a subject still strictly preferring the right block at $\beta = 1$ reveals $\lambda > \varepsilon/((1 - \varepsilon) \log n)$, and ε is raised. *Controls.* Orientation: $(q, K_o^{\text{id}}) \succ (q, K_o^0)$ is Axiom (A8) itself. Garbling: under the total record garbling of K_o^{id} , both blocks carry zero information and indifference must obtain; a surviving strict preference indicates that an uncontrolled feature of the display still traces the action. Transport: repeating Step 2 at another n , prior, or payoff level must return the same λ , by Proposition 15(ii); an estimate that declines in $\log n$ is the signature of the capped-information criterion.

Self-contained foundation for the trace component

This appendix contains the pure-record characterization on which Lemma 24 rests. It is included in full so that the present paper has no logical dependency on an earlier manuscript. In this auxiliary model, X is a finite visible-record alphabet, $P : A \rightarrow \Delta(X)$ is a record channel, and the primitive relation ranks action lotteries by the trace they leave in X . Conditions (P1)–(P5) below are exactly the restrictions induced on the constant-payoff subdomain by, respectively, (A1),(A8), (A2), (A3), (A4),(A5), and (A6) of the main model.

C The pure-trace characterization

Let A be a finite set of *actions* and X a finite set of *records*. A channel is a stochastic kernel $P : A \rightarrow \Delta(X)$. For each finite channel P , the primitive behavioural datum is a binary relation

$$\succeq_P \quad \text{on} \quad \Delta(A).$$

Condition (P1) below requires this relation to be a weak order. The interpretation is that $q \succeq_P q'$ means: in the fixed environment P , lottery q displays at least as much traceability as q' . The environment P is not an object of choice. A lottery is the procedure being evaluated; the channel is the fixed observation technology through which the realised act may be recorded by records.

Comparison environments. To compare lotteries carried by different channels without abandoning channel-indexed preferences, place the channels in labelled, disjoint blocks of one larger environment. For channels $P : A \rightarrow \Delta(X)$ and $Q : B \rightarrow \Delta(Y)$, write $P \sqcup Q$ for the block-diagonal channel with action alphabet $(A \times \{0\}) \cup (B \times \{1\})$ and record alphabet $(X \times \{0\}) \cup (Y \times \{1\})$. It sends $(a, 0)$ to $(x, 0)$ with probability $P(x | a)$ and $(b, 1)$ to $(y, 1)$ with probability $Q(y | b)$; all cross-block probabilities are zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, define lotteries on the tagged action alphabet by

$$\begin{aligned} q^0(a, 0) &:= q(a), & q^0(b, 1) &:= 0, \\ p^1(a, 0) &:= 0, & p^1(b, 1) &:= p(b), \end{aligned} \quad (a \in A, b \in B).$$

The superscript is a block label, not a power. Accordingly, $q^0 \succeq_{P \sqcup Q} p^1$ is an ordinary preference comparison inside one fixed environment.⁴ When the two blocks have the same action alphabet, q^0 and q^1 are the two tagged copies of the same lottery q . For a finite family $\{P_i : A_i \rightarrow \Delta(X_i)\}_{i \in K}$, write $\bigsqcup_{i \in K} P_i$ for the analogous finite block environment and q_i^i for the lottery that assigns $q_i(a)$ to each tagged action (a, i) and zero to every other block.

Pure-record conditions. The conditions are imposed directly on the family $\{\succeq_P\}_P$. Five conditions are imposed on the family. The first three are structural: each environment carries

⁴For a complete numerical example, let $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $X = \{x_0, x_1\}$, and $Y = \{y_0, y_1\}$. With rows indexed by actions and columns by records, take $P = \begin{pmatrix} 0.1 & 0.9 \\ 0.8 & 0.2 \end{pmatrix}$ and $Q = \begin{pmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{pmatrix}$. Order the tagged actions as $(a_1, 0), (a_2, 0), (b_1, 1), (b_2, 1)$ and the tagged records as $(x_0, 0), (x_1, 0), (y_0, 1), (y_1, 1)$. Then $P \sqcup Q = \text{diag}(P, Q)$: the first two actions generate only the first two records, and the last two actions only the last two. If $q = (0.3, 0.7)$ and $p = (0.8, 0.2)$, their embeddings are $q^0 = (0.3, 0.7; 0, 0)$ and $p^1 = (0, 0; 0.8, 0.2)$, where the semicolon separates the two blocks. Thus $q^0 \succeq_{P \sqcup Q} p^1$ says precisely that lottery q , observed through P , is at least as traceable as lottery p , observed through Q ; it does not treat either channel as an object of choice. Under the representation proved below, the two sides yield approximately 0.330 and 0.019 bits of mutual information, respectively, and hence the comparison is strict in this example.

a continuous weak order with a fixed orientation (P1, P2), and labelled blocks provide the comparison framework (P3). The substantive restrictions are two: garbling cannot increase traceability (P4), and sequential observation is evaluated branch by branch (P5).

(P1) Weak order and local non-triviality. For every finite channel $P : A \rightarrow \Delta(X)$, $\succeq_P$ is complete and transitive on $\Delta(A)$. The family is locally non-trivial: for every finite action set A with $|A| \geq 2$ and every full-support $q \in \Delta(A)$,

$$q^0 \succ_{\text{Id}_A \sqcup U_A} q^1,$$

where $\text{Id}_A : A \rightarrow \Delta(A)$ is full revelation and $U_A : A \rightarrow \Delta(\{*\})$ is the one-record uninformative channel.

This condition gives each observation environment an ordinary weak order over act lotteries and fixes the orientation of the comparison. When the realised act is genuinely uncertain, observing it directly is strictly more traceable than observing nothing about it.

(P2) Continuity. For every pair of finite alphabets (A, X) , the set

$$\{(P, q, q') \in \Delta(X)^A \times \Delta(A) \times \Delta(A) : q \succeq_P q'\}$$

is closed in the usual Euclidean product topology. Thus within-environment rankings vary continuously with both the lotteries and the fixed channel.

Continuity says that traceability comparisons are stable under small changes in the procedure and the environment. Small modelling perturbations should not create abrupt reversals. The condition is stated entirely in the primitives: it quantifies over channels and lotteries, not over derived Bayesian objects. Continuity of block comparisons in posterior-law convergence at a fixed full-support prior is not assumed; it is derived below (Lemma 40).

(P3) Block-comparison coherence. *Duplication:* for every channel $P : A \rightarrow \Delta(X)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \iff q^0 \succeq_{P \sqcup P} (q')^1.$$

Irrelevant blocks: for every finite block environment $\bigsqcup_{k \in K} P_k$, every ordered pair of distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff q_i^0 \succeq_{P_i \sqcup P_j} q_j^1.$$

Applied at the ordered pair $(1, 0)$ of a two-block environment, the second clause exchanges block labels; this is how reversed block comparisons are brought to canonical form throughout (Lemma 36(a)).

Labelled blocks are a way to place procedures carried by different channels in a common comparison problem. The labels should not themselves create traceability, alter the ranking inside a duplicated environment, or let unused blocks affect the comparison between active blocks.

(P4) Split data-processing aversion. Condition (P4) is the conjunction of the following two clauses. This is the formulation used by the formal theorem: the record and action coordinates are processed separately, and no convention is imposed on null reported actions.

(DP-rec) *Record post-processing.* For every channel $P : A \rightarrow \Delta(X)$, every stochastic $T : X \rightarrow \Delta(X')$, and every $q \in \Delta(A)$, define

$$(PT)(x' | a) := \sum_{x \in X} P(x | a) T(x' | x).$$

Then

$$q^0 \succeq_{P \sqcup PT} q^1.$$

(DP-act) *Bayesian action coarsening.* For every $P : A \rightarrow \Delta(X)$, every stochastic $S : A \rightarrow \Delta(A')$, every $q \in \Delta(A)$, write

$$(qS)(a') := \sum_{a \in A} q(a)S(a' | a).$$

For every channel $\hat{P} : A' \rightarrow \Delta(X)$ satisfying

$$(qS)(a')\hat{P}(x | a') = \sum_{a \in A} q(a)P(x | a)S(a' | a) \quad \text{for every } (a', x) \in A' \times X, \quad (\text{DP-act-law})$$

we have

$$q^0 \succeq_{P \sqcup \hat{P}} (qS)^1.$$

When qS has full support, (DP-act-law) determines $\hat{P}$ uniquely; otherwise the clause applies to every channel consistent with it. In particular, bijective relabellings are neutral by applying (DP-act) in both directions.

The act whose trace is evaluated may be reported through a possibly noisy description, and the visible record may be garbled after it is generated. Neither operation can create observable trace: the original procedure is weakly more traceable than any procedure obtained by either operation.

Remark 30. The two primitive clauses imply the convenient simultaneous form used in a few calculations. Let $S : A \rightarrow \Delta(A')$ and $T : X \rightarrow \Delta(X')$, and let $P' : A' \rightarrow \Delta(X')$ satisfy

$$(qS)(a')P'(x' | a') = \sum_{a \in A} \sum_{x \in X} q(a)P(x | a)S(a' | a)T(x' | x) \quad \text{for every } (a', x') \in A' \times X'. \quad (\text{DP})$$

First apply (DP-rec) to form PT , and then apply (DP-act) to (q, PT) through S . Equation (DP) says exactly that the supplied P' is an allowed Bayesian completion of this second step, including on null reported actions. Chain the two comparisons in a common three-block environment using Conditions (P1) and (P3). This yields $q^0 \succeq_{P \sqcup P'} (qS)^1$. Conversely, the simultaneous form specialises to each named clause. Thus the two presentations are equivalent under (P1) and (P3), but the proposition below takes the two named clauses as its literal premise. If $(qS)(a') = 0$, both (DP-act-law) and (DP) leave the corresponding channel row unrestricted; null reports require no convention.

Reached records and posteriors. For a lottery $q \in \Delta(A)$ and a channel $P : A \rightarrow \Delta(X)$, write

$$m_{q,P}(x) := \sum_{a \in A} q(a)P(x | a), \quad r_x^{q,P}(a) := \frac{q(a)P(x | a)}{m_{q,P}(x)} \quad \text{if } m_{q,P}(x) > 0$$

for, respectively, the probability of record x and the posterior belief about the realised action. Call x reached when $m_{q,P}(x) > 0$; all branch comparisons below are restricted to reached records.

(P5) Branchwise continuation monotonicity. Let $P_1 : A \rightarrow \Delta(X_1)$ be a first-stage channel and, after each first-stage record x , let $Q^x, R^x : A \rightarrow \Delta(X_2^x)$ be two alternative continuation channels. Write $P_1 \triangleright \{Q^x\}$, and analogously $P_1 \triangleright \{R^x\}$, for the compound channel with record alphabet $\bigcup_{x \in X_1} (\{x\} \times X_2^x)$ that reports both stages:

$$(P_1 \triangleright \{Q^x\})(x, z | a) := P_1(x | a)Q^x(z | a).$$

Fix $q \in \Delta(A)$ and let $m(x) := m_{q,P_1}(x)$ and $r_x := r_x^{q,P_1}$ for all branches with $m(x) > 0$.

If

$$r_x^0 \succeq_{Q^x \sqcup R^x} r_x^1 \quad \text{for every } x \text{ with } m(x) > 0,$$

then

$$q^0 \succeq_{(P_1 \triangleright \{Q^x\}) \sqcup (P_1 \triangleright \{R^x\})} q^1.$$

If at least one reached branch comparison is strict, then the aggregate comparison is strict.

Sequential observation is evaluated branch by branch. After a first-stage signal, if one continuation leaves at least as much trace as another at every posterior that can be reached, then replacing the continuation branch by branch should not reduce overall traceability. If the improvement is strict on a reached branch, the aggregate comparison is strict.

Remark 31. Condition (P5) plays in this system the role the sure-thing principle plays in Savage's: it is a dominance condition across the branches of a resolving observation, with zero-probability branches unconstrained exactly as Savage's null events are. It is also where all cardinal content resides. The other conditions are ordinal, environment-by-environment conditions; it is (P5), through its derived consequences—public-coin independence (Lemma 39), background inertness (Lemma 46), and the chain rule (Lemma 51)—that aligns the initially separate scales across priors and forces branch-additive aggregation. The strictness clause is not decorative: it is what allows the converse directions of Lemmas 39 and 46 to be extracted from a dominance condition. For example, a mutual-information index capped at a fixed level satisfies (P1)–(P4) but fails exactly (P5). Compound observation environments are accordingly the natural locus of experimental testing: they are where branch-additive and non-additive criteria come apart.

C.1 Auxiliary result

Throughout, logarithms are base two, so information is measured in bits; changing the base merely rescales all information values. For a finite distribution x , write

$$H(x) := - \sum_z x(z) \log x(z), \quad 0 \log 0 := 0,$$

and, for a prior–channel pair (q, P) , define

$$I_{q,P}(A; X) := H(m_{q,P}) - \sum_{a \in A} q(a) H(P(\cdot | a)).$$

When the alphabets are clear, write $I_{q,P}$; write I for the map $(q, P) \mapsto I_{q,P}$.

Proposition 32 (Pure-trace characterisation). *For a family $\{\succeq_P\}_P$ as described above, the following are equivalent.*

- (i) *The family satisfies Conditions (P1)–(P5).*
- (ii) *For every finite channel $P : A \rightarrow \Delta(X)$ and every $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$, $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff I_{q_i, P_i}(A_i; X_i) \geq I_{q_j, P_j}(A_j; X_j).$$

The proof of the equivalence of (i) and (ii) is in the Appendix; the moreover clause is immediate from (ii), since a block-supported lottery has constant block label. The sufficiency argument is the hard part and has three movements. First, *quotient*: block coherence, record post-processing, and a prior-specific Blackwell argument show that at each full-support prior only the induced law of posterior beliefs about the act matters, and that this quotient inherits closedness from Condition (P2) (Lemmas 36–40). Second, *cardinalisation*: public-coin independence—derived from Condition (P5), not assumed—delivers a continuous affine representative on each posterior-law fibre by the mixture-space theorem; derived background inertness of independent components yields a quasi-additive product form; and a tangent-space argument makes branch coefficients path-independent, giving a cocycle, a chain rule, and coherent scales across support faces (Lemmas 41–52). Third, *identification*: the key step is scale alignment. The conditions derive it through data processing and branchwise continuation, rather than through an assumption on product environments. The initially fibrewise affine indices are thereby placed on one cardinal scale; a two-grouping argument eliminates the possible product interaction, and Faddeev’s recursion then identifies the value of full revelation with Shannon entropy, hence the index with mutual information (Lemmas 53–55).

To state uniqueness, it is useful to make the common scale explicit. A real-valued index V on all finite prior–channel pairs is a *common-scale representation* if, for every channel $P : A \rightarrow \Delta(X)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \iff V(q, P) \geq V(q', P),$$

and, for every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff V(q_i, P_i) \geq V(q_j, P_j).$$

Call a common-scale representation *branch-additive* if, for every finite A , every $q \in \Delta(A)$, every first-stage channel $P_1 : A \rightarrow \Delta(X_1)$, and every continuation profile $\{Q^x : A \rightarrow \Delta(X_2^x)\}_{x \in X_1}$,

$$V(q, P_1 \triangleright \{Q^x\}) = V(q, P_1) + \sum_{x: m(x) > 0} m(x) V(r_x, Q^x),$$

where $m(x) = m_{q, P_1}(x)$ and $r_x = r_x^{q, P_1}$ on every reached branch.

Proposition 33 (Uniqueness). *Let $\{\succeq_P\}_P$ satisfy the equivalent conditions of Proposition 32. For any real-valued index V on all finite prior–channel pairs:*

- (i) *V is a common-scale representation if and only if $V = \varphi \circ I$ for a single strictly increasing $\varphi : [0, \infty) \rightarrow \mathbb{R}$.*
- (ii) *A common-scale representation V is branch-additive if and only if $V = aI$ for some $a > 0$.*

The primitive remains ordinal: part (i) permits any common strictly increasing transformation. Once block-supported procedures are required to be scored by their component-pair values, however, the block comparisons permit only one transformation across environments. Branch additivity is not another behavioural condition; it is an exact accounting requirement on a numerical index. Part (ii) shows that this requirement selects the linear gauge, just as mixture affinity selects the expected-utility index within its ordinal class. It fixes both the zero and the curvature of the information scale, leaving only a positive choice of unit, with bits and nats the standard base-two and base- e conventions.

This normalization is what makes the additive combination in the main model meaningful. If the trace component were represented by a nonlinear transformation $\varphi \circ I$, a coefficient multiplying it would have no stable interpretation. Proposition 33 shows that the branch-additive gauge is $V = aI$; the factor a is absorbed into the main coefficient λ . Once the payoff-utility scale and

the information unit are fixed, one effective trade-off parameter can therefore be carried across environments. The proof is at the end of this appendix.

Uniqueness concerns the scale; a complementary observation concerns the sign. The condition system splits into a structural part and an oriented part. Conditions (P2) and (P3) are equivalences or closedness conditions and are invariant under reversal of every preference in the family; Condition (P5) is self-dual, since branchwise monotonicity for the reversed family is the original condition with the two continuation profiles exchanged, and strictness survives the exchange. Only the orientation clause of Condition (P1) and the data-processing condition (P4) carry a sign.

Call an *orientation* of the system a choice, for each of the two signed clauses—the non-triviality clause of (P1) and Condition (P4)—of either its original or its reversed form, or the replacement of both by total indifference on the comparisons of the non-triviality clause.

Corollary 34 (Sign trichotomy). *Let $\{\succeq_P\}_P$ satisfy the weak-order clause of (P1) and Conditions (P2), (P3), (P5). Then:*

- (i) *under the original orientation of both signed clauses, the family is represented by $I_{q,P}$ (Proposition 32);*
- (ii) *under the reversal of both signed clauses, the family is represented by $-I_{q,P}$;*
- (iii) *the two mixed orientations are inconsistent: no family satisfies the original (P4) together with reversed non-triviality, or reversed (P4) together with original non-triviality.*

The totally indifferent family satisfies the structural conditions with both signed clauses replaced by indifference. Hence the structural conditions admit exactly three coherent orientations, represented by I , $-I$, and 0 .

Proof. For (ii), the reversed family $\{\preceq_P\}_P$ satisfies (P1)–(P5)—the structural conditions are reversal-invariant and (P5) is self-dual as noted above—so Proposition 32 represents it by $I_{q,P}$, hence the original family by $-I_{q,P}$. For (iii), fix a full-support q on a non-singleton A and apply (P4) to $P = \text{Id}_A$, $S = \text{Id}_A$, and a constant record kernel $T : A \rightarrow \Delta(\{*\})$. Then $qS = q$ and the processed channel is U_A , so the original orientation gives

$$q^0 \succeq_{\text{Id}_A \sqcup U_A} q^1,$$

contradicting reversed non-triviality. The reversed orientation of (P4) gives the opposite weak comparison, contradicting original non-triviality. The final claim is the necessity verification in the Appendix for (i), its reversal for (ii), and direct verification for the indifferent family. $\square$

Proposition 35 (Independence of the pure-record conditions). *None of Conditions (P1)–(P5) is implied by the other four: for each condition there is a family of weak orders that satisfies the remaining four and violates it.*

The five families are constructed in the Appendix. Two of them carry behavioural content of their own and mark the substantive frontier of the system: the Shannon-entropy-scaled criterion $(1 + H(q))I_{q,P}$ satisfies every condition except data processing, valuing the uncertainty of a procedure beyond its trace, and the satiation criterion $\min\{I_{q,P}, c\}$, for any fixed $c > 0$, satisfies every condition except branchwise continuation monotonicity, ceasing to value revelation once c bits are secured.

Taken together, the results measure the freedom the conditions leave, and it amounts to a sign and a unit. Behaviour selects the sign: a family satisfying the structural conditions together with an orientation of the two signed clauses is represented by I , by $-I$, or by total indifference (Corollary 34), and none of the five conditions securing this rigidity is redundant (Proposition 35). Convention selects the unit: within an orientation, the branch-additive representations are exactly

the positive multiples of I (Proposition 33). In this sense the formula is less the content of Proposition 32 than the trichotomy is: the conditions leave open not which functional measures traceable agency, but whether the agent seeks it, shuns it, or is indifferent to it. The two oriented arms are not symmetric in behavioural content, however; the Discussion takes up what the negative orientation does and does not deliver.

Embedding into the payoff–record model. Fix any $o_0 \in O$ and identify an auxiliary record channel $P : A \rightarrow \Delta(X)$ with the joint channel $K_P(o, x \mid a) = \mathbf{1}_{\{o=o_0\}}P(x \mid a)$. Main Axiom (A4) is (DP-rec), main Axiom (A5) is (DP-act), and main Axiom (A6) is (P5) because every payoff is constant. Thus Proposition 32 applies directly to the restriction used in Lemma 24.

C.2 Proof of the pure-trace characterization

C.2.1 Necessity

For every finite channel $P : A \rightarrow \Delta(X)$, define

$$q \succeq_P^I q' \iff I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

We verify that the family $\{\succeq_P^I\}_P$ satisfies Conditions (P1)–(P5): this is the implication (ii) $\Rightarrow$ (i) of Proposition 32, and it establishes the consistency of the condition system.

The *posterior law* induced by (q, P) is the finite distribution

$$\mu_{q,P} := \sum_{x: m_{q,P}(x) > 0} m_{q,P}(x) \delta_{r_x^{q,P}}$$

over posterior beliefs. Bayes plausibility gives $\int r d\mu_{q,P}(r) = q$. We repeatedly use the elementary identities

$$I_{q,P} = H(m_{q,P}) - \sum_{a \in A} q(a) H(P(\cdot \mid a)), \quad (\text{N1})$$

and

$$I_{q,P} = H(q) - \int_{\Delta(A)} H(r) d\mu_{q,P}(r). \quad (\text{N2})$$

Both formulae are read on the support of the relevant distributions; zero-probability rows and records do not affect the value.

Condition (P1). For each fixed P , the relation $\succeq_P^I$ is complete and transitive because it is represented by the real-valued function $q \mapsto I_{q,P}$. If $|A| \geq 2$ and q has full support, then full revelation gives

$$I_{q, \text{Id}_A} = H(q) > 0,$$

whereas the one-record uninformative channel U_A gives $I_{q, U_A} = 0$. Hence

$$q^0 \succ_{\text{Id}_A \sqcup U_A}^I q^1.$$

Condition (P2). For fixed finite alphabets, (N1) shows that $(P, q) \mapsto I_{q,P}$ is continuous, since Shannon entropy is continuous on every finite simplex. Therefore the set

$$\{(P, q, q') : q \succeq_P^I q'\} = \{(P, q, q') : I_{q,P} \geq I_{q',P}\}$$

is closed. Posterior-law continuity of block comparisons is not an condition; for the mutual-information family it also holds directly, by (N2) and continuity of H on the compact simplex $\Delta(A)$, consistently with its derivation from the conditions in Lemma 40 below.

Condition (P3). For every channel P and lotteries $q, q' \in \Delta(A)$,

$$I_{q^0, P \sqcup P} = I_{q,P}, \quad I_{(q')^1, P \sqcup P} = I_{q',P},$$

because a block-supported lottery puts probability one on its own block and the block label is constant under that lottery. Hence

$$q \succeq_P^I q' \iff q^0 \succeq_{P \sqcup P}^I (q')^1.$$

The same observation proves invariance to adding or deleting irrelevant blocks. If q_i^i and q_j^j are supported on blocks i and j , their mutual-information values in $\bigsqcup_{k \in K} P_k$ are exactly

$$I_{q_i, P_i} \quad \text{and} \quad I_{q_j, P_j},$$

independently of all other blocks.

Condition (P4). It is useful to verify the stronger simultaneous inequality from Remark 30, which immediately supplies both named clauses. Let $S : A \rightarrow \Delta(A')$ and $T : X \rightarrow \Delta(X')$ be stochastic, and let P' satisfy (DP). Generate $A \sim q$; then, conditionally independently given A , generate A' from $S(\cdot | A)$ and X from $P(\cdot | A)$; finally generate X' from $T(\cdot | X)$. The joint law of (A, A', X, X') is

$$q(a)S(a' | a)P(x | a)T(x' | x),$$

so A' is conditionally independent of (X, X') given A , and X' is conditionally independent of (A, A') given X . Hence

$$A' \longleftarrow A \longrightarrow X \longrightarrow X'$$

is a Markov chain in the sense that $A' \rightarrow A \rightarrow X \rightarrow X'$ satisfies the required conditional independences. Data processing applied to the fork $A' \leftarrow A \rightarrow X$ gives $I(A; X) \geq I(A'; X)$, and data processing applied to $A' \rightarrow X \rightarrow X'$ gives $I(A'; X) \geq I(A'; X')$. Therefore

$$I_{q, P}(A; X) \geq I(A'; X').$$

By construction the joint law of (A', X') is exactly $(qS)(a')P'(x' | a')$, the left-hand side of (DP), so $I(A'; X') = I_{qS, P'}(A'; X')$. Rows a' with $(qS)(a') = 0$ carry no mass under qS , so this value is the same for every P' satisfying (DP), however the null rows are completed. Using the block computation of (P3) above to evaluate the two blocks separately,

$$q^0 \succeq_{P \sqcup P'}^I (qS)^1.$$

The first named clause is the case $S = \text{Id}_A$ and gives $I_{q, P}(A; X) \geq I_{q, PT}(A; X')$. For later use, in the action-only case $T = \text{Id}_X$ write $S^q P$ for the conditional channel from the reported action to the record on every reached row:

$$S^q P(x | a') := \frac{\sum_{a \in A} q(a)S(a' | a)P(x | a)}{(qS)(a')} \quad \text{when } (qS)(a') > 0.$$

A *completion* of $S^q P$ is any channel $\hat{P} : A' \rightarrow \Delta(X)$ that agrees with this formula on reached rows; its values on null rows are unrestricted. The second named clause gives $I_{q, P}(A; X) \geq I_{qS, \hat{P}}(A'; X)$ for every completion $\hat{P}$ of $S^q P$.

Condition (P5). Fix a first-stage channel $P_1 : A \rightarrow \Delta(X_1)$. Let

$$m(x) = m_{q, P_1}(x), \quad r_x = r_x^{q, P_1}.$$

For a continuation profile $\mathcal{Q} = \{Q^x\}_x$, the chain rule for mutual information gives

$$I_{q, P_1 \triangleright \mathcal{Q}}(A; X_1, X_2) = I_{q, P_1}(A; X_1) + \sum_{x: m(x) > 0} m(x) I_{r_x, Q^x}(A; X_2^x). \quad (\text{N3})$$

Therefore, if

$$r_x^0 \succeq_{Q^x \sqcup R^x}^I r_x^1 \quad \text{for every positive-probability branch } x,$$

then $I_{r_x, Q^x} \geq I_{r_x, R^x}$ for every such x . Using (N3),

$$I_{q, P_1 \triangleright \{Q^x\}} \geq I_{q, P_1 \triangleright \{R^x\}},$$

so

$$q^0 \preceq_{(P_1 \triangleright \{Q^x\}) \sqcup (P_1 \triangleright \{R^x\})}^I q^1.$$

If one branch comparison is strict and that branch has positive probability, then the corresponding summand in (N3) is strictly larger, so the aggregate comparison is strict.

No product condition is imposed, so the verification ends here. For later use, write $q_1 \otimes q_2$ for the product lottery and define the product of channels by

$$(P_1 \otimes P_2)(x_1, x_2 \mid a_1, a_2) := P_1(x_1 \mid a_1)P_2(x_2 \mid a_2).$$

The mutual-information family satisfies the background-inertness property derived in Lemma 46 below, since

$$I_{q_1 \otimes q_2, P_1 \otimes R_2}(A_1, A_2; X_1, X_2) = I_{q_1, P_1}(A_1; X_1) + I_{q_2, R_2}(A_2; X_2), \quad (\text{N4})$$

so that the comparison of $P_1 \otimes R_2$ and $Q_1 \otimes R_2$ at $q_1 \otimes q_2$ reduces to the comparison of P_1 and Q_1 at q_1 , whatever the common background R_2 . Consistently with Lemma 46, this is a consequence of the chain rule (N3) and not an independent requirement.

C.2.2 Sufficiency

Assume Conditions (P1)–(P5). We prove the implication (i) $\Rightarrow$ (ii) in Proposition 32. The proof is written first for full-support lotteries. Boundary cases are obtained by exact restriction to the positive support and transport back to the original action set. The key point is that posterior-law sufficiency is not assumed; it is derived from block comparisons and record post-processing, and so is its continuity (Lemma 40).

How to read the proof. The argument has five interfaces. Each stage ends with exactly the object needed by the next one:

- Stage 1: replace channels by Bayes-plausible posterior laws and derive the needed quotient continuity from primitive closedness;
- Stage 2: cardinalise each fixed-prior quotient and make support, neutrality, and relabelling behaviour explicit;
- Stage 3: transport those affine values through independent products and into one cross-prior block scale, allowing provisionally a bilinear interaction;
- Stage 4: identify reached-branch coefficients, prove their cocycle, and align the scales on nested support faces;
- Stage 5: compare product and sequential decompositions, eliminate the interaction, and invoke Faddeev’s recursion to obtain Shannon entropy and hence mutual information.

The long posterior-grid, product-coefficient, and branch tangent-space calculations are technical subarguments. A reader interested in the logical spine may use their lemma statements as interfaces. The kernel proof chooses its canonical representative before the branch and product calculations and implements the branch interfaces using finite atomic signed posterior laws. The paper keeps product calibration before branch calibration because this makes the scalar algebra easier to read. The early full-revelation normalisation and exact relabelling lemma below supply the coherence needed by that presentation; the kernel proof reaches the same later interfaces in the opposite order.

Stage 1: posterior-law quotient and derived continuity. The first five lemmas establish a coherent order on posterior laws. In particular, posterior-law sufficiency and continuity are conclusions, not extra axioms; the barycentric spread–merge construction in Lemma 40 is the technical core of this stage.

Lemma 36 (Coherent block comparisons). *Assume Conditions (P1) and (P3).*

(a) *Let (q_1, P_1) , (q_2, P_2) , (q_3, P_3) be pairs of a lottery $q_k \in \Delta(A_k)$ and a channel $P_k : A_k \rightarrow \Delta(X_k)$, with action sets, record sets, and lotteries not necessarily equal and the lotteries not necessarily of full support. Then the two-block comparisons between such pairs are complete,*

$$q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1 \quad \text{or} \quad q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

satisfy the reverse-block equivalence

$$q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0 \iff q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

and are transitive:

$$q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1 \quad \text{and} \quad q_2^0 \succeq_{P_2 \sqcup P_3} q_3^1 \implies q_1^0 \succeq_{P_1 \sqcup P_3} q_3^1.$$

In particular, for a fixed full-support $q \in \Delta(A)$ the comparisons $q^0 \succeq_{P \sqcup Q} q^1$ define a weak order over channels on A at prior q , and pairwise comparisons between block-supported lotteries are independent of the block environment in which the two blocks are embedded.

(b) *Under (P2), the comparisons in (a) are also closed in the channels at fixed alphabets and fixed lotteries: if $P_n \rightarrow P$ in $\Delta(X_1)^{A_1}$, $Q_n \rightarrow Q$ in $\Delta(X_2)^{A_2}$, $q_1 \in \Delta(A_1)$ and $q_2 \in \Delta(A_2)$ are fixed, and*

$$q_1^0 \succeq_{P_n \sqcup Q_n} q_2^1 \quad \text{for all } n,$$

then

$$q_1^0 \succeq_{P \sqcup Q} q_2^1.$$

Fixedness of the alphabets is inherited from Condition (P2), which is stated at a fixed pair of alphabets.

Proof. (a) Completeness follows from Condition (P1) applied to the two-block environment $P_1 \sqcup P_2$: either $q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1$ or $q_2^0 \succeq_{P_1 \sqcup P_2} q_1^1$. In the second case, applying Condition (P3) to the two-block environment with blocks P_1 and P_2 , using the ordered pair $(i, j) = (1, 0)$, gives the canonical reverse comparison

$$q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0 \iff q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

which is the reverse-block equivalence.

For transitivity, form the three-block environment $\bigsqcup_{k=1}^3 P_k$. By the irrelevant-blocks clause of Condition (P3) with block pair $(1, 2)$, the first hypothesis is equivalent to

$$q_1^1 \succeq_{\bigsqcup_{k=1}^3 P_k} q_2^2,$$

and by the same clause with block pair $(2, 3)$, the second hypothesis is equivalent to

$$q_2^2 \succeq_{\bigsqcup_{k=1}^3 P_k} q_3^3.$$

Transitivity of the weak order given by Condition (P1) in this common environment gives

$$q_1^1 \succeq_{\bigsqcup_{k=1}^3 P_k} q_3^3,$$

and applying Condition (P3) again, now deleting the irrelevant middle block, yields $q_1^0 \succeq_{P_1 \sqcup P_3} q_3^1$. Nothing in these arguments requires the pairs to share alphabets or lotteries, or the lotteries to

have full support. The specialisation to a fixed full-support q and channels on a common A is immediate.

(b) The block channel $P_n \sqcup Q_n$ has fixed tagged copies of the alphabets A_1, A_2, X_1, X_2 and converges to $P \sqcup Q$ in the finite-dimensional product simplex. Since the block-supported lotteries q_1^0, q_2^1 are fixed, Condition (P2) applied at this pair of alphabets gives

$$q_1^0 \succeq_{P \sqcup Q} q_2^1. \quad \square$$

Lemma 37 (Prior-specific Blackwell equivalence). *Let $q \in \Delta(A)$ have full support. Let $P : A \rightarrow \Delta(X)$ and $Q : A \rightarrow \Delta(Y)$ be finite channels. If*

$$\mu_{q,P} = \mu_{q,Q},$$

then there are stochastic kernels $T : X \rightarrow \Delta(Y)$ and $T' : Y \rightarrow \Delta(X)$ such that

$$Q = PT, \quad P = QT'.$$

Proof. It is enough to construct T ; the construction of T' is symmetric. Let

$$m_P(x) := m_{q,P}(x), \quad m_Q(y) := m_{q,Q}(y).$$

If $m_P(x) = 0$, then $P(x | a) = 0$ for every $a \in A$, because q has full support. Such records do not affect PT , so define the corresponding row of T arbitrarily. The same observation applies to null records of Q .

Let $\mathcal{R}$ be the finite set of posterior vectors that occur with positive probability under either channel. For $r \in \mathcal{R}$, define

$$X_r := \{x \in X : m_P(x) > 0, r_x^{q,P} = r\}, \quad Y_r := \{y \in Y : m_Q(y) > 0, r_y^{q,Q} = r\}.$$

The sets $\{X_r\}_{r \in \mathcal{R}}$ partition the positive-probability records of P , and the sets $\{Y_r\}_{r \in \mathcal{R}}$ partition the positive-probability records of Q , by posterior vector. Equality of posterior laws implies that the total probability attached to posterior r is the same for the two channels:

$$\lambda_r := \sum_{x \in X_r} m_P(x) = \sum_{y \in Y_r} m_Q(y) > 0.$$

For $x \in X_r$ and $y \in Y_r$, set

$$T(y | x) := \frac{m_Q(y)}{\lambda_r},$$

and set $T(y | x) = 0$ when $x \in X_r$ and $y \notin Y_r$. This defines a stochastic row for every positive-probability record x , since the probabilities over Y_r sum to one. Rows corresponding to null records of P were already chosen arbitrarily.

Fix $a \in A$ and $y \in Y_r$. Since all records in X_r have posterior r , Bayes' rule gives

$$q(a)P(x | a) = m_P(x)r(a) \quad (x \in X_r).$$

Therefore

$$\sum_{x \in X_r} P(x | a) = \frac{\lambda_r r(a)}{q(a)}.$$

Using the definition of T ,

$$(PT)(y | a) = \sum_{x \in X_r} P(x | a) \frac{m_Q(y)}{\lambda_r} = \frac{m_Q(y)r(a)}{q(a)}.$$

But $y \in Y_r$ also has posterior r , so Bayes' rule for Q gives

$$Q(y | a) = \frac{m_Q(y)r(a)}{q(a)}.$$

Thus $(PT)(y | a) = Q(y | a)$ for every positive-probability y . If $m_Q(y) = 0$, then full support of q implies $Q(y | a) = 0$ for every a . No positive-probability record of P sends mass to such a y under the construction; rows of T indexed by null records of P may be chosen arbitrarily, but those records have $P(x | a) = 0$ for every a , so they contribute nothing to PT . Hence $(PT)(y | a) = 0 = Q(y | a)$ for every a , and therefore $Q = PT$. $\square$

Lemma 38 (Posterior-law sufficiency). *Fix a full-support $q \in \Delta(A)$. If $P : A \rightarrow \Delta(X)$ and $Q : A \rightarrow \Delta(Y)$ generate the same posterior law at q ,*

$$\mu_{q,P} = \mu_{q,Q},$$

then

$$q^0 \sim_{P \sqcup Q} q^1.$$

Consequently, by Lemma 36, at a fixed full-support prior the block comparison of channels depends only on the induced posterior law.

Proof. By Lemma 37, there are stochastic kernels T and T' such that $Q = PT$ and $P = QT'$. Clause (DP-rec) of Condition (P4) gives

$$q^0 \succeq_{P \sqcup PT} q^1, \quad q^0 \succeq_{Q \sqcup QT'} q^1.$$

Using $PT = Q$, the first comparison is

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Using $QT' = P$, the second comparison is

$$q^0 \succeq_{Q \sqcup P} q^1.$$

By the reverse-block equivalence in Lemma 36 (equivalently, Condition (P3) applied to $P \sqcup Q$ with ordered pair $(i, j) = (1, 0)$), this is equivalent to

$$q^1 \succeq_{P \sqcup Q} q^0.$$

Thus $q^0 \sim_{P \sqcup Q} q^1$. $\square$

For channels $P : A \rightarrow \Delta(X_P)$ and $R : A \rightarrow \Delta(X_R)$ and $\lambda \in [0, 1]$, write $\lambda P \oplus (1 - \lambda)R$ for the channel that publicly selects P with probability λ and R otherwise, and reports the selected channel together with its record. Thus

$$(\lambda P \oplus (1 - \lambda)R)((z, i) | a) = \begin{cases} \lambda P(z | a), & i = 0, \\ (1 - \lambda)R(z | a), & i = 1. \end{cases}$$

Because the coin is observed,

$$\mu_{q, \lambda P \oplus (1 - \lambda)R} = \lambda \mu_{q,P} + (1 - \lambda) \mu_{q,R}.$$

Lemma 39 (Derived public-coin independence). *Assume Conditions (P1) and (P3), clause (DP-rec) of Condition (P4), and Condition (P5). Fix a finite action set A , a full-support $q \in \Delta(A)$, and channels P, Q, R on A . For every $\lambda \in (0, 1)$,*

$$q^0 \succeq_{P \sqcup Q} q^1 \iff q^0 \succeq_{(\lambda P \oplus (1 - \lambda)R) \sqcup (\lambda Q \oplus (1 - \lambda)R)} q^1.$$

Proof. Throughout, for channels V, W on A write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$, and $V \succ_q W$ for the corresponding strict comparison. By Lemma 36 the comparison $\succeq_q$ is complete and transitive, and by Lemma 38 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$. In particular, channels with equal posterior laws at q may be substituted on either side of any comparison; we refer to this as *substitution*. Substitution also preserves strict comparisons: if $\mu_{q,V} = \mu_{q,V'}$ and $\mu_{q,W} = \mu_{q,W'}$, then $V \sim_q V'$ and $W \sim_q W'$ by Lemma 38, and transitivity of the complete relation $\succeq_q$ gives $V \succ_q W$ if and only if $V' \succ_q W'$.

Step 1: the public coin is an uninformative first stage. Define the constant channel $C : A \rightarrow \Delta(\{0, 1\})$ by

$$C(0 | a) := \lambda, \quad C(1 | a) := 1 - \lambda \quad (a \in A).$$

Since $\lambda \in (0, 1)$, both branches have positive mass: $m_{q,C}(0) = \lambda$ and $m_{q,C}(1) = 1 - \lambda$. Because the rows of C are constant, Bayes' rule gives, for each $x \in \{0, 1\}$,

$$r_x^{q,C}(a) = \frac{q(a)C(x | a)}{m_{q,C}(x)} = q(a),$$

so both branch posteriors equal the prior: $r_0 = r_1 = q$.

Step 2: padding the continuations to a common alphabet. Condition (P5) requires the two continuation channels on a given branch to share a record alphabet, whereas P and Q may not. Let $X_{PQ} := X_P \sqcup X_Q$ and define $\tilde{P}, \tilde{Q} : A \rightarrow \Delta(X_{PQ})$ by

$$\tilde{P}(z | a) := \begin{cases} P(z | a) & z \in X_P, \\ 0 & z \in X_Q, \end{cases} \quad \tilde{Q}(z | a) := \begin{cases} 0 & z \in X_P, \\ Q(z | a) & z \in X_Q. \end{cases}$$

Padding adds only zero-probability records, so the positive-mass records and their posteriors are unchanged: $\mu_{q,\tilde{P}} = \mu_{q,P}$ and $\mu_{q,\tilde{Q}} = \mu_{q,Q}$. By substitution,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff \tilde{P} \succeq_q \tilde{Q},$$

and the same equivalence holds for the strict comparisons.

Step 3: the compounds implement the public-coin mixtures. Consider the two-stage channels with first stage C , continuation pair $\tilde{P}$ versus $\tilde{Q}$ on branch 0, and the common continuation R on branch 1:

$$C \triangleright \{\tilde{P}, R\} \quad \text{and} \quad C \triangleright \{\tilde{Q}, R\},$$

where the profile notation lists the branch-0 and branch-1 continuations in order. By the definition of two-stage composition,

$$(C \triangleright \{\tilde{P}, R\})(0, z | a) = \lambda \tilde{P}(z | a), \quad (C \triangleright \{\tilde{P}, R\})(1, z | a) = (1 - \lambda) R(z | a).$$

A record $(0, z)$ has marginal mass $\lambda m_{q,\tilde{P}}(z)$, and when this is positive the constant factor λ cancels in Bayes' rule, so its posterior is $r_z^{q,\tilde{P}}$; symmetrically for records $(1, z)$. Hence

$$\mu_{q,C \triangleright \{\tilde{P}, R\}} = \lambda \mu_{q,\tilde{P}} + (1 - \lambda) \mu_{q,R} = \lambda \mu_{q,P} + (1 - \lambda) \mu_{q,R} = \mu_{q,\lambda P \oplus (1-\lambda)R},$$

using the public-coin mixing identity, and likewise $\mu_{q,C \triangleright \{\tilde{Q}, R\}} = \mu_{q,\lambda Q \oplus (1-\lambda)R}$. By substitution, the right-hand comparison of the lemma is therefore equivalent to

$$C \triangleright \{\tilde{P}, R\} \succeq_q C \triangleright \{\tilde{Q}, R\}. \quad (*)$$

Step 4: forward implication. Assume $q^0 \succeq_{P \sqcup Q} q^1$. We verify the branch hypotheses of Condition (P5) for the first stage C and the continuation profiles $\{\tilde{P}, R\}$ versus $\{\tilde{Q}, R\}$. Both branches have positive mass and both branch posteriors equal q (Step 1). On branch 0 the

hypothesis is $q^0 \succeq_{\tilde{P} \sqcup \tilde{Q}} q^1$, which holds by Step 2. On branch 1 the two profiles share the continuation R , and the hypothesis $q^0 \succeq_{R \sqcup R} q^1$ holds because Condition (P1) gives $q \succeq_R q$ and the duplication clause of Condition (P3) converts it into the two-block form. Condition (P5) then yields $(*)$, hence the mixture comparison by Step 3.

Step 5: converse, via the strictness clause. Assume the mixture comparison holds, so that $(*)$ holds by Step 3, and suppose towards a contradiction that $q^0 \succeq_{P \sqcup Q} q^1$ fails. By completeness and the reverse-block equivalence of Lemma 36, $Q \succ_q P$, and Step 2 gives $\tilde{Q} \succ_q \tilde{P}$. Now apply Condition (P5) with the roles of the two profiles exchanged: the branch-0 comparison $\tilde{Q} \succ_q \tilde{P}$ is strict on a branch of mass $\lambda > 0$, and the branch-1 comparison $R \succeq_q R$ holds as in Step 4. The strictness clause of Condition (P5) yields

$$C \triangleright \{\tilde{Q}, R\} \succ_q C \triangleright \{\tilde{P}, R\},$$

which contradicts $(*)$. Hence $q^0 \succeq_{P \sqcup Q} q^1$. $\square$

The lemma shows that public-coin independence is not an independent behavioural commitment: a public coin is exactly a first-stage observation that reveals nothing about the act, so invariance under common public mixtures is branchwise continuation monotonicity applied through trivial branch posteriors. The statement is given at full-support priors, which is the only case used below (Lemma 41).

For a finite action set A , let $\mathbf{X}_A$ be the set of finite-support probability measures on $\Delta(A)$. Write

$$b(\mu) := \int_{\Delta(A)} r \, d\mu(r), \quad \mathcal{M}_q := \{\mu \in \mathbf{X}_A : b(\mu) = q\}.$$

Thus $\mathcal{M}_q$ is the set of finite posterior laws with prior q as their mean, and $\mu_{q,P} \in \mathcal{M}_q$ for every channel P .

Lemma 40 (Posterior-law continuity). *Fix a full-support $q \in \Delta(A)$. Let P_n, Q_n, P, Q be finite channels on A with*

$$\mu_{q,P_n} \Rightarrow \mu_{q,P}, \quad \mu_{q,Q_n} \Rightarrow \mu_{q,Q}.$$

If

$$q^0 \succeq_{P_n \sqcup Q_n} q^1 \quad \text{for all } n,$$

then

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Proof. Throughout the proof, for channels V, W on A write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$. By Lemma 36 this comparison is complete and transitive, and by Lemma 38 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$; in particular, channels with equal posterior laws at q may be substituted on either side of any comparison. We refer to this as *substitution*.

For a finite-support Bayes-plausible law

$$\mu = \sum_{k \in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

with atoms listed so that $\lambda_k > 0$, define the canonical implementing channel $C_\mu : A \rightarrow \Delta(K)$ by

$$C_\mu(k \mid a) := \frac{\lambda_k r_k(a)}{q(a)}.$$

Full support of q and $\sum_k \lambda_k r_k = q$ make C_μ a channel with $\mu_{q,C_\mu} = \mu$.

Step 1: merging posteriors moves weakly down. Let $\mu = \sum_{k \in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q$ and let $\{K_1, \dots, K_m\}$ partition K into nonempty groups. The merged law is

$$\text{mrg}(\mu) := \sum_{j=1}^m w_j \delta_{s_j}, \quad w_j := \sum_{k \in K_j} \lambda_k, \quad s_j := \frac{1}{w_j} \sum_{k \in K_j} \lambda_k r_k.$$

Barycentres are preserved, so $\text{mrg}(\mu) \in \mathcal{M}_q$. Let $T : K \rightarrow \Delta(\{1, \dots, m\})$ be the deterministic kernel with $T(j \mid k) = 1$ if and only if $k \in K_j$. Then

$$(C_\mu T)(j \mid a) = \sum_{k \in K_j} \frac{\lambda_k r_k(a)}{q(a)} = \frac{w_j s_j(a)}{q(a)} = C_{\text{mrg}(\mu)}(j \mid a),$$

so clause (DP-rec) of Condition (P4) gives

$$C_\mu \succeq_q C_{\text{mrg}(\mu)}.$$

Conversely, say that $\mu' \in \mathcal{M}_q$ is a *spread* of μ if μ' is obtained by replacing each atom (λ_k, r_k) of μ by finitely many atoms $(\lambda_k w_{kj}, v_{kj})_j$ with $w_{kj} \geq 0$, $\sum_j w_{kj} = 1$, and $\sum_j w_{kj} v_{kj} = r_k$. Then $\mu = \text{mrg}(\mu')$ for the grouping by k , so

$$C_{\mu'} \succeq_q C_\mu.$$

Atoms with $w_{kj} = 0$ are deleted from the list; and if the same posterior appears in several list entries, any two listings of the same measure induce the same posterior law at q , so the corresponding canonical channels are interchangeable by substitution. Merging records can only obscure which act was realised; splitting an atom into a barycentric family is the reverse operation.

Step 2: bounded supports. We claim the lemma holds under the additional hypothesis that all laws μ_{q, P_n} and μ_{q, Q_n} have support of size at most K , for some fixed K . Write

$$\mu_{q, P_n} = \sum_{k=1}^K \lambda_{n,k} \delta_{r_{n,k}},$$

padding with zero-mass atoms ($\lambda_{n,k} = 0$, $r_{n,k} := q$) so that exactly K terms appear, and define the K -record channel $C_n : A \rightarrow \Delta(\{1, \dots, K\})$ by

$$C_n(k \mid a) := \frac{\lambda_{n,k} r_{n,k}(a)}{q(a)}.$$

Each row sums to one because $\sum_k \lambda_{n,k} r_{n,k} = q$, zero-mass columns contribute nothing, and $\mu_{q, C_n} = \mu_{q, P_n}$. Define D_n from μ_{q, Q_n} in the same way. By substitution,

$$C_n \succeq_q D_n \quad \text{for all } n.$$

The parameter sequences of C_n and D_n jointly live in the compact metric space $([0, 1] \times \Delta(A))^{2K}$; pass to a single subsequence along which $\lambda_{n,k} \rightarrow \lambda_k^*$ and $r_{n,k} \rightarrow r_k^*$ for every k , simultaneously for both channels. Then $C_n \rightarrow C^*$ entrywise, where $C^*(k \mid a) = \lambda_k^* r_k^*(a)/q(a)$; each row of C^* sums to one as an entrywise limit of rows summing to one, so C^* is a channel. For every continuous f on $\Delta(A)$,

$$\int f d\mu_{q, P_n} = \sum_{k=1}^K \lambda_{n,k} f(r_{n,k}) \longrightarrow \sum_{k=1}^K \lambda_k^* f(r_k^*),$$

so by uniqueness of weak limits

$$\mu_{q, P} = \sum_{k=1}^K \lambda_k^* \delta_{r_k^*} = \mu_{q, C^*},$$

where the second equality holds because record k of C^* has probability λ_k^* and, when $\lambda_k^* > 0$, posterior r_k^* . Similarly $D_n \rightarrow D^*$ entrywise with $\mu_{q,D^*} = \mu_{q,Q}$. The closedness clause, Lemma 36(b), applied along the subsequence gives

$$C^* \succeq_q D^*,$$

and substitution returns $q^0 \succeq_{P \sqcup Q} q^1$. This proves the bounded-support case; note that only the primitive closedness of Condition (P2) has been used, through Lemma 36.

Step 3: the general case, by an ε -grid sandwich. Write $\mu_n := \mu_{q,P_n}$, $\nu_n := \mu_{q,Q_n}$, $\mu := \mu_{q,P}$, $\nu := \mu_{q,Q}$. The only remaining gap is that the supports of μ_n, ν_n need not be bounded. Fix $\varepsilon > 0$. Two finite partitions of $\Delta(A)$ are used, one for each surrogate; they play independent roles and need not be related. For the spread, fix a finite triangulation $\mathcal{T}$ of $\Delta(A)$ into simplices of diameter at most ε , together with a measurable partition $\{\tilde{S} : S \in \mathcal{T}\}$ of $\Delta(A)$ with $\tilde{S} \subseteq S$, assigning each shared face to exactly one incident cell. For the merge, fix a finite Borel partition $\{E_1, \dots, E_M\}$ of $\Delta(A)$ into sets of diameter at most ε whose relative boundaries carry no ν -mass: cover $\Delta(A)$ by finitely many open balls of radius at most $\varepsilon/2$, choosing the radii so that no atom of ν lies on a bounding sphere (ν has finitely many atoms, so all but finitely many radii work), and disjointify.

For finite-support $\rho \in \mathcal{M}_q$ define two bounded-support surrogates. Both remain in $\mathcal{M}_q$: the spread operation replaces each posterior by a barycentric decomposition of itself, while the merge operation replaces a group of posteriors by their conditional mean, weighted by the total mass of the group. Cells with zero mass are ignored (or, equivalently, padded by an arbitrary posterior with zero weight when a fixed record alphabet is convenient). The *grid spread* $\text{sp}_\varepsilon(\rho)$ replaces each atom (λ, β) with $\beta \in \tilde{S}$ by the atoms $(\lambda w_v(\beta), v)_v$, where v ranges over the vertices of S and $(w_v(\beta))_v$ are the barycentric coordinates of β in S . It is a spread of ρ in the sense of Step 1, is supported on the fixed vertex set of $\mathcal{T}$, and satisfies $C_{\text{sp}_\varepsilon(\rho)} \succeq_q C_\rho$. The *cell merge* $\text{mg}_\varepsilon(\rho)$ merges the atoms of ρ within each class E_j of the Borel partition; it has at most one atom per class and satisfies $C_\rho \succeq_q C_{\text{mg}_\varepsilon(\rho)}$ by Step 1, since Step 1 allows any grouping of atoms.

By substitution and transitivity (Lemma 36), the hypothesis $P_n \succeq_q Q_n$ gives the chain

$$C_{\text{sp}_\varepsilon(\mu_n)} \succeq_q C_{\mu_n} \succeq_q C_{\nu_n} \succeq_q C_{\text{mg}_\varepsilon(\nu_n)}.$$

Both surrogate sequences converge weakly on fixed alphabets. For the spread side, let f be continuous on $\Delta(A)$. The triangulation defines a continuous piecewise-affine interpolant $I_{\mathcal{T}}f$ by

$$I_{\mathcal{T}}f(r) := \sum_{v \in \text{vert}(S)} w_v(r) f(v) \quad \text{for } r \in S.$$

This is well-defined: on a shared face the barycentric coordinates computed in the two incident simplices put zero weight on vertices outside the face and coincide with the intrinsic barycentric coordinates on that face. Hence $I_{\mathcal{T}}f$ is continuous on all of $\Delta(A)$. Moreover, for every finite-support ρ ,

$$\int f d\text{sp}_\varepsilon(\rho) = \int I_{\mathcal{T}}f d\rho,$$

because each posterior is replaced by the barycentric lottery over the vertices of its simplex. Thus $\mu_n \Rightarrow \mu$ gives $\text{sp}_\varepsilon(\mu_n) \Rightarrow \text{sp}_\varepsilon(\mu)$. For the merge side, each class E_j has relative boundary contained in the chosen bounding spheres, which carry no ν -mass, so E_j is a ν -continuity set; therefore $\nu_n(E_j) \rightarrow \nu(E_j)$ and, since the truncated map $r \mapsto r \mathbf{1}_{E_j}(r)$ is bounded and its discontinuity set lies in the relative boundary of E_j , which is ν -null, $\int_{E_j} r d\nu_n \rightarrow \int_{E_j} r d\nu$ for every class. Classes with $\nu(E_j) > 0$ thus have converging merged masses and merged locations. If $\nu(E_j) = 0$, then the merged mass tends to zero, so the chosen location in that class is immaterial for weak convergence; padding such a class by any fixed posterior only changes a zero-weight atom. Hence $\text{mg}_\varepsilon(\nu_n) \Rightarrow \text{mg}_\varepsilon(\nu)$.

The laws $\text{sp}_\varepsilon(\mu_n)$ and $\text{mg}_\varepsilon(\nu_n)$ have supports of size at most $N(\varepsilon)$, the maximum of the number of vertices of $\mathcal{T}$ and the number M of classes of the Borel partition. Step 2 applied to the chain's endpoints therefore yields

$$C_{\text{sp}_\varepsilon(\mu)} \succeq_q C_{\text{mg}_\varepsilon(\nu)}.$$

Finally, send $\varepsilon \rightarrow 0$ along a sequence $\varepsilon_m \downarrow 0$ with triangulations and Borel partitions chosen as above. The law $\text{sp}_{\varepsilon_m}(\mu)$ has at most $k_P |A|$ atoms, where k_P is the number of atoms of μ : each atom splits among the $|A|$ vertices of one cell. Each new atom lies within ε_m of the atom it came from and group masses are preserved, so $\text{sp}_{\varepsilon_m}(\mu) \Rightarrow \mu$. The law $\text{mg}_{\varepsilon_m}(\nu)$ has at most k_Q atoms, where k_Q is the number of atoms of ν : only classes meeting the support of ν carry mass. Once ε_m is smaller than the minimal distance between distinct atoms of ν , each class contains at most one atom of ν and $\text{mg}_{\varepsilon_m}(\nu) = \nu$ exactly. The comparisons $C_{\text{sp}_{\varepsilon_m}(\mu)} \succeq_q C_{\text{mg}_{\varepsilon_m}(\nu)}$ hold for every m , the supports are uniformly bounded, and the laws converge weakly to μ and ν ; one further application of Step 2 gives $C_\mu \succeq_q C_\nu$. Substitution returns

$$q^0 \succeq_{P \sqcup Q} q^1. \quad \square$$

The lemma shows that continuity of block comparisons in posterior-law convergence is not an independent behavioural commitment: it is the primitive closedness of Condition (P2) seen through the quotient that Condition (P3) and clause (DP-rec) of Condition (P4) impose, with the convex order supplying compactness where record alphabets grow without bound. Only weak preference is closed in the limit; no strictness claim is made or used.

Stage 2: affine fibres and canonical boundary behaviour. Public-coin independence and quotient continuity now permit Herstein–Milnor cardinalisation. The next lemmas record the finite integral form, normalize no information to zero, and make action coarsening, exact support restriction, and undetectable distinctions available before values are compared across priors. In the formal development these facts are packaged by a full-revelation-normalized selected representative. The paper now makes that same initial selection explicitly; a later coherent product gauge is declared when it changes the representative.

Lemma 41 (Posterior-separable representation). *Fix a full-support $q \in \Delta(A)$. There is a continuous affine functional*

$$F_q : \mathcal{M}_q \rightarrow \mathbb{R}$$

such that, for any two channels P, Q on A ,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

Moreover, for some continuous $\phi_q : \Delta(A) \rightarrow \mathbb{R}$,

$$F_q(\mu) = \int_{\Delta(A)} \phi_q(r) d\mu(r),$$

and therefore

$$F_q(\mu_{q,P}) = \sum_{x: m_{q,P}(x) > 0} m_{q,P}(x) \phi_q(r_x^{q,P}).$$

The affine functional is unique up to positive affine transformations. The integrand ϕ_q is not pointwise unique; adding an affine function of r leaves its integral on the fibre $\mathcal{M}_q$ changed only by a constant.

Proof. First note that every finite-support Bayes-plausible law is implemented by a finite channel. If

$$\mu = \sum_{k=1}^n \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

define

$$P(k | a) = \frac{\lambda_k r_k(a)}{q(a)}.$$

Since q has full support and $\sum_k \lambda_k r_k = q$, this is a channel. Its record probability at k is λ_k , and the posterior after k is r_k ; hence $\mu_{q,P} = \mu$.

Define a comparison on $\mathcal{M}_q$ as follows: for $\mu, \nu \in \mathcal{M}_q$, choose any channels P, Q implementing them and compare the two block-supported copies of q in $P \sqcup Q$. This comparison is independent of the chosen implementations. Indeed, if P, P' both implement μ and Q, Q' both implement ν , then Lemma 38 gives

$$q^0 \sim_{P \sqcup P'} q^1, \quad q^0 \sim_{Q \sqcup Q'} q^1.$$

Placing the relevant blocks in common block environments and using Lemma 36, transitivity gives

$$q^0 \succeq_{P \sqcup Q} q^1 \iff q^0 \succeq_{P' \sqcup Q'} q^1.$$

Thus the quotient comparison on $\mathcal{M}_q$ is well defined; denote it by $\succeq_q$. This construction is only at the fixed full-support prior q , and every law in $\mathcal{M}_q$ is finite-support and, by the preceding construction, implemented by a finite channel. Lemma 36 also gives completeness and transitivity of this quotient comparison.

The set $\mathcal{M}_q$ is a mixture space under ordinary convex combination. It is convex because barycentres are preserved by convex combination. The quotient relation $\succeq_q$ is a weak order by Lemma 36. Public-coin mixtures of implementing channels correspond exactly to convex mixtures of posterior laws:

$$\mu_{q, \lambda P \oplus (1-\lambda)R} = \lambda \mu_{q,P} + (1-\lambda) \mu_{q,R}.$$

Therefore the derived public-coin independence property (Lemma 39) gives mixture independence: for every $\mu, \nu, \rho \in \mathcal{M}_q$ and $\lambda \in (0, 1)$,

$$\mu \succeq_q \nu \iff \lambda \mu + (1-\lambda) \rho \succeq_q \lambda \nu + (1-\lambda) \rho.$$

The relative weak topology on $\mathcal{M}_q$ is inherited from the weak topology on probability measures over the compact metric simplex $\Delta(A)$, and convex-mixture paths are continuous in this topology. Lemma 40 gives relatively closed upper and lower contour sets. Indeed, if $\mu_n \Rightarrow \mu$ with $\mu_n, \mu, z \in \mathcal{M}_q$ and $\mu_n \succeq_q z$, choose finite implementing channels for these laws and apply Lemma 40 (with the constant sequence implementing z) to get $\mu \succeq_q z$; the argument for $\{\mu : z \succeq_q \mu\}$ is the same. Therefore, for fixed $\mu, \nu, \rho \in \mathcal{M}_q$, the sets

$$\{\lambda \in [0, 1] : \lambda \mu + (1-\lambda) \nu \succeq_q \rho\} \quad \text{and} \quad \{\lambda \in [0, 1] : \rho \succeq_q \lambda \mu + (1-\lambda) \nu\}$$

are closed.

By the Herstein–Milnor mixture-space theorem [30], there is a mixture-preserving representation $F_q : \mathcal{M}_q \rightarrow \mathbb{R}$:

$$\mu \succeq_q \nu \iff F_q(\mu) \geq F_q(\nu),$$

and

$$F_q(\lambda \mu + (1-\lambda) \nu) = \lambda F_q(\mu) + (1-\lambda) F_q(\nu).$$

If the quotient order is nonconstant, this representative is unique up to positive affine transformations. If the quotient order is constant, take $F_q \equiv 0$.

Moreover, F_q is continuous in the relative weak topology. For any $z \in \mathcal{M}_q$, the sets

$$\{\mu : F_q(\mu) \geq F_q(z)\} = \{\mu : \mu \succeq_q z\} \quad \text{and} \quad \{\mu : F_q(\mu) \leq F_q(z)\} = \{\mu : z \succeq_q \mu\}$$

are relatively closed by the preceding Lemma 40 argument. Since F_q is affine, its range on the convex set $\mathcal{M}_q$ is an interval. Hence every upper and lower level set of F_q is relatively closed: levels outside the range are trivial, and levels inside the range equal $F_q(z)$ for some $z \in \mathcal{M}_q$. Thus F_q is both upper and lower semicontinuous, and therefore continuous. For any channels P, Q on A ,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

The no-information posterior law at q is δ_q . Write

$$\chi_q := \sum_{a \in A} q(a) \delta_{e_a}$$

for the full-revelation posterior law, where e_a assigns probability one to action a .

It remains to write the affine functional in posterior-separable form, without leaving the finite-support domain. Let $X := \Delta(A)$. Choose

$$0 < \varepsilon < \min_{a \in A} q(a).$$

For each $r \in X$, define

$$s_r := \frac{q - \varepsilon r}{1 - \varepsilon} \in \Delta(A)$$

and

$$\eta_r := \varepsilon \delta_r + (1 - \varepsilon) \sum_{a \in A} s_r(a) \delta_{e_a}.$$

Then $\eta_r \in \mathcal{M}_q$, because its barycentre is

$$\varepsilon r + (1 - \varepsilon) s_r = q.$$

Choose constants c_a such that

$$\sum_{a \in A} q(a) c_a = F_q(\chi_q),$$

for instance $c_a = F_q(\chi_q)$ for every a . Define

$$\phi_q(r) := \frac{F_q(\eta_r) - (1 - \varepsilon) \sum_{a \in A} s_r(a) c_a}{\varepsilon}.$$

The map $r \mapsto \eta_r$ is weakly continuous and F_q is continuous, so ϕ_q is continuous.

Now take any finite-support law

$$\mu = \sum_{k=1}^n \lambda_k \delta_{r_k} \in \mathcal{M}_q.$$

Since $\sum_k \lambda_k r_k = q$, we have

$$\sum_{k=1}^n \lambda_k s_{r_k} = q.$$

Therefore

$$\sum_{k=1}^n \lambda_k \eta_{r_k} = \varepsilon \mu + (1 - \varepsilon) \chi_q.$$

By affinity of F_q ,

$$\sum_{k=1}^n \lambda_k F_q(\eta_{r_k}) = F_q(\varepsilon\mu + (1-\varepsilon)\chi_q) = \varepsilon F_q(\mu) + (1-\varepsilon)F_q(\chi_q).$$

Using $\sum_k \lambda_k s_{r_k} = q$ and $\sum_a q(a)c_a = F_q(\chi_q)$ gives

$$\sum_{k=1}^n \lambda_k \phi_q(r_k) = F_q(\mu).$$

This is exactly

$$F_q(\mu) = \int_X \phi_q(r) d\mu(r)$$

for every finite-support $\mu \in \mathcal{M}_q$. Applying this to $\mu_{q,P}$ gives the stated finite-sum formula. The affine representative F_q is unique up to positive affine transformations by Herstein–Milnor. The integrand is not pointwise unique on a fixed barycentre fibre: if ℓ is affine in r , then $\int \ell(r) d\mu(r) = \ell(q)$ for every $\mu \in \mathcal{M}_q$. $\square$

Lemma 42 (Refinement monotonicity and canonical normalisation). *Fix a full-support $q \in \Delta(A)$ and let F_q, ϕ_q be as in Lemma 41. If a finite law $\mu \in \mathcal{M}_q$ is obtained from another finite law $\nu \in \mathcal{M}_q$ by replacing each posterior of ν by a finite Bayes-plausible continuation experiment, then*

$$F_q(\mu) \geq F_q(\nu).$$

Consequently, ϕ_q may be chosen convex. After replacing ϕ_q by an observationally equivalent representative and subtracting a constant from F_q , one may impose

$$\phi_q(q) = F_q(\delta_q) = 0, \quad \phi_q(r) \geq 0 \quad \text{for every } r \in \Delta(A),$$

and therefore

$$F_q(\mu) \geq 0 \quad \text{for every } \mu \in \mathcal{M}_q.$$

If $|A| \geq 2$, the representative can and will be selected uniquely by the additional full-revelation normalisation

$$F_q(\chi_q) = 1.$$

If $|A| = 1$, then $\mathcal{M}_q = \{\delta_q\}$ and we select $F_q \equiv 0$.

Proof. Write

$$\nu = \sum_{i=1}^n m_i \delta_{r_i}, \quad \mu = \sum_{i=1}^n m_i \nu_i, \quad \nu_i \in \mathcal{M}_{r_i}.$$

Choose a channel $P : A \rightarrow \Delta(\{1, \dots, n\})$ implementing ν at prior q . For each i , choose a finite continuation channel $Q^i : A \rightarrow \Delta(Z_i)$ implementing ν_i at prior r_i ; on actions outside the support of r_i , define the rows arbitrarily. Those rows are never reached under prior r_i and do not affect ν_i ; moreover, if $r_i(a) = 0$, then $q(a)P(i | a) = m_i r_i(a) = 0$, so they do not affect the compound posterior law at q . The compound channel

$$C := P \triangleright \{Q^i\}_{i=1}^n$$

implements μ at prior q . Let $T : \bigsqcup_{i=1}^n (\{i\} \times Z_i) \rightarrow \Delta(\{1, \dots, n\})$ be the kernel $T(i | (i, z)) = 1$; then $CT = P$. Clause (DP-rec) of Condition (P4) applied to C and T gives

$$q^0 \succeq_{C \sqcup P} q^1.$$

By Lemma 41,

$$F_q(\mu) \geq F_q(\nu).$$

To prove convexity, take $r, s_1, \dots, s_k \in \Delta(A)$ with

$$r = \sum_{j=1}^k \lambda_j s_j, \quad \lambda_j \geq 0, \quad \sum_j \lambda_j = 1.$$

Choose $0 < \varepsilon < \min_a q(a)$ and set

$$t := \frac{q - \varepsilon r}{1 - \varepsilon}.$$

This belongs to $\Delta(A)$: for each a , $q(a) - \varepsilon r(a) \geq q(a) - \varepsilon > 0$, and the coordinates sum to one. The law

$$\varepsilon \sum_{j=1}^k \lambda_j \delta_{s_j} + (1 - \varepsilon) \delta_t$$

is a refinement of

$$\varepsilon \delta_r + (1 - \varepsilon) \delta_t,$$

and both belong to $\mathcal{M}_q$. Refinement monotonicity and the integral representation give

$$\varepsilon \sum_{j=1}^k \lambda_j \phi_q(s_j) + (1 - \varepsilon) \phi_q(t) \geq \varepsilon \phi_q(r) + (1 - \varepsilon) \phi_q(t).$$

Cancelling the common term and dividing by ε yields

$$\phi_q(r) \leq \sum_{j=1}^k \lambda_j \phi_q(s_j),$$

so ϕ_q is convex.

Since q is in the relative interior of the finite-dimensional simplex, the supporting-hyperplane theorem for finite convex functions gives a relative subgradient $g \in \mathbb{R}^A$ of the continuous convex function ϕ_q at q . Define

$$\tilde{\phi}_q(r) := \phi_q(r) - g \cdot (r - q).$$

For every $\mu \in \mathcal{M}_q$,

$$\int g \cdot (r - q) d\mu(r) = g \cdot (b(\mu) - q) = 0,$$

so $\tilde{\phi}_q$ represents the same affine functional on $\mathcal{M}_q$. The subgradient inequality gives

$$\tilde{\phi}_q(r) \geq \tilde{\phi}_q(q) \quad \text{for all } r \in \Delta(A).$$

Finally replace

$$F_q(\mu) \quad \text{by} \quad F_q(\mu) - F_q(\delta_q)$$

and replace $\tilde{\phi}_q$ by

$$\tilde{\phi}_q - \tilde{\phi}_q(q).$$

Before this final constant shift, the representation with $\tilde{\phi}_q$ gives

$$F_q(\delta_q) = \int \tilde{\phi}_q(r) d\delta_q(r) = \tilde{\phi}_q(q).$$

Thus the same constant is subtracted from both sides of the integral representation, and subtracting a constant from F_q preserves all ordinal comparisons on $\mathcal{M}_q$. The final normalisation gives

$$\phi_q(q) = F_q(\delta_q) = 0,$$

and makes ϕ_q non-negative on the whole simplex. Hence

$$F_q(\mu) = \int \phi_q(r) d\mu(r) \geq 0$$

for every $\mu \in \mathcal{M}_q$.

Suppose now that $|A| \geq 2$. Full revelation implements χ_q and no information implements δ_q . The strict local-orientation clause of Condition (P1), transported through the quotient comparison, gives

$$F_q(\chi_q) > F_q(\delta_q) = 0.$$

Divide both F_q and ϕ_q by the positive number $F_q(\chi_q)$. This preserves the integral representation, all comparisons, convexity, and non-negativity, and gives $F_q(\chi_q) = 1$. These two anchor values make the selected affine representative unique. If $|A| = 1$, every posterior law with barycentre q is δ_q , so the zero representative is the only relevant choice. $\square$

For a finitely supported posterior law $\mu = \sum_i \lambda_i \delta_{r_i}$ and a bijection $\pi : A \rightarrow A'$, write

$$\mu\pi := \sum_i \lambda_i \delta_{r_i\pi}$$

for the posterior law obtained by relabelling actions through π .

Lemma 43 (Exact cross-fibre relabelling). *Action coarsening implies ordinal cross-fibre equivalence for bijections. The zero/full-revelation selection in Lemma 42 makes that equivalence exact: for every bijection $\pi : A \rightarrow A'$ and full-support $q \in \Delta(A)$,*

$$F_{q\pi}(\mu\pi) = F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$. In particular, no-information experiments have the same selected value zero on every finite action set.

Proof. Ordinal equivalence. Let $\pi : A \rightarrow A'$ be a bijection. View π as the deterministic kernel $S(a' | a) = \mathbf{1}_{a'=\pi(a)}$. For any channel $P : A \rightarrow \Delta(X)$ and full-support $q \in \Delta(A)$, the Bayesian pushforward satisfies $S^q P = P\pi^{-1}$ (explicitly: $(S^q P)(x | a') = P(x | \pi^{-1}(a'))$). Clause (DP-act) of Condition (P4) gives

$$q^0 \succeq_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

For the reverse inequality, apply (DP-act) to the bijection $\pi^{-1} : A' \rightarrow A$, the prior $q\pi \in \Delta(A')$, and the channel $P\pi^{-1} : A' \rightarrow \Delta(X)$. Then $(\pi^{-1})^{q\pi}(P\pi^{-1}) = P$ (the inverse bijection undoes the relabeling), and $(q\pi)\pi^{-1} = q$. Hence

$$(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1.$$

By the irrelevant-blocks clause of Condition (P3), block comparisons depend only on the pair of channels involved, not on numeric labels. The comparison $(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1$ asks whether $(q\pi, P\pi^{-1})$ is at least as good as (q, P) . Swapping the block labels (so that P becomes block 0 and $P\pi^{-1}$ becomes block 1) gives the equivalent statement $q^0 \preceq_{P \sqcup P\pi^{-1}} (q\pi)^1$. Combining with the first inequality gives

$$q^0 \sim_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

If P implements $\mu \in \mathcal{M}_q$, then $P\pi^{-1}$ implements $\mu\pi \in \mathcal{M}_{q\pi}$: for every positive-probability record x ,

$$m_{q\pi, P\pi^{-1}}(x) = m_{q, P}(x), \quad r_x^{q\pi, P\pi^{-1}} = r_x^{q, P} \pi.$$

Now take arbitrary $\mu, \nu \in \mathcal{M}_q$, and choose channels P, Q implementing them at prior q . Let E be the four-block environment with blocks $P, P\pi^{-1}, Q$, and $Q\pi^{-1}$. Write x, x', y, y' for the block-supported lotteries $q, q\pi, q$, and $q\pi$ on these four blocks, respectively. By Lemma 36, the

two two-block indifferences give $x \sim_E x'$ and $y \sim_E y'$. Hence, by transitivity of the weak order on E ,

$$x \succeq_E y \iff x' \succeq_E y'.$$

Applying Lemma 36 again to delete irrelevant blocks gives

$$q^0 \succeq_{P \sqcup Q} q^1 \iff (q\pi)^0 \succeq_{P\pi^{-1} \sqcup Q\pi^{-1}} (q\pi)^1.$$

By Lemma 41 on the two fibres, this is equivalent to

$$F_q(\mu) \geq F_q(\nu) \iff F_{q\pi}(\mu\pi) \geq F_{q\pi}(\nu\pi).$$

The map $\mu \mapsto \mu\pi$ is an affine bijection from $\mathcal{M}_q$ to $\mathcal{M}_{q\pi}$, with inverse induced by π^{-1} . Therefore $G(\mu) := F_{q\pi}(\mu\pi)$ is a continuous affine representative of the same weak order on $\mathcal{M}_q$ as F_q . Herstein–Milnor uniqueness gives $G = aF_q + b$ with $a > 0$. Since $\delta_q\pi = \delta_{q\pi}$ and Lemma 42 gives

$$F_q(\delta_q) = F_{q\pi}(\delta_{q\pi}) = 0,$$

we have $b = 0$. If $|A| = 1$, both representatives vanish identically. If $|A| \geq 2$, relabelling sends full revelation to full revelation, $\chi_q\pi = \chi_{q\pi}$, while the canonical selection gives $F_q(\chi_q) = F_{q\pi}(\chi_{q\pi}) = 1$. Substitution in $G = aF_q$ therefore gives $a = 1$. Hence

$$F_{q\pi}(\mu\pi) = F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$.

No-information equivalence. By Lemma 42, $F_q(\delta_q) = 0$ for every full-support q on every finite action set, independently of cardinality. $\square$

Lemma 44 (Undetectable distinctions neutrality). *Let $S : A \rightarrow A'$ be an onto deterministic partition map, and let $P : A \rightarrow \Delta(X)$ be S -measurable:*

$$S(a) = S(b) \implies P(\cdot | a) = P(\cdot | b).$$

Then for every full-support $q \in \Delta(A)$,

$$q^0 \sim_{P \sqcup S^q P} (qS)^1.$$

In particular, if $G_S : A \rightarrow \Delta(A')$ reveals the block $S(a)$, then

$$(q, G_S) \sim (qS, \text{Id}_{A'}).$$

Proof. Since S is onto and q has full support, qS has full support on A' . Thus the Bayesian pushforward $S^q P$ has no zero-row ambiguity in this lemma. View the deterministic map S as its associated stochastic kernel. One direction is exactly clause (DP-act) of Condition (P4):

$$q^0 \succeq_{P \sqcup S^q P} (qS)^1.$$

For the reverse direction, define the Bayesian refinement kernel $R : A' \rightarrow \Delta(A)$ by

$$R(a | a') = \begin{cases} q(a)/(qS)(a'), & S(a) = a', \\ 0, & \text{otherwise.} \end{cases}$$

For each $a' \in A'$, $R(\cdot | a')$ is stochastic because

$$\sum_{a \in A} R(a | a') = \frac{\sum_{a: S(a)=a'} q(a)}{(qS)(a')} = 1.$$

Moreover, for every $a \in A$,

$$((qS)R)(a) = \sum_{a' \in A'} (qS)(a')R(a | a') = (qS)(S(a)) \frac{q(a)}{(qS)(S(a))} = q(a) > 0.$$

Thus $(qS)R = q$.

Since P is S -measurable, the row of $S^q P$ at each block a' equals the common P -row on that block: if $S(a) = a'$, then

$$S^q P(x | a') = \frac{\sum_{b: S(b)=a'} q(b)P(x | b)}{(qS)(a')} = P(x | a).$$

Now pushing $S^q P$ back through R recovers P . For each $a \in A$,

$$(R^{qS}(S^q P))(x | a) = \frac{\sum_{a' \in A'} (qS)(a')R(a | a')S^q P(x | a')}{((qS)R)(a)} = S^q P(x | S(a)) = P(x | a).$$

Therefore

$$R^{qS}(S^q P) = P.$$

Apply (DP-act) to $(qS, S^q P)$ with the refinement kernel R :

$$(qS)^0 \succeq_{S^q P \sqcup P} q^1.$$

By Condition (P3) applied to the two-block environment $P \sqcup S^q P$ with ordered pair $(1, 0)$, this is equivalent to

$$(qS)^1 \succeq_{P \sqcup S^q P} q^0.$$

Together the two inequalities give the desired indifference. $\square$

For a nonempty $B \subseteq A$ and a finite law μ whose barycentre is supported on B , non-negativity implies that every atom p of μ is supported on B . Define its support projection by

$$\pi_B(\mu) := \sum_{p \in \text{supp}(\mu)} \mu(\{p\}) \delta_{p|_B}.$$

It is a finite law on $\Delta(B)$ with barycentre $b(\mu)|_B$.

Lemma 45 (Support restriction). *Let $r \in \Delta(A)$ have support $B := \text{supp}(r) \subsetneq A$. Then:*

- (i) *Posterior-law reduction. For any channel $P : A \rightarrow \Delta(X)$, the posterior law $\mu_{r,P}$ depends only on the rows $P(\cdot | b)$ for $b \in B$. Rows $P(\cdot | a)$ with $a \notin B$ have zero contribution because $r(a) = 0$.*
- (ii) *Preference equivalence. Two channels $P, Q : A \rightarrow \Delta(X)$ that agree on B (i.e. $P(\cdot | b) = Q(\cdot | b)$ for all $b \in B$) are indifferent at prior r .*
- (iii) *Face identification. For every channel $P : A \rightarrow \Delta(X)$,*

$$(r, P) \sim (r|_B, P|_B)$$

as a two-block comparison; consequently the comparison between channels P, Q at prior r agrees with the comparison between $P|_B, Q|_B$ at the full-support prior $r|_B \in \Delta(B)$. This is proved by projecting A to B and embedding B back into A , using the all-completions form of clause (DP-act) of Condition (P4) for the embedding.

- (iv) *Representative. Define $F_r : \mathcal{M}_r \rightarrow \mathbb{R}$ by $F_r(\mu) := F_{r|_B}^B(\pi_B(\mu))$, where π_B projects posterior laws from $\Delta(A)$ to $\Delta(B)$ and $F_{r|_B}^B$ is the Lemma 41 representative on $\Delta(B)$. This F_r represents continuation comparisons at prior r in action set A .*

Proof. In this proof, $(s, R) \succeq (t, T)$ abbreviates the block comparison $s^0 \succeq_{R \sqcup T} t^1$, and $\sim$ denotes both weak inequalities. These comparisons involve different priors and different action sets; they chain by transitivity in Lemma 36(a), which is stated across such pairs.

Part (i): The marginal probability $m_{r,P}(x) = \sum_a r(a)P(x | a)$ and the Bayes numerator $r(a)P(x | a)$ are both zero for $a \notin B$. Hence $\mu_{r,P}$ is determined by $\{P(\cdot | b) : b \in B\}$.

Part (ii): Let P and $\tilde{P}$ be channels on A that agree on B . We first show $(r, P) \sim (r, \tilde{P})$. Fix $b_0 \in B$, and let $S : A \rightarrow B$ be the deterministic map with $S(b) = b$ for $b \in B$ and $S(a) = b_0$ for $a \notin B$. Then $rS = r|_B$ and the Bayesian pushforward S^*P is $P|_B$ on B . Clause (DP-act) gives

$$(r, P) \succeq (r|_B, P|_B).$$

Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$, and the Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on B and unconstrained on $A \setminus B$. By the all-completions convention in (DP-act), choose the completion to be $\tilde{P}$. Clause (DP-act) gives

$$(r|_B, P|_B) \succeq (r, \tilde{P}).$$

Thus $(r, P) \succeq (r, \tilde{P})$. Applying the same projection/inclusion argument with $\tilde{P}$ as the original channel and choosing P as the inclusion completion gives $(r, \tilde{P}) \succeq (r, P)$. Hence $(r, P) \sim (r, \tilde{P})$.

If P, Q agree on B , choose a common extension $\tilde{P} = \tilde{Q}$ of this common restriction to all of A (for instance, use one fixed off-support row). The preceding paragraph gives $(r, P) \sim (r, \tilde{P})$ and $(r, Q) \sim (r, \tilde{Q})$; since $\tilde{P} = \tilde{Q}$, common-environment transitivity gives $(r, P) \sim (r, Q)$.

Part (iii): We first prove $(r, P) \sim (r|_B, P|_B)$ for every channel $P : A \rightarrow \Delta(X)$.

Forward direction ((DP-act) projection). Let $S : A \rightarrow B$ map all $a \notin B$ to some fixed $b_0 \in B$. The coarsened prior is $rS = r|_B$ (since $r(a) = 0$ for $a \notin B$), and the Bayesian pushforward S^*P equals $P|_B$ on B . By (DP-act),

$$(r, P) \succeq (r|_B, P|_B).$$

Reverse direction ((DP-act) inclusion with completion). Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$. The Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on B and arbitrary on $A \setminus B$. Choose the (DP-act) completion to be any extension $\tilde{P} : A \rightarrow \Delta(X)$ of $P|_B$. Then

$$(r|_B, P|_B) \succeq (r, \tilde{P}).$$

By Part (ii), $(r, \tilde{P}) \sim (r, P)$ because $\tilde{P}$ and P agree on $B = \text{supp}(r)$. Hence $(r|_B, P|_B) \succeq (r, P)$. Together the two directions give $(r, P) \sim (r|_B, P|_B)$.

Comparison equivalence. For any P, Q : $(r, P) \succeq (r, Q)$ iff (by the above indifferences $(r, P) \sim (r|_B, P|_B)$, $(r, Q) \sim (r|_B, Q|_B)$, and Condition (P1) transitivity) $(r|_B, P|_B) \succeq (r|_B, Q|_B)$.

Part (iv): If $\mu \in \mathcal{M}_r$, every posterior in the finite support of μ is supported on B ; otherwise the barycentre r would assign positive mass to $A \setminus B$. Hence $\pi_B(\mu) \in \mathcal{M}_{r|_B}$ is well defined. By (iii), continuation comparisons at r are isomorphic to those at $r|_B$. The Lemma 41 representative $F_{r|_B}^B$ on $\Delta(B)$ therefore induces a representative on $\mathcal{M}_r$ via π_B . $\square$

Stage 3: product transport and the cross-prior bridge. An independent background is first shown to be ordinally inert. Separate affinity then yields the most general compatible product expression, $x + y + \kappa xy$, after a coherent positive gauge. Crucially, Lemma 49 is proved before any coarse-revelation value is used: product structure and face scales alone do not identify cross-prior block comparisons.

Lemma 46 (Derived background inertness). *Assume Conditions (P1) and (P3)–(P5). Let A_1, A_2 be finite action sets and let $q_1 \in \Delta(A_1)$ and $q_2 \in \Delta(A_2)$ have full support. Then, for all channels P_1, Q_1 on A_1 and every background channel R_2 on A_2 ,*

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \iff q_1^0 \succeq_{P_1 \sqcup Q_1} q_1^1, \quad (\text{BI})$$

and symmetrically for comparisons in the second component with a common first-component background. In particular the left-hand comparison does not depend on the common background: for all backgrounds R_2, S_2 on A_2 ,

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \iff (q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes S_2) \sqcup (Q_1 \otimes S_2)} (q_1 \otimes q_2)^1.$$

Proof. Throughout, for a prior s on an action set A and channels V, W on A write $V \succeq_s W$ for the block comparison $s^0 \succeq_{V \sqcup W} s^1$, and $(s, V) \sim (t, W)$ for the two-block indifference between the pairs. By Lemma 36(a) these comparisons form a weak order for each fixed prior and chain transitively across pairs with different priors and action sets. By Lemma 38, at a full-support prior a channel may be replaced by any channel with the same posterior law, in weak and in strict comparisons alike; we call this *substitution*.

Preliminary reduction: a common foreground alphabet. Condition (P5) requires the two continuation channels used on a given branch to share a record alphabet. If $P_1 : A_1 \rightarrow \Delta(X)$ and $Q_1 : A_1 \rightarrow \Delta(X')$, replace them by their zero-padded versions on $X \sqcup X'$, as in Step 2 of Lemma 39. Padding adds only null records, so it changes neither μ_{q_1} , nor $\mu_{q_1 \otimes q_2, \cdot \otimes R_2}$; both sides of (BI) are therefore unaffected by substitution at the full-support priors q_1 and $q_1 \otimes q_2$. Assume from now on that P_1 and Q_1 share the record alphabet X_1 .

For a channel $V : A_1 \rightarrow \Delta(X_1)$ and a finite set A_2 , write $\tilde{V}$ for its lift to the product action set,

$$\tilde{V}(z \mid (a_1, a_2)) := V(z \mid a_1),$$

and let $\tilde{R}_2(x_2 \mid (a_1, a_2)) := R_2(x_2 \mid a_2)$ be the lift of the background.

Step 1: a product channel is a two-stage channel with background first stage. Take $\tilde{R}_2$ as first stage on the action set $A_1 \times A_2$ and the constant continuation profile $\tilde{V}$ in every branch, so that every branch uses the same continuation alphabet X_1 and the compound record set $\bigsqcup_{x_2 \in X_2} (\{x_2\} \times X_1)$ is $X_2 \times X_1$. By the definition of two-stage composition,

$$(\tilde{R}_2 \triangleright \{\tilde{V}\}_{x_2})(x_2, z \mid (a_1, a_2)) = R_2(x_2 \mid a_2) V(z \mid a_1) = (V \otimes R_2)(z, x_2 \mid (a_1, a_2)).$$

The two channels therefore differ only by the bijection $(x_2, z) \mapsto (z, x_2)$ of record labels, which leaves every record probability and every posterior unchanged. Hence they induce the same posterior law at $q_1 \otimes q_2$, and substitution makes them interchangeable in every comparison at that prior. Applying this to $V = P_1$ and $V = Q_1$, the left-hand side of (BI) is equivalent to

$$\tilde{R}_2 \triangleright \{\tilde{P}_1\} \succeq_{q_1 \otimes q_2} \tilde{R}_2 \triangleright \{\tilde{Q}_1\}, \quad (*)$$

and likewise for the strict comparisons.

Step 2: every reached branch posterior is a product with first marginal q_1 . Write $q := q_1 \otimes q_2$, $m(x_2) := m_{q, \tilde{R}_2}(x_2)$ and, for $m(x_2) > 0$, $r_{x_2} := r_{x_2}^{q, \tilde{R}_2}$. Since the rows of $\tilde{R}_2$ do not depend on a_1 ,

$$m(x_2) = \sum_{a_2 \in A_2} q_2(a_2) R_2(x_2 \mid a_2) = m_{q_2, R_2}(x_2),$$

and Bayes' rule gives

$$r_{x_2}(a_1, a_2) = \frac{q_1(a_1) q_2(a_2) R_2(x_2 \mid a_2)}{m(x_2)} = q_1(a_1) q_2^{x_2}(a_2), \quad q_2^{x_2} := r_{x_2}^{q_2, R_2}.$$

Thus $r_{x_2} = q_1 \otimes q_2^{x_2}$: the background signal may change beliefs about a_2 , but in every reached branch the two components stay independent and the first marginal is still q_1 . At least one branch has positive probability.

Step 3: a lifted foreground channel is evaluated as on its own component. We claim that for every $s_2 \in \Delta(A_2)$ and every channel $V : A_1 \rightarrow \Delta(X_1)$,

$$(q_1 \otimes s_2, \tilde{V}) \sim (q_1, V). \quad (\dagger)$$

First suppose s_2 has full support, so that $q_1 \otimes s_2$ has full support on $A_1 \times A_2$. Let $S : A_1 \times A_2 \rightarrow A_1$ be the coordinate projection, an onto deterministic map, and note that $\tilde{V}$ is S -measurable, since $S(a) = S(b)$ implies $\tilde{V}(\cdot | a) = \tilde{V}(\cdot | b)$. Moreover $(q_1 \otimes s_2)S = q_1$ and, for every a_1 ,

$$(S^{q_1 \otimes s_2} \tilde{V})(z | a_1) = \frac{\sum_{a_2 \in A_2} q_1(a_1) s_2(a_2) V(z | a_1)}{q_1(a_1)} = V(z | a_1),$$

so the Bayesian pushforward of $\tilde{V}$ under S is V itself. Undetectable-distinctions neutrality (Lemma 44) gives $(\dagger)$.

If s_2 is not full support, let $B_2 := \text{supp}(s_2)$, so that $B := A_1 \times B_2$ is the support of $q_1 \otimes s_2$ and $B \subsetneq A_1 \times A_2$. The set B is the product action set $A_1 \times B_2$, so no relabelling is involved: the restricted prior is $q_1 \otimes (s_2|_{B_2})$ and the restricted channel $\tilde{V}|_B$ is the lift of V to $A_1 \times B_2$. Face identification (Lemma 45(iii)) gives $(q_1 \otimes s_2, \tilde{V}) \sim (q_1 \otimes s_2|_{B_2}, \tilde{V}|_B)$, and the full-support case just proved, applied on the action set $A_1 \times B_2$, gives $(q_1 \otimes s_2|_{B_2}, \tilde{V}|_B) \sim (q_1, V)$. Transitivity in a common block environment yields $(\dagger)$ in general.

Fix a branch with $m(x_2) > 0$ and apply $(\dagger)$ at $s_2 = q_2^{x_2}$ to $V = P_1$ and to $V = Q_1$, using $r_{x_2} = q_1 \otimes q_2^{x_2}$ from Step 2. Place the four pairs $(r_{x_2}, \tilde{P}_1)$, (q_1, P_1) , $(r_{x_2}, \tilde{Q}_1)$, (q_1, Q_1) as blocks of one finite block environment (Condition (P3), Lemma 36). The two indifferences and transitivity of the weak order in that environment (Condition (P1)) give

$$\tilde{P}_1 \succeq_{r_{x_2}} \tilde{Q}_1 \iff P_1 \succeq_{q_1} Q_1, \quad (\ddagger)$$

and, since the relation is complete, the same equivalence for the strict comparisons. In particular the branch comparison is the same in every reached branch and does not involve R_2 .

Step 4: forward implication. Assume $P_1 \succeq_{q_1} Q_1$. By $(\ddagger)$, the branch hypothesis $r_{x_2}^0 \succeq_{\tilde{P}_1 \sqcup \tilde{Q}_1} r_{x_2}^1$ of Condition (P5) holds for every branch with $m(x_2) > 0$, with the common first stage $\tilde{R}_2$ and the two constant continuation profiles. Condition (P5) yields $(*)$, hence the left-hand side of (BI) by Step 1.

Step 5: converse, via the strictness clause. Assume the left-hand side of (BI), so that $(*)$ holds by Step 1, and suppose $P_1 \succeq_{q_1} Q_1$ fails. By completeness and the reverse-block equivalence of Lemma 36, $Q_1 \succ_{q_1} P_1$, so by the strict form of $(\ddagger)$ every reached branch satisfies $\tilde{Q}_1 \succ_{r_{x_2}} \tilde{P}_1$. At least one such branch exists and has positive probability, so the strictness clause of Condition (P5), applied with the two profiles exchanged, gives

$$\tilde{R}_2 \triangleright \{\tilde{Q}_1\} \succ_{q_1 \otimes q_2} \tilde{R}_2 \triangleright \{\tilde{P}_1\},$$

contradicting $(*)$ because $\succeq_{q_1 \otimes q_2}$ is a weak order (Lemma 36). Hence $P_1 \succeq_{q_1} Q_1$, proving (BI).

The second-component clause is the same argument with the roles of the two coordinates exchanged: the common first-component background is used as the first stage and the second-component channels as the continuations. Finally, since both product comparisons in the last display of the lemma are equivalent to the single background-free comparison $P_1 \succeq_{q_1} Q_1$, they are equivalent to each other. $\square$

The lemma shows that separability across independent components is not an independent behavioural commitment. A product channel is a two-stage channel whose first stage is the background experiment: it can teach the observer about the background action, but it leaves the foreground component independent of it and its marginal unchanged, so in every reached branch the continuation comparison is the foreground comparison itself. Branchwise monotonicity then transmits that single comparison to the compound, and its strictness clause transmits it back. The derived statement is stronger than mere invariance to changing the background: it identifies the product comparison with the comparison of the foreground channels alone, which is the form used in Lemma 47 below.

For finite posterior laws $\mu = \sum_i m_i \delta_{r_i} \in \mathbf{X}_A$ and $\nu = \sum_j n_j \delta_{s_j} \in \mathbf{X}_B$, define

$$\mu \otimes \nu := \sum_{i,j} m_i n_j \delta_{r_i \otimes s_j}.$$

Bayes' rule then gives the product factorisation

$$\mu_{q_1 \otimes q_2, P_1 \otimes P_2} = \mu_{q_1, P_1} \otimes \mu_{q_2, P_2}. \quad (\text{PL})$$

Lemma 47 (Coherent product quasi-additivity). *There is a choice of positive rescalings of the zero-normalised fibrewise representatives F_q , still denoted F_q , and a scalar $\kappa \in \mathbb{R}$ such that, for all full-support priors $p \in \Delta(A)$ and $r \in \Delta(B)$ and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$,*

$$F_{p \otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) + \kappa F_p(\mu) F_r(\nu). \quad (\text{QA})$$

If one factor is a singleton, the same formula holds with its identically zero representative. Moreover, for every nondegenerate r and every $\nu \in \mathcal{M}_r$,

$$1 + \kappa F_r(\nu) > 0. \quad (\text{POS})$$

Proof. All representatives in this proof are zero-normalised: $F_q(\delta_q) = 0$. The proof first treats nondegenerate action sets, meaning action sets with at least two elements.

Step 1: product-coordinate independence. Fix full-support priors p and r on nondegenerate finite action sets and set

$$L_{p,r}(\mu, \nu) := F_{p \otimes r}(\mu \otimes \nu).$$

This map is continuous and separately affine. To identify a first-coordinate slice, let P, Q implement $\mu, \mu' \in \mathcal{M}_p$ and let R implement $\nu \in \mathcal{M}_r$. By the product factorisation (PL), $P \otimes R$ implements $\mu \otimes \nu$ and $Q \otimes R$ implements $\mu' \otimes \nu$ at the full-support prior $p \otimes r$. Hence Lemma 41, applied on the fibre $\mathcal{M}_{p \otimes r}$ and on the fibre $\mathcal{M}_p$, turns background inertness (BI) of Lemma 46 into

$$L_{p,r}(\mu, \nu) \geq L_{p,r}(\mu', \nu) \iff F_p(\mu) \geq F_p(\mu').$$

The symmetric clause of Lemma 46 identifies every second-coordinate slice with the F_r -order. Local non-triviality, the fibrewise representation, and the zero normalisation from Lemma 42 make both slice orders nonconstant: $F_p(\chi_p) > 0$ and $F_r(\chi_r) > 0$.

Step 2: the pair-specific bilinear form. Put $x(\mu) := F_p(\mu)$ and $y(\nu) := F_r(\nu)$. For fixed ν , affine-utility uniqueness gives

$$L_{p,r}(\mu, \nu) = \alpha(\nu)x(\mu) + \gamma(\nu), \quad \alpha(\nu) > 0.$$

Taking $\mu = \delta_p$ shows $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$. This is a continuous affine representative of the F_r -order and vanishes at δ_r , so

$$\gamma(\nu) = B_{p,r}y(\nu)$$

for some $B_{p,r} > 0$. Likewise, the slice at $\nu = \delta_r$ gives

$$L_{p,r}(\mu, \delta_r) = A_{p,r}x(\mu)$$

for some $A_{p,r} > 0$.

Let $\bar{\mu} := \chi_p$ and $h_p := F_p(\chi_p) > 0$. The function $\nu \mapsto L_{p,r}(\bar{\mu}, \nu)$ is again a continuous affine representative of the F_r -order. Hence there are $D_{p,r} > 0$ and a constant $E_{p,r}$ such that

$$L_{p,r}(\bar{\mu}, \nu) = D_{p,r}y(\nu) + E_{p,r}.$$

At $\nu = \delta_r$, $E_{p,r} = A_{p,r}h_p$. Comparing this identity with $L_{p,r}(\bar{\mu}, \nu) = \alpha(\nu)h_p + B_{p,r}y(\nu)$ gives

$$\alpha(\nu) = A_{p,r} + C_{p,r}y(\nu), \quad C_{p,r} := \frac{D_{p,r} - B_{p,r}}{h_p}.$$

Therefore

$$L_{p,r}(\mu, \nu) = A_{p,r}x(\mu) + B_{p,r}y(\nu) + C_{p,r}x(\mu)y(\nu). \quad (\text{BIL})$$

The coordinate-slice slopes are positive:

$$A_{p,r} + C_{p,r}y(\nu) > 0, \quad B_{p,r} + C_{p,r}x(\mu) > 0.$$

Step 3: normalising the two linear coefficients. Let

$$\alpha : ((A \times B) \times C) \longrightarrow A \times (B \times C), \quad \alpha((a, b), c) = (a, (b, c)),$$

be the canonical associating bijection. It carries the left-grouped product prior and posterior law to the right-grouped ones. Exact cross-fibre relabelling (Lemma 43) therefore makes the two evaluations equal, with no unknown affine factor. Expanding this common value under the two groupings and comparing the coefficients of x , y , and z gives

$$A_{p \otimes r, s} A_{p, r} = A_{p, r \otimes s}, \quad (\text{C1})$$

$$A_{p \otimes r, s} B_{p, r} = B_{p, r \otimes s} A_{r, s}, \quad (\text{C2})$$

$$B_{p \otimes r, s} = B_{p, r \otimes s} B_{r, s}. \quad (\text{C3})$$

Coefficient comparison is legitimate because the mixtures $(1-t)\delta_p + t\chi_p$ make x range over a nondegenerate interval, and similarly in the other coordinates.

The coordinate swap is another bijection, so exact cross-fibre relabelling gives

$$F_{r \otimes p}(\nu \otimes \mu) = F_{p \otimes r}(\mu \otimes \nu).$$

Comparing coefficients in (BIL),

$$B_{r,p} = A_{p,r}, \quad A_{r,p} = B_{p,r}, \quad C_{r,p} = C_{p,r}.$$

Thus, with $\rho(p, r) := B_{p,r}/A_{p,r}$,

$$\rho(r, p) = \rho(p, r)^{-1}. \quad (\text{SW})$$

Divide (C2) by (C1):

$$\rho(p, r) = \rho(p, r \otimes s) A_{r, s}.$$

Fix a nondegenerate reference prior q_0 on a finite action set A_0 and put $g(r) := \rho(q_0, r)$. Then

$$A_{r, s} = \frac{g(r)}{g(r \otimes s)}. \quad (\text{G})$$

The coefficients $A_{p,r}$, $B_{p,r}$, $C_{p,r}$ are unchanged when either prior is transported by a bijection. Indeed, transport the product law by the product bijection, use Lemma 43 on the two input fibres and the product fibre, and compare the unique coefficients in (BIL). Hence ρ , and therefore g , is invariant under relabelling. In particular it takes the same value on the two associations of a product prior.

Rescale every representative by $F_p^* := g(p)F_p$. Under this positive rescaling,

$$A_{p,r}^* = A_{p,r} \frac{g(p \otimes r)}{g(p)}, \quad B_{p,r}^* = B_{p,r} \frac{g(p \otimes r)}{g(r)}, \quad C_{p,r}^* = C_{p,r} \frac{g(p \otimes r)}{g(p)g(r)}.$$

Because g is relabelling-invariant, these rescaled representatives retain the exact relabelling equality of Lemma 43. Thus the associator and swap comparisons used below still carry no scalar defect. Equation (G) gives $A_{p,r}^* = 1$ for every pair. The transformed swap identity gives $B_{p,r}^* = A_{r,p}^* = 1$. Thus, after the positive gauge rescaling, $A_{p,r} = B_{p,r} = 1$ for every

nondegenerate pair (we now drop the stars). Positivity of the slice slopes is preserved because the rescaling factors are positive.

For a boundary prior $r \in \Delta(A)$ with $B = \text{supp}(r)$, redefine the post-gauge boundary representative by exact support transport,

$$F_r^A(\mu) := F_{r|_B}^B(\pi_B(\mu)),$$

where the representative on the right is the newly gauged full-support one; singleton support faces remain identically zero. Thus the support identity of Lemma 45 remains definitional after the gauge change.

Step 4: the interaction coefficient is universal. Write the remaining coefficient as $\kappa_{p,r}$. Comparing the xy , xz , yz , and xyz coefficients in the two associative expansions gives

$$\kappa_{p,r} = \kappa_{p,r \otimes s}, \quad (\text{K1})$$

$$\kappa_{p \otimes r, s} = \kappa_{p, r \otimes s}, \quad (\text{K2})$$

$$\kappa_{p \otimes r, s} = \kappa_{r, s}, \quad (\text{K3})$$

$$\kappa_{p \otimes r, s} \kappa_{p, r} = \kappa_{p, r \otimes s} \kappa_{r, s}. \quad (\text{K4})$$

The last equation follows from the first three, but it records the full coefficient comparison. After $A = B = 1$, exact invariance under the coordinate swap and comparison of the bilinear coefficient give

$$\kappa_{p,r} = \kappa_{r,p}. \quad (\text{KS})$$

Using (K1)–(K3) with the reference prior q_0 ,

$$\kappa_{p,r} = \kappa_{p,r \otimes q_0} = \kappa_{p \otimes r, q_0} = \kappa_{r, q_0}.$$

Hence $\kappa_{p,r}$ is independent of its first argument. By symmetry it is also independent of its second argument. Denote the common value by κ . This proves (QA) for nondegenerate factors.

The positive slice-slope condition now reads

$$1 + \kappa F_r(\nu) > 0,$$

which is (POS).

Step 5: singleton factors. Let $q_* = \delta_*$ be the prior on a singleton. Its zero-normalised representative is identically zero. If p is nondegenerate, the canonical bijection $A \rightarrow A \times \{*\}$ and exact relabeling give

$$F_{p \otimes q_*}(\mu \otimes \delta_{q_*}) = F_p(\mu).$$

The product gauge preserves this equality because $p \otimes q_*$ is a relabelling of p . Since $F_{q_*} = 0$, equation (QA) follows immediately. The reversed order is identical, and the two-singleton case is $0 = 0$. $\square$

Corollary 48 (Exact relabelling after the product gauge). *For every bijection $\pi : A \rightarrow A'$ and every full-support $q \in \Delta(A)$,*

$$F_{q\pi}(\mu\pi) = F_q(\mu) \quad (\mu \in \mathcal{M}_q).$$

Proof. Before the product gauge this is Lemma 43. The proof of Lemma 47 shows that the gauge factor $g(q)$ is unchanged by every bijection, so multiplication by that factor preserves the equality. The singleton case is immediate because both representatives vanish. $\square$

Lemma 49 (Cross-prior block representation). *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels $P : A \rightarrow \Delta(X)$ and $Q : B \rightarrow \Delta(Y)$,*

$$q^0 \succeq_{P \sqcup Q} r^1 \iff F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}). \quad (\text{BC})$$

Boundary representatives are interpreted by Lemma 45.

Proof. First suppose q and r have full support, allowing singleton action sets. At prior $q \otimes r$, quasi-additivity and zero normalisation give

$$F_{q \otimes r}(\mu_{q,P} \otimes \delta_r) = F_q(\mu_{q,P}), \quad F_{q \otimes r}(\delta_q \otimes \mu_{r,Q}) = F_r(\mu_{r,Q}).$$

These values are in the same fibre $\mathcal{M}_{q \otimes r}$, so Lemma 41 gives

$$(q \otimes r)^0 \succeq_{(P \otimes U_B) \sqcup (U_A \otimes Q)} (q \otimes r)^1 \iff F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}).$$

It remains to transfer the product-lifted comparison back to the original two blocks. For the A -side pair, Clause (DP-act) of Condition (P4) applied to the projection $\pi_A : A \times B \rightarrow A$ gives

$$(q \otimes r, P \otimes U_B) \succeq (q, P)$$

in block comparisons. Conversely, let $T : A \rightarrow \Delta(A \times B)$ be

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b).$$

Then $qT = q \otimes r$, and the Bayesian pushforward $T^q P$ has row $P(\cdot \mid a)$ at every (a, b) . Thus $T^q P$ is $P \otimes U_B$ after the canonical one-record identification of the B -experiment, and (DP-act) gives the reverse comparison. Equivalently, if the harmless record identification is not made, Lemma 38 identifies $T^q P$ and $P \otimes U_B$ because they induce the same posterior law at $q \otimes r$. Hence $(q \otimes r, P \otimes U_B)$ and (q, P) are indifferent in every relevant block comparison. The same projection/embedding argument identifies $(q \otimes r, U_A \otimes Q)$ with (r, Q) .

Place the four pairs (q, P) , $(q \otimes r, P \otimes U_B)$, (r, Q) , and $(q \otimes r, U_A \otimes Q)$ as blocks of one finite block environment. Lemma 36 and transitivity transfer the two pairwise indifferences to the two-block comparison, proving (BC) for full-support priors.

For arbitrary priors, let $A_+ := \text{supp}(q)$ and $B_+ := \text{supp}(r)$. Lemma 45 identifies (q, P) with $(q|_{A_+}, P|_{A_+})$, identifies (r, Q) with $(r|_{B_+}, Q|_{B_+})$, and identifies the two values $F_q(\mu_{q,P})$ and $F_r(\mu_{r,Q})$ with their support-face values. Applying the full-support bridge on A_+ and B_+ and pulling back by Lemma 45 proves the claim. Singleton supports use the same zero-representative convention as Lemma 47. $\square$

Stage 4: branch coefficients, cocycle, and support faces. Insertion into a reached branch is affine. Its linear part is evaluated on finite atomic signed posterior laws; the spanning argument shows that the resulting positive coefficient depends only on the prior and reached posterior. Path independence gives the cocycle and the normalized chain rule. The subsequent cardinal correction aligns auxiliary chain scales on nested faces; it does not add a behavioural assumption or change the ordinal representative.

Lemma 50 (Branch aggregation). *Let $P_1 : A \rightarrow \Delta(X_1)$ be a first-stage channel at full-support q , with branch probabilities $m(x)$ and posteriors r_x . There are positive branch coefficients $\beta(q, r_x)$, depending only on q and the reached posterior r_x , such that for every continuation profile $\{Q^x\}_{x:m(x)>0}$,*

$$F_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q, P_1}) + \sum_{x:m(x)>0} m(x) \beta(q, r_x) F_{r_x}(\mu_{r_x, Q^x}).$$

If r_x has full support, $\beta(q, r_x)$ is the full-support tangent coefficient. If $1 < |\text{supp}(r_x)| < |A|$, it is the full-to-face coefficient. If $|\text{supp}(r_x)| = 1$, the value F_{r_x} is identically zero and $\beta(q, r_x)$ is an arbitrary positive convention.

Proof. Fix the first-stage channel P_1 and its branch structure $(m(x), r_x)_{x:m(x)>0}$. For any profile of continuation channels $\{Q^x\}$, the compound posterior law at prior q is

$$\mu_{q, P_1 \triangleright \{Q^x\}} = \sum_{x:m(x)>0} m(x) \mu_{r_x, Q^x}.$$

Setup: linear part of affine functional. Since F_q is affine on $\mathcal{M}_q$ (Lemma 41), it extends uniquely to an affine functional on the affine hull of $\mathcal{M}_q$. Write L_q for the linear part: for any $\mu, \mu' \in \mathcal{M}_q$,

$$F_q(\mu) - F_q(\mu') = L_q(\mu - \mu').$$

Branch-affine structure. Fix a positive-probability branch $\bar{x}$ and fix continuation channels $H^x : A \rightarrow \Delta(Z^x)$ in all other branches $x \neq \bar{x}$. Let $r := r_{\bar{x}}$ and $m := m(\bar{x})$. If $\text{supp}(r)$ is a singleton, then $\mathcal{M}_r = \{\delta_r\}$ and $F_r(\delta_r) = 0$ by the singleton zero-normalisation convention; such branches contribute zero and their coefficient convention is recorded in Case 1 below. For the branch-affine argument in this paragraph, assume $|\text{supp}(r)| \geq 2$. If r is a boundary posterior, F_r and continuation comparisons at r are understood through Lemma 45.

Let

$$\rho_H := \sum_{x \neq \bar{x}} m(x) \mu_{r_x, H^x}.$$

For any $\nu \in \mathcal{M}_r$, the aggregate posterior law produced by using branch law ν in branch $\bar{x}$ and the fixed continuations elsewhere is

$$T_H(\nu) := m\nu + \rho_H \in \mathcal{M}_q.$$

Define $g : \mathcal{M}_r \rightarrow \mathbb{R}$ by

$$g(\nu) := F_q(T_H(\nu)).$$

Thus g depends only on ν , and $g = F_q \circ T_H$ is continuous and affine.

We now show that g and F_r represent the same weak order on $\mathcal{M}_r$. For $\nu, \nu' \in \mathcal{M}_r$, choose finite continuation channels implementing them at prior r ; when r is boundary these are implemented on $B = \text{supp}(r)$ and extended arbitrarily to A , as licensed by Lemma 45. Padding record alphabets if needed, Condition (P5) compares the two continuation profiles that differ only in branch $\bar{x}$. If $F_r(\nu) > F_r(\nu')$, Condition (P5) gives $g(\nu) > g(\nu')$. Conversely, if $g(\nu) \geq g(\nu')$ but $F_r(\nu) < F_r(\nu')$, applying the strict part of Condition (P5) to the swapped branch comparison gives $g(\nu') > g(\nu)$, a contradiction. Hence $g(\nu) \geq g(\nu')$ iff $F_r(\nu) \geq F_r(\nu')$.

Since $|\text{supp}(r)| \geq 2$, local non-triviality on the support face and Lemmas 45 and 42 imply F_r is nonconstant. The continuous affine functions g and F_r are therefore nonconstant affine representatives of the same weak order on $\mathcal{M}_r$. Herstein–Milnor uniqueness gives

$$g(\nu) = \alpha F_r(\nu) + \gamma$$

for some $\alpha > 0$ and $\gamma \in \mathbb{R}$.

The coefficient α is independent of the fixed continuations. Write $g^{(1)}$ and $g^{(2)}$ for the functionals arising from two different choices $\{H^x\}$ and $\{\tilde{H}^x\}$ in the other branches. Both represent the same order as F_r , so $g^{(1)} = \alpha_1 F_r + \gamma_1$ and $g^{(2)} = \alpha_2 F_r + \gamma_2$ with $\alpha_1, \alpha_2 > 0$. To show $\alpha_1 = \alpha_2$, consider the difference $g^{(1)} - g^{(2)} = (\alpha_1 - \alpha_2)F_r + (\gamma_1 - \gamma_2)$. For any $\nu \in \mathcal{M}_r$,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = F_q\left(m(\bar{x})\nu + \sum_{x \neq \bar{x}} m(x) \mu_{r_x, H^x}\right) - F_q\left(m(\bar{x})\nu + \sum_{x \neq \bar{x}} m(x) \mu_{r_x, \tilde{H}^x}\right).$$

Both arguments are posterior laws in $\mathcal{M}_q$, differing only in the fixed non- $\bar{x}$ components. Since L_q has already been defined on the affine-hull difference space,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = L_q(\rho_1 - \rho_2),$$

where ρ_i is the fixed non- $\bar{x}$ background law for $g^{(i)}$. This difference is independent of ν . Since F_r is nonconstant in the present nondegenerate case, $\alpha_1 = \alpha_2$.

Path-independence of the branch coefficient via tangent-space decomposition. We now prove that $\alpha/m(\bar{x})$ depends only on the pair $(q, r_{\bar{x}})$, not on the first-stage experiment. This is the key step that converts ordinal branchwise monotonicity into a cardinal aggregation formula.

Tangent space at a posterior. For a posterior $r \in \Delta(A)$ with support $B := \text{supp}(r)$, define the *tangent space*:

$$V_r := \text{span}\{\nu - \nu' : \nu, \nu' \in \mathcal{M}_r\}.$$

Elements of V_r are signed measures with total mass zero and barycentre zero.

Tangent-space identification. For any r with full support, we claim

$$V_r = W := \{\eta \in \text{span } \Delta_f(\Delta(A)) : \eta(\Delta(A)) = 0, \int p d\eta(p) = 0\},$$

the space of finite-support signed measures on $\Delta(A)$ with zero total mass and zero barycentre. Indeed, if $\nu, \nu' \in \mathcal{M}_r$, then $\nu - \nu'$ has total mass $1 - 1 = 0$ and barycentre $r - r = 0$, so $V_r \subset W$. Conversely, given any $\eta \in W$: if $\eta = 0$ then $\eta \in V_r$ trivially. Otherwise, since η has finite support, write its Jordan decomposition $\eta = \eta^+ - \eta^-$ where $\eta^+, \eta^- \geq 0$ are finite-support positive measures with equal total mass $c > 0$. Normalise $\zeta^+ := \eta^+/c$ and $\zeta^- := \eta^-/c$; both are finite-support probability measures on $\Delta(A)$ with the same barycentre $b := \int p d\zeta^+(p) = \int p d\zeta^-(p)$. Since $b \in \Delta(A)$, the set $\{a : b(a) > 0\}$ is nonempty. Because r has full support,

$$\theta := \min\left\{1, \min_{a:b(a)>0} \frac{r(a)}{b(a)}\right\} > 0.$$

Choose $0 < \lambda < \theta$ and set

$$\tau := \frac{r - \lambda b}{1 - \lambda}.$$

Then $1 - \lambda > 0$, $\sum_a \tau(a) = 1$, and $\tau(a) \geq 0$ for every a : if $b(a) > 0$ this follows from $\lambda < r(a)/b(a)$, while if $b(a) = 0$ the numerator is $r(a) > 0$. Hence $\tau \in \Delta(A)$. Define

$$\nu := \lambda \zeta^+ + (1 - \lambda) \delta_\tau, \quad \nu' := \lambda \zeta^- + (1 - \lambda) \delta_\tau.$$

Both are finite-support probability measures on $\Delta(A)$ with barycentre $\lambda b + (1 - \lambda) \tau = r$, so $\nu, \nu' \in \mathcal{M}_r$. Then $\nu - \nu' = \lambda(\zeta^+ - \zeta^-) = (\lambda/c)\eta$, hence $\eta = (c/\lambda)(\nu - \nu') \in V_r$. Thus $W \subset V_r$ and $V_r = W$. Since W depends only on the full-support property (not on the specific prior), $V_q = V_r = V_s = W$ for all full-support priors.

Case 1: Degenerate branch posterior ($|B| = 1$). If $r = \delta_a$ for some action a , then $\mathcal{M}_{\delta_a} = \{\delta_{\delta_a}\}$ is a singleton: the only Bayes-plausible law with barycentre δ_a is the point mass at δ_a itself. Hence $V_r = \{0\}$, and every continuation experiment at prior δ_a gives the same posterior law δ_{δ_a} . We set $F_{\delta_a}(\delta_{\delta_a}) = 0$. For such branches, the term $m(x)\beta(q, r_x)F_{r_x}(\mu_{r_x, Q^x})$ is zero under any convention for $\beta(q, r_x)$.

For the aggregation formula below, assign any positive value to $\beta(q, \delta_a)$; the choice is not behaviourally identified because the corresponding continuation value is always zero.

Case 2: Non-full-support branch posterior ($1 < |B| < |A|$). If r has support $B \subsetneq A$ with $|B| \geq 2$, apply Lemma 45. Continuation comparisons at prior r reduce to those at $r|_B \in \Delta(B)$, with representative $F_r := F_{r|_B}^B \circ \pi_B$.

For any finite action set C , write

$$W_C := \{\sigma \in \text{span } \Delta_f(\Delta(C)) : \sigma(\Delta(C)) = 0, \int_{\Delta(C)} p d\sigma(p) = 0\}.$$

Let $j_B : \Delta(B) \rightarrow \Delta(A)$ be the face inclusion, extending each posterior by zero outside B , and let $i_B := (j_B)_\#$ act on finite signed posterior laws. By Lemma 45(iv), $F_r^A = F_{r|_B}^B \circ \pi_B$, so on face directions

$$L_r^A(i_B \sigma) = L_{r|_B}^B(\sigma) \quad (\sigma \in W_B).$$

Since $|B| \geq 2$ and $r|_B$ is full support on B , local non-triviality and Lemma 42 give

$$L_{r|_B}^B(\chi_{r|_B} - \delta_{r|_B}) > 0,$$

so $L_{r|B}^B \neq 0$.

Full-to-face scalar. Fix any full-support $q \in \Delta(A)$. The boundary posterior r is reachable from q : choose

$$0 < \varepsilon < \min\left\{1, \min_{b \in B} \frac{q(b)}{r(b)}\right\}$$

and set $P_1(1 | a) = \varepsilon r(a)/q(a)$, with $r(a) = 0$ for $a \notin B$. Then branch 1 has positive probability ε and posterior r . For any nonzero $\sigma \in W_B$, the tangent-space identification on the face B gives $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}_{r|B}^B$. Implement ν, ν' by continuation channels on B and extend them arbitrarily to A . Lemma 45 identifies their comparison at the boundary prior r with their comparison at $r|_B$.

Condition (P5) applied to the branch reaching r , with all other branches fixed, gives the positive sign direction for feasible differences. The negative direction follows by swapping the two continuation laws, and equality follows by applying the weak part of Condition (P5) in both directions. Extending by positive scalar multiples from feasible differences to all of W_B , $L_q^A \circ i_B$ and $L_{r|B}^B$ have the same sign partition on W_B . Since $L_{r|B}^B \neq 0$, both are nonzero and there is a unique $\beta_A(q, r) > 0$ such that

$$L_q^A(i_B \sigma) = \beta_A(q, r) L_{r|B}^B(\sigma) \quad (\sigma \in W_B).$$

This is the ambient full-to-face coefficient used for boundary branches.

Case 3: Full-support branch posterior ($B = A$). This is the main case. Suppose r has full support on A .

Claim 1: aggregate change from branch variation. If a first-stage branch reaches posterior r with probability m , and the branch continuation law changes from ν to ν' (both in $\mathcal{M}_r$), then the aggregate posterior law changes by $m(\nu - \nu') \in m \cdot V_r$. This difference depends only on (m, ν, ν', r) —not on which experiment produced the branch, nor on the other branches.

Proof of Claim 1. The aggregate changes from $m\nu + \rho$ to $m\nu' + \rho$, where ρ collects the other branches. The difference is $m(\nu - \nu')$.

Claim 2: $L_q|_{V_r}$ is a positive scalar multiple of $L_r|_{V_r}$.

Proof of Claim 2. We first show that every direction in V_r is proportional to a feasible difference. By the Tangent-space identification above, any nonzero $\sigma \in V_r$ can be written as $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}_r$. Hence every nonzero $\sigma \in V_r$ is a positive scalar multiple of some feasible difference.

By Condition (P5) (branchwise monotonicity), if $\nu \succ_r \nu'$ (i.e. $F_r(\nu) > F_r(\nu')$, equivalently $L_r(\nu - \nu') > 0$), then the aggregate comparison ranks the ν -continuation strictly higher. Since the aggregate value difference is $L_q(m(\nu - \nu')) = mL_q(\nu - \nu')$ with $m > 0$, we have $L_q(\nu - \nu') > 0$ whenever $L_r(\nu - \nu') > 0$.

Reverse and equality directions. If $L_r(\nu - \nu') < 0$, completeness of $\succeq_r$ gives $\nu' \succ_r \nu$, and Condition (P5) applied to the reversed comparison gives $L_q(\nu' - \nu) > 0$, i.e. $L_q(\nu - \nu') < 0$. If $L_r(\nu - \nu') = 0$, then $\nu \sim_r \nu'$, so both $\nu \succeq_r \nu'$ and $\nu' \succeq_r \nu$; applying Condition (P5) weakly in both directions gives $L_q(\nu - \nu') \geq 0$ and $L_q(\nu - \nu') \leq 0$, hence $L_q(\nu - \nu') = 0$.

Since every nonzero $\sigma \in V_r$ is proportional to some feasible difference, the sign agreement extends from feasible differences to all of V_r : for any $\sigma \in V_r$, $L_q(\sigma) > 0$ iff $L_r(\sigma) > 0$, $L_q(\sigma) < 0$ iff $L_r(\sigma) < 0$, and $L_q(\sigma) = 0$ iff $L_r(\sigma) = 0$.

Two nonzero linear functionals on a vector space that have the same positive half-space must be positive scalar multiples of each other. (If $L_r \neq 0$, then $\ker L_q = \ker L_r$ and they lie in the same half-space, so $L_q = cL_r$ for some $c > 0$.) Since r has full support and local non-triviality gives $F_r(\chi_r) > F_r(\delta_r) = 0$, we have $L_r \neq 0$. Hence there exists $\beta(q, r) > 0$ such that

$$L_q(\sigma) = \beta(q, r) L_r(\sigma) \quad \text{for all } \sigma \in V_r.$$

Path-independence. Consider any first-stage experiment reaching posterior r in some branch with probability m . Changing the branch continuation from ν to ν' changes the aggregate value

by

$$L_q(m(\nu - \nu')) = m \beta(q, r) L_r(\nu - \nu') = m \beta(q, r) (F_r(\nu) - F_r(\nu')).$$

The coefficient $m \beta(q, r)$ depends only on (q, r, m) . In particular, $\beta(q, r)$ is independent of which experiment reached r and independent of the other branches.

Comparing with the earlier notation $g(\nu) = \alpha F_r(\nu) + \gamma$: the value difference is $g(\nu) - g(\nu') = \alpha(F_r(\nu) - F_r(\nu'))$, so $\alpha = m \beta(q, r)$ and hence $\alpha/m = \beta(q, r)$ universally. The coefficient $\beta(q, r_{\bar{x}})$ is defined as the scalar from $L_q|_{V_{r_{\bar{x}}}} = \beta(q, r_{\bar{x}}) L_{r_{\bar{x}}}|_{V_{r_{\bar{x}}}}$; it depends only on q and $r_{\bar{x}}$, not on the experiment that produced the branch.

Aggregation formula. Let

$$\mu^0 := \sum_{x:m(x)>0} m(x) \delta_{r_x} = \mu_{q, P_1}.$$

For each positive-probability branch let $B_x := \text{supp}(r_x)$. If $B_x = A$, set $v_x := \mu_{r_x, Q^x} - \delta_{r_x} \in W_A$. If $1 < |B_x| < |A|$, set

$$\bar{v}_x := \pi_{B_x}(\mu_{r_x, Q^x}) - \delta_{r_x|_{B_x}} \in W_{B_x}, \quad v_x := i_{B_x} \bar{v}_x \in W_A.$$

If $|B_x| = 1$, set $v_x := 0$. Then

$$\mu_{q, P_1 \triangleright \{Q^x\}} = \mu^0 + \sum_{x:m(x)>0} m(x) v_x.$$

Since $V_q = W_A$, affinity of F_q gives

$$F_q(\mu_{q, P_1 \triangleright \{Q^x\}}) - F_q(\mu^0) = \sum_{x:m(x)>0} m(x) L_q^A(v_x).$$

For a full-support branch $B_x = A$, Case 3 gives

$$L_q^A(v_x) = \beta(q, r_x) L_{r_x}^A(v_x) = \beta(q, r_x) F_{r_x}(\mu_{r_x, Q^x}).$$

For a nondegenerate boundary branch $1 < |B_x| < |A|$, Case 2 gives

$$L_q^A(v_x) = L_q^A(i_{B_x} \bar{v}_x) = \beta_A(q, r_x) L_{r_x|_{B_x}}^{B_x}(\bar{v}_x) = \beta_A(q, r_x) F_{r_x}(\mu_{r_x, Q^x}).$$

The last equalities use zero normalisation: Lemma 42 in full support, Lemma 45 plus Lemma 42 on the support face in the boundary case, and the singleton convention when $|B_x| = 1$. Therefore

$$F_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q, P_1}) + \sum_{x:m(x)>0} m(x) \beta(q, r_x) F_{r_x}(\mu_{r_x, Q^x}),$$

where $\beta(q, r_x)$ denotes the Case 3 coefficient when $B_x = A$, the Case 2 full-to-face coefficient when $1 < |B_x| < |A|$, and any positive singleton convention when $|B_x| = 1$. $\square$

Lemma 51 (Branch-coefficient cocycle and normalised chain rule). *Let the full-support and full-to-face coefficients be those constructed in Lemma 50, applied in the relevant ambient simplex or intrinsically on the relevant support face. If q has full support and r, s are nondegenerate priors with*

$$\text{supp}(s) \subseteq \text{supp}(r) \subseteq \text{supp}(q),$$

then the coefficients satisfy the cocycle identity

$$\beta(q, s) = \beta(q, r) \beta(r, s).$$

Hence $\beta(q, r) = a_q/a_r$ for positive scales a_q , with singleton scales fixed by convention. The rescaled values

$$\hat{F}_q(\mu) := \frac{F_q(\mu)}{a_q}$$

satisfy the exact chain rule: for every first-stage channel P_1 at full-support q and every continuation profile $\{Q^x\}_{x:m(x)>0}$,

$$\hat{F}_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = \hat{F}_q(\mu_{q, P_1}) + \sum_{x:m(x)>0} m(x) \hat{F}_{r_x}(\mu_{r_x, Q^x}).$$

Proof. Full-support cocycle. First consider reachable full-support $q, r, s \in \text{int } \Delta(A)$ with $|A| \geq 2$. For such priors, all three tangent spaces coincide. By the tangent-space identification in Lemma 50, $V_q = V_r = V_s = W$ for all full-support priors. The full-support coefficient identities constructed there share a common domain W :

$$L_q|_W = \beta(q, r)L_r|_W, \quad L_r|_W = \beta(r, s)L_s|_W.$$

Substituting the second into the first:

$$L_q|_W = \beta(q, r)\beta(r, s)L_s|_W.$$

Comparing with $L_q|_W = \beta(q, s)L_s|_W$ and using $L_s \neq 0$:

$$\beta(q, s) = \beta(q, r)\beta(r, s).$$

Reachability. The above uses the full-support coefficient construction for the pairs (q, r) , (r, s) , and (q, s) . Each invocation requires the pair to be reachable: the second prior must be a possible posterior of some experiment at the first prior. For any full-support q, r , construct a two-record experiment P_1 with $P_1(1 | a) = \varepsilon r(a)/q(a)$ for $0 < \varepsilon < \min\{1, \min_a q(a)/r(a)\}$. Record 1 has probability ε and posterior r ; record 0 has posterior $(q - \varepsilon r)/(1 - \varepsilon)$. Hence any full-support r is reachable from any full-support q , so the construction applies to all full-support pairs. The cocycle therefore holds for all full-support triples.

Fix a full-support basepoint q_0 , set $a_{q_0} = 1$, and for full-support q define

$$a_q := \frac{1}{\beta(q_0, q)}.$$

By the full-support cocycle, this is equivalently $a_q = \beta(q, q_0)$, and

$$\beta(q, r) = \frac{\beta(q, q_0)}{\beta(r, q_0)} = \frac{a_q}{a_r}.$$

Boundary coefficients and face scales. If r is nondegenerate with support $B \subsetneq A$, Lemma 50 constructs the ambient full-to-face coefficient $\beta_A(q, r) > 0$ by

$$L_q^A i_B = \beta_A(q, r) L_r^B|_B.$$

When the ambient action set is clear, write this coefficient as $\beta(q, r)$.

When nondegenerate boundary priors r, s have common support B , $\beta_A(r, s)$ means the unique $c > 0$ satisfying

$$L_r^A(i_B \sigma) = c L_s^A(i_B \sigma) \quad (\sigma \in W_B).$$

This is a scalar on the common face, not on all of W_A . Since

$$L_r^A \circ i_B = L_r^B|_B, \quad L_s^A \circ i_B = L_s^B|_B,$$

and the intrinsic face scalar satisfies $L_{r|B}^B(\sigma) = \beta_B(r|B, s|B)L_{s|B}^B(\sigma)$, we have

$$\beta_A(r, s) = \beta_B(r|B, s|B).$$

Fix the same full-support basepoint q_0 on $\Delta(A)$. If r_0 is any nondegenerate face basepoint with support B , then for every nondegenerate r with support B ,

$$\beta_A(q_0, r) = \beta_A(q_0, r_0) \beta_B(r_0|B, r|B).$$

Indeed, both sides are the coefficient multiplying $L_{r|B}^B$ to obtain $L_{q_0}^A \circ i_B$, and $L_{r|B}^B \neq 0$. Define the ambient boundary scale by

$$a_r := \frac{1}{\beta_A(q_0, r)}.$$

If the face scale is based at r_0 with $a_{r_0} := 1/\beta_A(q_0, r_0)$, then

$$a_{r|B}^B := \frac{a_{r_0}}{\beta_B(r_0|B, r|B)} = \frac{1}{\beta_A(q_0, r_0)\beta_B(r_0|B, r|B)} = \frac{1}{\beta_A(q_0, r)} = a_r.$$

Thus the ambient and intrinsic scales agree: treating r as a boundary posterior in $\Delta(A)$ or as a full-support prior on $\Delta(B)$ yields the same normalised representative $\hat{F}_r$.

Extension to boundary posteriors. If r is nondegenerate with support $B \subsetneq A$, Lemma 50 defines $\beta(q, r) = \beta_A(q, r)$ by

$$L_q^A i_B = \beta_A(q, r) L_{r|B}^B.$$

Using the full-support identity $L_q^A = \beta(q, q_0) L_{q_0}^A$ and restricting to $i_B(W_B)$,

$$L_q^A i_B = \beta(q, q_0) L_{q_0}^A i_B = \beta(q, q_0) \beta_A(q_0, r) L_{r|B}^B.$$

Comparing with the definition of $\beta_A(q, r)$ and using $L_{r|B}^B \neq 0$ gives

$$\beta(q, r) = \beta(q, q_0) \beta(q_0, r) = \frac{a_q}{a_r},$$

where $a_r = 1/\beta(q_0, r)$ by the boundary-scale definition above.

Singleton scale conventions. If $r = \delta_a$ is a singleton prior, set $a_{\delta_a} := a_{q_0} = 1$ and $\hat{F}_{\delta_a} \equiv 0$. Define $\beta(q, \delta_a) := a_q/a_{\delta_a}$ as a convention. No behavioural branch coefficient is identified at a singleton target, but singleton terms in the aggregation and chain formulas are always zero.

Mixed-support cocycle. Let r, s be nondegenerate with nested supports $C := \text{supp}(s) \subseteq B := \text{supp}(r)$, and write $s|_B$ for s viewed as a boundary prior on the action set B . Interpret

$$\beta(r, s) := \beta_B(r|B, s|B),$$

the intrinsic full-to-face coefficient on action set B . Let $i_C^A : W_C \rightarrow W_A$ and $i_C^B : W_C \rightarrow W_B$ be the face inclusions. For every $\sigma \in W_C$,

$$L_q^A i_C^A \sigma = \beta(q, r) L_{r|B}^B i_C^B \sigma = \beta(q, r) \beta(r, s) L_{s|C}^C \sigma.$$

The direct full-to-face definition from Lemma 50 for the pair (q, s) gives

$$L_q^A i_C^A \sigma = \beta(q, s) L_{s|C}^C \sigma.$$

Since $L_{s|C}^C \neq 0$, $\beta(q, s) = \beta(q, r) \beta(r, s)$. If $\text{supp}(s) \not\subseteq \text{supp}(r)$, then $\beta(r, s)$ is not behaviourally defined; singleton targets are also excluded from the behavioural cocycle because their scales are conventions only.

Chain rule. Applying Lemma 50 and substituting $\beta(q, r) = a_q/a_r$ gives

$$F_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q, P_1}) + \sum_{x: m(x) > 0} m(x) \frac{a_q}{a_{r_x}} F_{r_x}(\mu_{r_x, Q^x}).$$

Dividing by a_q and writing $\widehat{F}_q(\mu) := F_q(\mu)/a_q$ yields

$$\widehat{F}_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = \widehat{F}_q(\mu_{q, P_1}) + \sum_{x: m(x) > 0} m(x) \widehat{F}_{r_x}(\mu_{r_x, Q^x}). \quad \square$$

Lemma 52 (Coherent relabelling and face scales). *The scales produced by Lemma 51 can be chosen simultaneously across all finite action sets so that:*

1. for every bijection $\pi : A \rightarrow A'$,

$$a_{q\pi}^{A'} = a_q^A;$$

2. if $\iota : B \hookrightarrow A$ is an injection with $|B| \geq 2$, r has full support on B , and q has full support on A , then the full-to-face branch coefficient for reaching the face $\iota(B)$ is

$$\beta_\iota(q, r) = \frac{a_q^A}{a_r^B}. \quad (\text{FS})$$

Singleton target scales remain a convention because every continuation value on a singleton fibre is zero.

Proof. For every nondegenerate finite action set A , first choose the full-support scales supplied by the cocycle part of Lemma 51, so that

$$\beta_A(q, q') = \frac{a_q^A}{a_{q'}^A} \quad (q, q' \in \text{int } \Delta(A)).$$

They are unique up to multiplication by one positive constant depending on A . Normalise initially by $a_{u_A}^A = 1$, where u_A is uniform. Exact relabelling invariance of the representatives implies relabelling invariance of the uniquely determined branch coefficients. Since bijections carry uniform priors to uniform priors, the initial scales are exactly invariant under bijections.

Let $\iota : B \hookrightarrow A$ be an injection between nondegenerate finite sets. By relabelling B onto its image $\iota(B)$ and using Corollary 48, it is enough to define the coefficient for the inclusion of the face $\iota(B)$; the notation below keeps the original symbol ι . Let i_ι denote the induced map on signed posterior-law tangent spaces. For full-support $q \in \Delta(A)$ and $r \in \Delta(B)$, let $\beta_\iota(q, r) > 0$ be the full-to-face scalar characterised by

$$L_q^A \circ i_\iota = \beta_\iota(q, r) L_r^B \quad \text{on } W_B.$$

The number

$$K(\iota) := \beta_\iota(q, r) \frac{a_r^B}{a_q^A} \quad (\text{K})$$

is independent of both q and r . Independence of q follows by comparing two full-support ambient priors and using $L_q^A = (a_q^A/a_{q'}^A) L_{q'}^A$ on W_A . Independence of r follows from same-face compatibility: if r, s have full support on B , then $L_r^B = (a_r^B/a_s^B) L_s^B$, and hence $\beta_\iota(q, s) = \beta_\iota(q, r) a_r^B/a_s^B$.

The defects multiply under composition. If $C \xrightarrow{j} B \xrightarrow{\iota} A$ and all three sets in the comparison are nondegenerate, then restricting first to B and then to C gives, for full-support q, r, s on A, B, C respectively,

$$\beta_{\iota \circ j}(q, s) = \beta_\iota(q, r) \beta_j(r, s).$$

Substituting the definition of K and cancelling the intermediate scale a_r^B gives

$$K(\iota \circ j) = K(\iota)K(j). \quad (\text{KC})$$

Exact relabelling invariance shows that $K(\iota)$ depends only on $n := |A|$ and $m := |B|$; write it as $K_{n,m}$. Bijections give $K_{n,n} = 1$, and

$$K_{n,\ell} = K_{n,m}K_{m,\ell} \quad (2 \leq \ell \leq m \leq n).$$

Put $t_n := K_{n,2}$, with $t_2 = 1$. Then $K_{n,m} = t_n/t_m$. Finally replace every scale on an n -point action set by

$$\tilde{a}_q^A := t_n a_q^A.$$

Under this change, the defect in (K) becomes

$$\widetilde{K}_{n,m} = K_{n,m} \frac{t_m}{t_n} = 1.$$

The rescaling is common to all priors on a fixed action set, so it leaves all within-set cocycle ratios unchanged; because t_n depends only on cardinality, it also preserves exact relabelling invariance. Dropping tildes gives (FS). Singleton target scales may be set arbitrarily because all singleton continuation terms are zero. $\square$

Stage 5: compatibility collapse and entropy identification. The product and sequential evaluations of full revelation must agree. Their compatibility produces a positive grouping-weight equation; evaluating one distribution by two groupings forces the product interaction to vanish and the chain scale to be universal. Entropy reduction and Faddeev's recursion then identify the remaining functional.

Lemma 53 (Interaction collapse and universal chain scale). *The universal interaction coefficient in (QA) is zero. Moreover, there exists a $a > 0$ such that the coherently chosen chain scales satisfy $a_q = a$ for every nondegenerate full-support prior on every finite action set. The same value may be assigned to singleton priors. Consequently,*

$$F_{p \otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) \quad (\text{PA})$$

and the branch formula has a universal chain scale.

Proof. Define, for every finite prior q read on its support,

$$H(q) := F_q(\chi_q), \quad Z(q) := 1 + \kappa H(q).$$

For a nondegenerate prior, local non-triviality gives $H(q) > 0$, and (POS) gives

$$Z(q) > 0. \quad (\text{Z+})$$

For a singleton, $H(q) = 0$ and $Z(q) = 1$.

Step 1: product revelation links the chain scale to Z . Let $q \in \Delta(A)$ and $r \in \Delta(B)$ be nondegenerate and full support. Quasi-additivity applied to full revelation gives

$$H(q \otimes r) = H(q) + H(r) + \kappa H(q)H(r). \quad (\text{QH})$$

Compute the same value sequentially. First reveal the A -coordinate and then reveal the B -coordinate in every branch. The first-stage value is $F_{q \otimes r}(\chi_q \otimes \delta_r) = H(q)$ by (QA). After observing a , the posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify its continuation full-revelation value with $H(r)$. For the injection

$$\iota_a : B \hookrightarrow A \times B, \quad b \mapsto (a, b),$$

Lemma 52 gives the branch coefficient

$$\beta_{\iota_a}(q \otimes r, r) = \frac{a_{q \otimes r}^{A \times B}}{a_r^B}.$$

The branch weights are $q(a)$ and the same coefficient and continuation value appear in every branch, so $\sum_a q(a) = 1$. Lemma 50 therefore yields

$$H(q \otimes r) = H(q) + \frac{a_{q \otimes r}}{a_r} H(r). \quad (\text{SA})$$

Comparing (QH) and (SA), and using $H(r) > 0$,

$$\frac{a_{q \otimes r}}{a_r} = Z(q). \quad (\text{R1})$$

Revealing B first gives symmetrically

$$\frac{a_{q \otimes r}}{a_q} = Z(r). \quad (\text{R2})$$

Hence $a_r Z(q) = a_q Z(r)$, so there is a universal constant $C > 0$ such that

$$a_q = CZ(q) \quad (\text{AS})$$

for every nondegenerate full-support prior on every finite action set. Assign $a_q := C$ at singleton priors, where $Z(q) = 1$ and all continuation values are zero.

If $\kappa = 0$, then $Z \equiv 1$, equation (AS) already gives universal scales, and (QA) is exact product additivity. For the remaining argument suppose $\kappa \neq 0$.

Step 2: the induced weight equation. Let q be full support on A , let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition into nonempty cells, and put

$$p_i := q(A_i), \quad q^i := q(\cdot \mid A_i), \quad p := (p_1, \dots, p_m).$$

Let $G_{\mathcal{P}}$ reveal the block. Undetectable-distinctions neutrality gives $(q, G_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}})$. The early cross-prior representation Lemma 49, importantly for the *unscaled* F 's, therefore gives

$$F_q(\mu_{q, G_{\mathcal{P}}}) = H(p). \quad (\text{BG})$$

Append full revelation in every branch. The compound posterior law is χ_q , so the chain formula and (AS) give

$$H(q) = H(p) + Z(q) \sum_{i=1}^m p_i \frac{H(q^i)}{Z(q^i)}. \quad (\text{GR})$$

Define

$$w(q) := \frac{1}{Z(q)} > 0.$$

Using $\kappa H(x) = Z(x) - 1$ in (GR) gives

$$\frac{Z(p)}{Z(q)} = \sum_{i=1}^m p_i \frac{1}{Z(q^i)}.$$

Equivalently,

$$w(q) = w(p) \sum_{i=1}^m p_i w(q^i). \quad (\text{W})$$

No continuity assertion is needed here.

Step 3: a two-grouping argument forces $w \equiv 1$. All distributions in this paragraph are read on their positive-probability supports, so the weight equation applies after support restriction. First, (W) applied to a product distribution gives

$$w(u \otimes v) = w(u)w(v), \quad (\text{M})$$

because the first-coordinate block probabilities are u and every conditional distribution is v .

Take arbitrary finite probability vectors u and v , possibly on different alphabets, and write

$$x := w(u), \quad y := w(v), \quad p := (1/2, 1/2).$$

On the labelled disjoint union $(U \times U)^0 \sqcup (V \times V)^1$, form the distribution

$$T := \frac{1}{2}(u \otimes u)^0 \sqcup \frac{1}{2}(v \otimes v)^1.$$

Grouping first by the top block and using (M),

$$w(T) = w(p) \frac{x^2 + y^2}{2}. \quad (\text{E1})$$

Now group first by the pair consisting of the top-block label and the first within-block coordinate. Its marginal is

$$S := \frac{1}{2}u^0 \sqcup \frac{1}{2}v^1,$$

and (W) gives

$$w(S) = w(p) \frac{x + y}{2}.$$

Conditional on a point in the first part of S , the remaining coordinate has distribution u ; conditional on a point in the second part, it has distribution v . A second application of (W) therefore gives

$$w(T) = w(S) \frac{x + y}{2} = w(p) \left(\frac{x + y}{2} \right)^2. \quad (\text{E2})$$

Since $w(p) > 0$, comparison of (E1) and (E2) yields

$$\frac{x^2 + y^2}{2} = \left(\frac{x + y}{2} \right)^2,$$

so $(x - y)^2 = 0$. As u and v were arbitrary, w is constant on all finite probability vectors. At a singleton prior, $H = 0$, hence $w = 1$. Therefore $w(q) \equiv 1$.

Thus $Z(q) = 1$ and $\kappa H(q) = 0$ for every q . Local non-triviality gives a nondegenerate q with $H(q) > 0$, hence $\kappa = 0$, contradicting the temporary alternative unless the interaction coefficient is zero. Equation (AS) now gives $a_q = C$ universally. Finally (QA) reduces to exact product additivity (PA). $\square$

Lemma 54 (Entropy reduction and Faddeev identification). *By Lemma 53, $a_q \equiv a$ is universal. Define*

$$\hat{F}_q := F_q/a, \quad H(q) := \hat{F}_q(\chi_q),$$

the rescaled value of full revelation. Then, for every channel P ,

$$\hat{F}_q(\mu_{q,P}) = H(q) - \sum_{x: m_{q,P}(x) > 0} m_{q,P}(x) H(r_x^{q,P}). \quad (\text{ER})$$

Moreover H satisfies Faddeev's recursion, so

$$H(q) = \alpha H(q)$$

for some $\alpha > 0$. Consequently,

$$\hat{F}_q(\mu_{q,P}) = \alpha I_{q,P}(A; X).$$

Proof. Entropy-reduction formula. Boundary values of H are read through support restriction: if q has support B , then $H(q) := H(q|_B)$, and $H(\delta_a) = 0$. Let $P : A \rightarrow \Delta(X)$ be any channel at full-support prior q . Append full revelation $\text{Id}_A : A \rightarrow \Delta(A)$ after P : the compound $P \triangleright \{\text{Id}_A\}_x$ (using Id_A in every branch) has $\mu_{r_x, \text{Id}_A} = \chi_{r_x}$ and the overall posterior law is full revelation χ_q . Applying the chain rule (Lemma 51) gives

$$\widehat{F}_q(\chi_q) = \widehat{F}_q(\mu_{q,P}) + \sum_{x:m(x)>0} m(x) \widehat{F}_{r_x}(\chi_{r_x}).$$

Hence

$$\widehat{F}_q(\mu_{q,P}) = H(q) - \sum_{x:m(x)>0} m(x) H(r_x).$$

If q is on the boundary, restrict q and P to $B = \text{supp}(q)$, apply the full-support formula on $\Delta(B)$, and pull the value and comparison back by Lemma 45; zero-probability rows and records do not affect either side of the displayed identity.

Scaled cross-prior bridge. Lemma 49 was proved for the unscaled values F_q . Since Lemma 53 has made $a_q \equiv a$ universal, it is equivalently valid for

$$V(q, P) := \widehat{F}_q(\mu_{q,P}) = F_q(\mu_{q,P})/a :$$

for all finite priors and channels,

$$q^0 \succeq_{P \sqcup Q} r^1 \iff V(q, P) \geq V(r, Q).$$

Faddeev recursion. First let q be full support and let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition into nonempty cells. Then $p_i := q(A_i) > 0$ for every i . Let $G_{\mathcal{P}} : A \rightarrow \Delta(\{1, \dots, m\})$ reveal only the block index: $G_{\mathcal{P}}(i | a) = \mathbf{1}_{a \in A_i}$. Write

$$p_i := q(A_i), \quad q^i := q(\cdot | A_i),$$

and let $p = (p_1, \dots, p_m) \in \Delta(\{1, \dots, m\})$. Consider the deterministic kernel $S : A \rightarrow \Delta(\{1, \dots, m\})$ given by $S(i | a) = \mathbf{1}_{a \in A_i}$, so $qS = p$.

By Lemma 44, since $G_{\mathcal{P}}$ is S -measurable,

$$(q, G_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}})$$

in the two-block comparison. Applying the scaled form of Lemma 49 to the forward comparison gives $V(q, G_{\mathcal{P}}) \geq V(p, \text{Id}_{\{1, \dots, m\}})$. Applying the reverse-block form from Lemma 36 and then the same bridge to the reverse comparison gives $V(p, \text{Id}_{\{1, \dots, m\}}) \geq V(q, G_{\mathcal{P}})$. Therefore

$$\widehat{F}_q(\mu_{q, G_{\mathcal{P}}}) = \widehat{F}_p(\chi_p) = H(p).$$

The entropy-reduction formula applied to $G_{\mathcal{P}}$ gives

$$H(q) = \widehat{F}_q(\chi_q) = \widehat{F}_q(\mu_{q, G_{\mathcal{P}}}) + \sum_{i=1}^m p_i \widehat{F}_{q^i}(\chi_{q^i}) = H(p) + \sum_{i=1}^m p_i H(q^i).$$

This is Faddeev's recursion. For boundary q , restrict to $\text{supp}(q)$, apply the same full-support recursion to the nonempty positive-mass intersections, and pull back by Lemma 45; equivalently, the recursion is read over positive-mass blocks only.

We now verify that H is continuous on each $\Delta_n := \{q \in \Delta(A) : |A| = n\}$.

Binary continuity via erasure calibrators. Define the binary entropy function $h : [0, 1] \rightarrow \mathbb{R}$ by $h(t) := H(t, 1 - t)$. We show h is continuous using prior-independent calibrators. By Lemma 42, support restriction, and $a > 0$, $H(q) \geq 0$ for every q , so $h(t) \geq 0$ on $[0, 1]$. If $c < 0$, then

$$\{t : h(t) \geq c\} = [0, 1], \quad \{t : h(t) \leq c\} = \emptyset,$$

both closed. Now fix $c \geq 0$. By local non-triviality, there exists a binary prior $\theta = (s, 1 - s)$ with $u := H(\theta) > 0$. Choose $n \in \mathbb{N}$ and $\lambda \in [0, 1]$ such that $c = \lambda nu$ (take n large enough that $c \leq nu$, then set $\lambda := c/(nu)$).

Let $w_n := \theta^{\otimes n}$ be the n -fold product prior on $B_n := \{0, 1\}^n$. By product additivity, $H(w_n) = nH(\theta) = nu$. Define the erasure experiment $E_{n,\lambda} : B_n \rightarrow \Delta(\{*, 1, \dots, 2^n\})$ which reveals the state fully with probability λ and reveals nothing with probability $1 - \lambda$. By the entropy-reduction formula,

$$\widehat{F}_{w_n}(\mu_{w_n, E_{n,\lambda}}) = \lambda H(w_n) = \lambda nu = c.$$

Now consider the block action space $A_c := \{0, 1\} \sqcup B_n$ and the block-diagonal channel $K_c := \text{Id}_{\{0,1\}} \sqcup E_{n,\lambda}$. For $t \in [0, 1]$, let q_t^0 be the prior $(t, 1 - t)$ supported on block $\{0, 1\}$, and let w_n^1 be w_n supported on block B_n . By the scaled form of Lemma 49 (which extends to boundary priors via Lemma 45),

$$q_t^0 \succeq_{K_c} w_n^1 \iff V(q_t, \text{Id}_{\{0,1\}}) = h(t) \geq c = V(w_n, E_{n,\lambda}).$$

Since K_c is fixed and the map $t \mapsto q_t^0$ is continuous in ℓ_1 , Condition (P2) implies that the set $\{t \in [0, 1] : h(t) \geq c\}$ is closed. Similarly, $\{t \in [0, 1] : h(t) \leq c\}$ is closed. Thus all real superlevel and sublevel sets of h are closed, so h is both upper and lower semicontinuous on $[0, 1]$, hence continuous.

Continuity on higher simplices by induction. For $n \geq 3$, write any $q \in \Delta_n$ as $q = (q_1, (1 - q_1)\bar{q})$ where $\bar{q} := (q_2/(1 - q_1), \dots, q_n/(1 - q_1)) \in \Delta_{n-1}$ for $q_1 < 1$. The Faddeev recursion with the binary partition $\{\{1\}, \{2, \dots, n\}\}$ gives

$$H(q) = h(q_1) + (1 - q_1)H(\bar{q}).$$

On the region $0 < q_1 < 1$, this is continuous: h is continuous by binary continuity, $H|_{\Delta_{n-1}}$ is continuous by the induction hypothesis, and $q \mapsto (q_1, \bar{q})$ is continuous. If $q_1 = 0$, support restriction, equivalently the positive-mass-block convention in the recursion, gives $H(q) = H(\bar{q})$ and $h(0) = 0$, so continuity at that face follows from the induction hypothesis. At $q_1 = 1$, the induction hypothesis and compactness of Δ_{n-1} imply H is bounded on Δ_{n-1} , so $(1 - q_1)H(\bar{q}) \rightarrow 0$ as $q_1 \rightarrow 1$. Thus $H(q) \rightarrow h(1) = 0 = H(\delta_1)$. The other boundary faces are handled by support restriction, relabelling on the support, and pulling back. This completes the induction.

We have verified the hypotheses of the strong-additivity form of Faddeev's finite entropy characterisation, as stated for example by Baez–Fritz–Leinster [8, Theorem 6], following Faddeev's original theorem [24]: a function on finite probability vectors that is continuous on each closed simplex, invariant under bijections, expansive under adjoining zero-probability states, zero on point masses, and strongly additive is a nonnegative multiple of Shannon entropy. Here invariance under bijections follows from Corollary 48, with boundary cases obtained by restricting to the support and pulling back; expansibility follows from support restriction; and the grouping recursion above is precisely strong additivity, read over positive-mass blocks. Faddeev's theorem therefore gives $H(q) = \alpha H(q)$ for some $\alpha \geq 0$. Local non-triviality (Condition (P1)) forces $\alpha > 0$.

Mutual information. Substituting $H = \alpha H$ into the entropy-reduction formula yields

$$\widehat{F}_q(\mu_{q,P}) = \alpha H(q) - \sum_{x: m_{q,P}(x) > 0} m_{q,P}(x) \alpha H(r_x^{q,P}) = \alpha (H(q) - \mathbb{E}[H(r_X)]) = \alpha I_{q,P}(A; X). \quad \square$$

Lemma 55 (Fixed-environment mutual-information representation). *For every finite channel $P : A \rightarrow \Delta(X)$ and all $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

Proof. Let

$$V(s, R) := \widehat{F}_s(\mu_{s,R}),$$

with boundary priors interpreted by Lemma 45. Lemma 49, divided by the universal scale from Lemma 53, gives the cross-prior block-comparison bridge: for finite action sets, priors s, t , and channels R, S ,

$$s^0 \succeq_{R \sqcup S} t^1 \iff V(s, R) \geq V(t, S),$$

including boundary priors by support restriction. Lemma 54 gives, for every finite prior/channel pair,

$$V(s, R) = \alpha I_{s,R}$$

with $\alpha > 0$.

Fix $P : A \rightarrow \Delta(X)$ and $q, q' \in \Delta(A)$. By the duplication clause of Condition (P3),

$$q \succeq_P q' \iff q^0 \succeq_{P \sqcup P} (q')^1.$$

Applying this bridge with $(s, R) = (q, P)$ and $(t, S) = (q', P)$ gives

$$q^0 \succeq_{P \sqcup P} (q')^1 \iff V(q, P) \geq V(q', P).$$

Substituting $V = \alpha I$ and using $\alpha > 0$ yields

$$q \succeq_P q' \iff I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

If q or q' is on the boundary, the bridge is read after restricting each pair to its positive-probability support and then pulling the comparison back by Lemma 45. Deleting zero-probability actions does not change mutual information, and the Condition (P3) equivalence remains the ambient fixed-channel comparison for P . $\square$

Proof of Proposition 32. That (ii) implies (i) is the necessity verification at the start of this appendix. That (i) implies (ii) is Lemma 55. For the moreover clause, apply (ii) to the fixed block environment $\bigsqcup_{k \in K} P_k$: a lottery supported on block i has constant block label, so

$$I_{q_i, \bigsqcup_{k \in K} P_k} = I_{q_i, P_i}(A_i; X_i),$$

and similarly for block j . $\square$

C.3 Uniqueness

Proof of Proposition 33. First note that the range of $(q, P) \mapsto I_{q,P}$ is $[0, \infty)$. For $q \in \Delta(A)$ and $t \in [0, 1]$, let $E_t : A \rightarrow \Delta(A \sqcup \{*\})$ reveal the realised action with probability t and otherwise report $*$:

$$E_t(a' \mid a) = t \mathbf{1}_{a'=a}, \quad E_t(* \mid a) = 1 - t.$$

Then $I_{q, E_t} = tH(q)$. In particular, if u_n is uniform on $n \geq 2$ actions, the values at u_n fill $[0, \log n]$. These intervals are unbounded, so every non-negative value is attained; mutual information is non-negative, so there are no other values.

For part (i), let V be a common-scale representation and define $\varphi : [0, \infty) \rightarrow \mathbb{R}$ by $\varphi(v) := V(q, P)$ for any pair satisfying $I_{q,P} = v$. The definition is independent of the chosen pair. Indeed, if $I_{q,P} = I_{r,Q}$, the moreover clause of Proposition 32 gives $q^0 \sim_{P \sqcup Q} r^1$, and the common-scale property gives $V(q, P) = V(r, Q)$. If $I_{q,P} > I_{r,Q}$, the block comparison is strict—the reverse weak comparison would require the opposite information inequality—so $V(q, P) > V(r, Q)$. Thus φ is strictly increasing and $V = \varphi \circ I$. Conversely, Proposition 32 shows that every $\varphi \circ I$ with φ strictly increasing is a common-scale representation.

For part (ii), every aI with $a > 0$ is a common-scale representation by part (i) and is branch-additive by the mutual-information chain rule (N3). Conversely, let the common-scale representation $V = \varphi \circ I$ be branch-additive. Composing the one-record uninformative channel with itself gives $\varphi(0) = 2\varphi(0)$, hence $\varphi(0) = 0$.

Fix $n \geq 2$, put $q = u_n$, and write $h_n := H(u_n) = \log n$. For $0 < \lambda < 1$, let the first stage be a public coin with two records and identical rows $(\lambda, 1 - \lambda)$. Its information is zero, each posterior is q , and its branch weights are λ and $1 - \lambda$. For any $x \in [0, h_n]$, use the erasure channel E_{x/h_n} in the first branch and the uninformative channel in the second. The chain rule (N3) makes the compound information λx , while branch additivity gives

$$\varphi(\lambda x) = \varphi(0) + \lambda\varphi(x) + (1 - \lambda)\varphi(0) = \lambda\varphi(x).$$

The same identity holds at $\lambda = 0, 1$ because $\varphi(0) = 0$. Taking $x = h_n$ and $\lambda = z/h_n$ therefore shows that

$$\varphi(z) = \frac{\varphi(h_n)}{h_n} z \quad (0 \leq z \leq h_n).$$

The intervals $[0, h_n]$ are nested and unbounded; since they carry the same function φ , their slopes agree. Hence $\varphi(z) = az$ on $[0, \infty)$ for a single constant a , and $a > 0$ because φ is strictly increasing. $\square$

C.4 Independence of the pure-record conditions

Proof of Proposition 35. For each condition, we construct a family of weak orders that satisfies the other four and violates it.

Condition (P1). Let every pair of lotteries be indifferent in every environment. The graphs in (P2) are then the whole parameter space, and (P3)–(P5) hold trivially; the strict clause of (P5) is vacuous. Local non-triviality fails.

Condition (P2). A discontinuous refinement of mutual information gives the required family. Choose an additive map $g : \mathbb{R} \rightarrow \mathbb{R}$ that is not of the form $g(x) = cx$, and, for a finite distribution p , set

$$H_g(p) := - \sum_{a:p(a)>0} p(a)g(\log p(a)).$$

Because $g(x + y) = g(x) + g(y)$, this entropy obeys the grouping identity and is additive on products, although it need not be continuous, concave, or non-negative. Define the associated information reduction by

$$I_{q,P}^g := H_g(q) - \sum_{x:m_{q,P}(x)>0} m_{q,P}(x)H_g(r_x^{q,P}),$$

and compare procedures lexicographically:

$$q \succeq_P q' \iff (I_{q,P}, I_{q,P}^g) \geq_{\text{lex}} (I_{q',P}, I_{q',P}^g).$$

This is a weak order, and the first coordinate gives local non-triviality. A block-supported lottery has the same two coordinates on its own block, so (P3) holds. Both coordinates obey the chain rule; hence branchwise lexicographic improvements aggregate, with strictness, and (P5) holds.

For (P4), let (A, X, A', X') carry the joint law in (DP). Shannon data processing settles the comparison whenever $I(A; X) > I(A'; X')$. In the equality case,

$$I(A; X) - I(A'; X') = I(A; X \mid A') + I(A'; X \mid X') = 0.$$

Thus $A \perp X \mid A'$ and $A' \perp X \mid X'$. Product additivity makes the corresponding two conditional I^g terms zero as well, so $I^g(A; X) = I^g(A'; X')$. The second coordinate therefore ties whenever the first one ties, which proves (P4).

It remains to choose g so that continuity fails. Let $p = (1/2, 1/4, 1/8, 1/8)$, and choose an interior $r \in \Delta(\{1, 2, 3, 4\})$ on the same Shannon-entropy level set such that $\log 2, \log r_1, \dots, \log r_4$ are linearly independent over $\mathbb{Q}$; generic r on that level set has this property. Extend these numbers to a Hamel basis and prescribe

$$g(\log 2) = 0, \quad g(\log r_1) = -1/r_1, \quad g(\log r_i) = 0 \quad (i = 2, 3, 4).$$

Then $H(p) = H(r)$ but $H_g(p) = 0 < 1 = H_g(r)$. Under full revelation, therefore, $p \not\prec_{\text{Id}_4} r$. If $p_n \rightarrow p$ moves from p toward the uniform distribution, strict concavity gives $H(p_n) > H(p) = H(r)$, so $p_n \succ_{\text{Id}_4} r$ for every n . The graph in (P2) is not closed.

Condition (P3). The labelled copies used in block environments make a simple label-dependent family possible. Treat labels as syntactic tags, with the outermost tag taking precedence, and define

$$w(a^0) = 1, \quad w(a^i) = 0 \quad (i \neq 0),$$

with $w(a) = 0$ for untagged actions. For a fixed $\varepsilon > 0$, represent each environment by

$$V_\varepsilon(q, P) := I_{q,P} + \varepsilon \sum_a q(a)w(a).$$

This continuous criterion defines a weak order and satisfies local non-triviality. Every comparison in (P4) places the original procedure in block 0 and the processed procedure in block 1, so its value difference is

$$I(A; X) - I(A'; X') + \varepsilon > 0.$$

For (P5), write $\Delta_x := I_{r_x, Q^x} - I_{r_x, R^x}$. A branch comparison is weak precisely when $\Delta_x + \varepsilon \geq 0$, while the aggregate value difference is

$$\sum_x m(x)\Delta_x + \varepsilon = \sum_x m(x)(\Delta_x + \varepsilon).$$

The weak and strict conclusions of (P5) follow immediately. Duplication coherence fails: take an untagged binary action set, $P = \text{Id}_2$, a degenerate q , and a non-degenerate q' close enough to q that $0 < H(q') < \varepsilon$. Then $q \not\prec_P q'$, whereas $q^0 \succ_{P \sqcup P} (q')^1$ because the former receives the block premium.

Condition (P4). Represent every environment by $V(q, P) := (1 + H(q))I_{q,P}$. Continuity and block coherence are immediate. In each branch comparison the entropy factor is common to the two continuations, so the mutual-information chain rule gives (P5), including strictness. To violate (P4), take $A = \{1, 2\}$, $q = (0.99, 0.01)$, $P = \text{Id}_A$, and split action 1 uniformly between two synonymous reported labels while sending action 2 to a third. The reported act still determines the original act, so mutual information is unchanged, but $H(qS) > H(q)$. The processed procedure is strictly preferred, contrary to (DP-act).

Condition (P5). Let $V(q, P) := \min\{I_{q,P}, c\}$ for a fixed $c > 0$. This continuous, weakly increasing transform of mutual information satisfies (P1)–(P4). For $c = 1$ bit, take q uniform on four actions and let the first stage reveal which of two pairs contains the realised act. Full revelation within each pair strictly improves every branch from zero to one bit, but the two compound procedures carry respectively one and two bits before capping and are therefore indifferent. The strict conclusion of (P5) fails. $\square$